GE Medical Systems

Technical Publications

Direction 5133311

Revision 9

Signa® EXCITE HD 1.5T System Installation

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1	Nov 2, 2004	Initial Release.
2	May 12, 2005	Sec 2 – Added installation instruction for CAN terminator at J155 of pen panel (MRIhc05764, MRIhc05770). Sec 4 – Revised the circuit breaker DIP switch settings. Sec 6 – Updated the cable map by adding optional remote MRU module, cables, and connections. Updated instructions for installing clamp blocks. Added note to not over–tighten gradient cable strain–reliefs at Penetration Panel. Corrected Times Microwave part numbers. Sec 8 – Added view of HP8200 computer. Updated SCIM cable installation. Added Power On information. Sec 10 – Updated cable specifications for several Run numbers.
3	Nov 30, 2005	Sec 6 – Added Mobile system interconnect cable maps. Section 7 – Removed instruction for cutting notch in front Top Cover and added instruction for how to adjust to vertical.
4	Aug 31, 2006	Revised the installation flow chart per Lean Team, Mechanical Installers, and Feild Engineers input. Section 6: Updated, added, and clarified system and cable installation procedures as a result of feedback from the above personnel. Clarified the dressing and tie–wrapping of cables in Rear Pedestal routed from LPCA cable track (MRIhc17577). Section 7: Reversed two procedures. Section 8: Revised wording on seismic anchoring. Section 10: Added table that provides extra details for the Installation Flow Chart in Section 1.
5	Jan 15, 2007	Sec 6: Corrected sub–Section numbers 6–3–9–B thru 6–3–9–D. Appendix: Section 10–3–1, corrected voltage rating for ground cable Runs 040 & 1254, and power cable Run 1183 (MRIhc19292).
6	Aug 16, 2007	Section 3: Added installation step for attaching the Canadian Registration Label. (PSR 13098263).
7	Oct 11, 2007	Section 1: Added sub–section 1–4 for non–English label installation instructions. Section 6: Added information for connection of Mobile system Heat Exchanger power cable, Run 973, to PDU in cable maps and added Heat Exchanger power cable terminal connection in PDU to table in sub–section 6–2–3–C (PSR 13120695).
8	Jun 17, 2009	Section 7: Removed section on installation of rating plates (PQR 13246367). Added English label and information on attaching non–English labels in Section 7–4 (PQR 13246299).
9	Nov 10, 2010	Sections 1 & 10: Revised Ferrous Material Handling warning, Section 6: Revised Ferrous Material Handling warnings and instructions for installing blower box. Section 9: Added step to checklist for confirming anchoring of blower box. (CAPA 1859293)

LIST OF EFFECTIVE PAGES

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

SECTION 1 – GETTING STARTED

TABLE OF CONTENTS

1–1 INTRODUCTION

This manual provides instructions to complete the installation of Singa EXCITE HD 1.5T System.

Additional options are installed according to the installation manual delivered with the option.

1–2 SITE READY CHECK FOR MECHANICAL INSTALLATION

Before equipment is delivered that is to be installed by procdures in this manual, the following must be complete to avoid delays and confusion.

Note

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

- ☐ Pre–installation work must be complete. Refer to *Direction 5133301, Signa EXCITE HD 1.5T Pre–installation*, Preinstallation Checklist tab.
- ☐ Magnet Delivery and Installation complete. Refer to Direction 2192624, *GE 1.5T & 1.0T Active Shield Magnet Delivery and Installation*.
- ☐ Magnet Subsystem should be installed. Refer to Direction 2192625, *GE 1.5T & 1.0T Active Shield Magnet and Cryogen Subsystem,* Section 1.

1–3 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
   http://gein2.med.ge.com/gib
   The FE Site Verification consists of three components that are available on the web from the main menu. They are:
   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–4 NON–ENGLISH LABEL INSTALLATION

Components and accessories delivered to this site may have English labels attached to them. For non–English sites, optional multi–language labels have also been supplied.

If required, non–English labels should be attached over the English labels.

- For Surface Coils and Coil Cables – Apply the appropriate label over the English label. The label is included with the coil component.
- For all other labels, follow the label update procedures in the manual or on the applicable multi–language label sheet.

Verify at the end of the installation that the original English labels are covered by new labels representing the native language of the customer.

1–5 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1–6 PRODUCT DELIVERY INSTRUCTIONS

The "Shipping Document" lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator's Room
  Note: It is important to make sure the Console cable Runs 1081, 1082, 1083, 1085, and 1255 are long enough to install and connect.
- Orange – Magnet Room
- Yellow – Post–installation (Phantoms, Coils).

"Product Delivery Instructions" (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M3333TA is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summery of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1–7 DELIVERY RECEIPT PROCESS

The following Delivery Receipt information was published in the GE Healthcare Technologies Field Service Procedures Manual, Rev 14.

1–7–1 ORDER STATUS

The PMI will communicate via the laptop, utilizing web based key date tracking tools, to communicate the site readiness status to the Customer Fulfillment Center for proper scheduling of the equipment delivery.

1–7–2 PREREQUISITE TO DELIVERY

Prior to delivery, the Service Engineer will download shipping information with the SmartScan application (if applicable) so they have a detailed box–level list of what equipment has been shipped. The Service Engineer will also request a download of GIB information from the SmartScan application to be retrieved later.

1–7–3 EQUIPMENT DELIVERY

The equipment will be delivered to the site via appropriate distribution processes of the specific modality. The modality maintains responsibility of the equipment until delivery is made at customer site. Service Engineer should be on site to receive what was delivered with the SmartScan bar–code reader and note if there are any damages on drivers shipping document. If a SmartScan bar–code reader is not available, the the Service Engineer is responsible for manually receiving in all equipment associated with the order.

- If there are damaged parts upon receipt, the Service Engineer should enter this information into the SmartScan application for record retention. If SmartScan is not utilized for equipment receipt, the Service Engineer should call the Missing at Site Claims number, 1–800–548–3366, to report the damaged parts.
- The customer will sign for the equipment, however it is understood that due to logistics issues and delivery schedules that this may not be possible in every situation.

After the initial equipment receipt, the scans from the bar–code scanner would be imported into the SmartScan application and compared with the ship list from Oracle. Any discrepancies (overages or underages) should be taken care of through EDQ. The GON should then be uploaded back to oracle for record retention.

It is the policy of GEHC to not install customer–supplied property.

In the event that a certain item on the order did not ship with the system, or an incorrect part was shipped, the Service Engineer will place an order for the correct part through FEMC. If the item is not available in FEMC, the Service Engineer will report the item using the Missing at Site Claims number, 1–800–548–3366.

1–8 REQUIRED TOOLS

WARNING!

FERROUS MATERIAL HAZARD! IF MAGNET IS ENERGIZED, THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Tools to install the Signa EXCITE HD 1.5T system are listed below:

TABLE 1–1
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	——	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	——	Wrecking bar
3	——	Claw hammer, 3/4 lb
4	46–271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	——	4 foot or equivalent carpenter level
6	2319156	Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
7	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586–2 Stripping tool replacement blade • 2134587 Cable slicer • 46–282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839–2 3/8 inch terminals
8	46–320273G3 or G4	Non–Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non–Magnetic Tool Kits needed. May substitute both of the following kits: • 46–320273G1 Non–Magnetic Tool Kit – Metric • 46–320273G2 Non–Magnetic Tool Kit – Inch
9	46–301450G1	Fiber optic connector repair kit (See Note 2)
10	2384858	Times Microwave LMR 1200 Stripping Tool (See Note 1)
11	2352193	Times Microwave LMR 600 Stripping Tool (See Note 1)
12	5111565	Times Microwave cutting Tool (See Note 1)
13	46–198094P1	Wrist grounding strap
14	——	Volt Meter
15	——	Extension cords, with ground conductor
16	——	Power strip, grounded type, with minimum of five outlets
17	——	Plastic or aluminum flashlight
18	——	Assorted crimp tools.
19	——	Non–magnetic level
20	——	Non–magnetic tape rule, 12 ft
21	——	Assorted drill bits
22	——	Inspection mirror
23	——	Hobby and utility knives

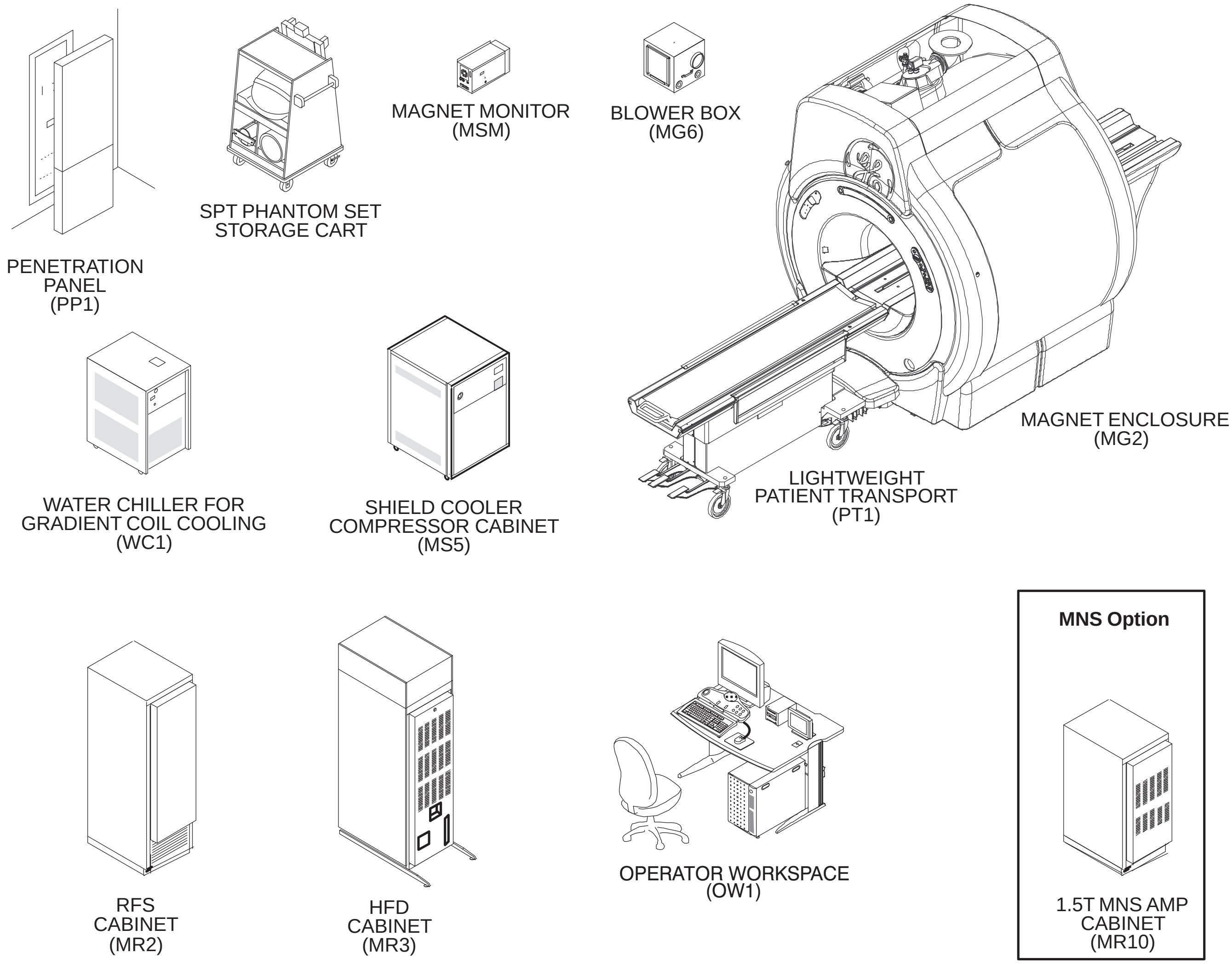
Note 1 Supplied as part of Signa.
2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.

1–9 BASIC SYSTEM

The basic Signa EXCITE HD 1.5T system, Illustration 1–1, consists of the following major equipment:

- System cooling equipment (See Illustration 1–2)
- 1.5T (30 kilogauss) LCC Magnet with Magnet Enclosure and Magnet Accessories
- Shield/Cryo Cooler Compressor Cabinet
- TwinSpeed Gradient (TRM) and RF body coils and 1.5T General Purpose Head Coil
- Main Disconnect Panel (GE supplied or customer supplied)
- System electronics cabinets:
  - –HFD Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480 Volt, 50/60 Hz with power filter
  - –EXCITE RFS Cabinet containing the Multi Generational Data Acquisition (MGD) chassis with EXCITE 8 or 16 Channel and ReFlex400 or ReFLex800, power supplies for the Magnet Enclosure system components.
  - –Narrow Band (NB) RF Amplifier Cabinet
  - –Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure
- Operator Workspace equipment: GOC Computer Cabinet with Linux PC, Mouse and Mouse Pad, LCD panel, and chair
- Pneumatic Patient Alert System
- Patient Transport Table and cradle
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows

1–9 BASIC SYSTEM (Continued)



SIGNA EXCITE 1.5T SYSTEM
ILLUSTRATION 1–1

1–10 SYSTEM OPTIONS

Purchased options may also be installed concurrently with the system upgrade. Flow charts provide steering to the Option Installation Direction that is delivered with the system.

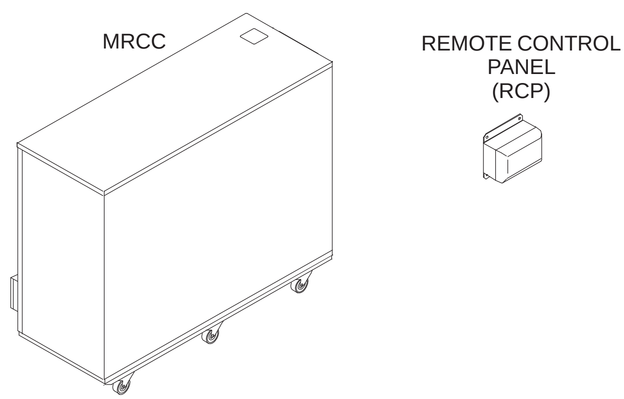
Refer to manual shipped with the option for installation of Surface Coil, Advantage Window, Camera Options, and Hardware & Software options.

Refer to installation manual shipped with the option for the following:

- M1060KM – Oxygen Monitor: *Direction 15336, Oxygen Monitor Subsystem*
- M1090TL – Second Patient Transport: *Direction 15477, Second Patient Transport Option Installation*
- M3100LA – 1.5T 2kW MNS Base Option Kit

1–11 FACILITY OPTIONS

- MR Signa Main Disconnect Panel with shield cooler compressor auto restart accessory (E4503AT) for 480/277 VAC wye 5 wire or grounded delta 4 wire, surface/semi–flush mount.
- MR Common Chiller (MRCC) (M3088TL)
  The MRCC is a single–loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller and can display supply water temperature, chiller setpoints and alarms. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor. See Illustration 1–2.



MR COMMON CHILLER (M3088TL)
ILLUSTRATION 1–2

- Signa System Seismic Anchorage Service (R4390JA).

Note

Magnet Seismic anchoring is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems.

- Oxygen Monitor Kit (M1060KM) which includes Oxygen Monitor and Remote Oxygen Monitor Module.

Note

The Oxygen Monitor does not bear a CE monogram and therefore may not be acceptable in all European countries.

1–12 INSTALLATION PROCEDURE

1–12–1 Installation Flow

The flow chart on the next page should be followed for an orderly and efficient installation of the Signa EXCITE HD 1.5T system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

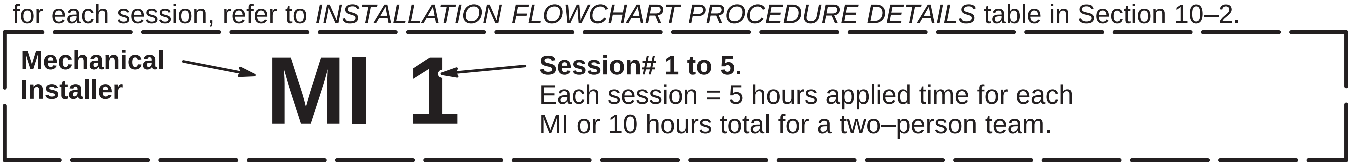
The flow chart references the Tab number and/or Section number for the respective installation procedures. These procedures are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The Procedures must be returned to the binder for future reference upon completion of the installation.

This installation flowchart has been developed assuming that all Signa EXCITE HD 1.5T equipment has been delivered together. Make sure that every part required is available before starting.

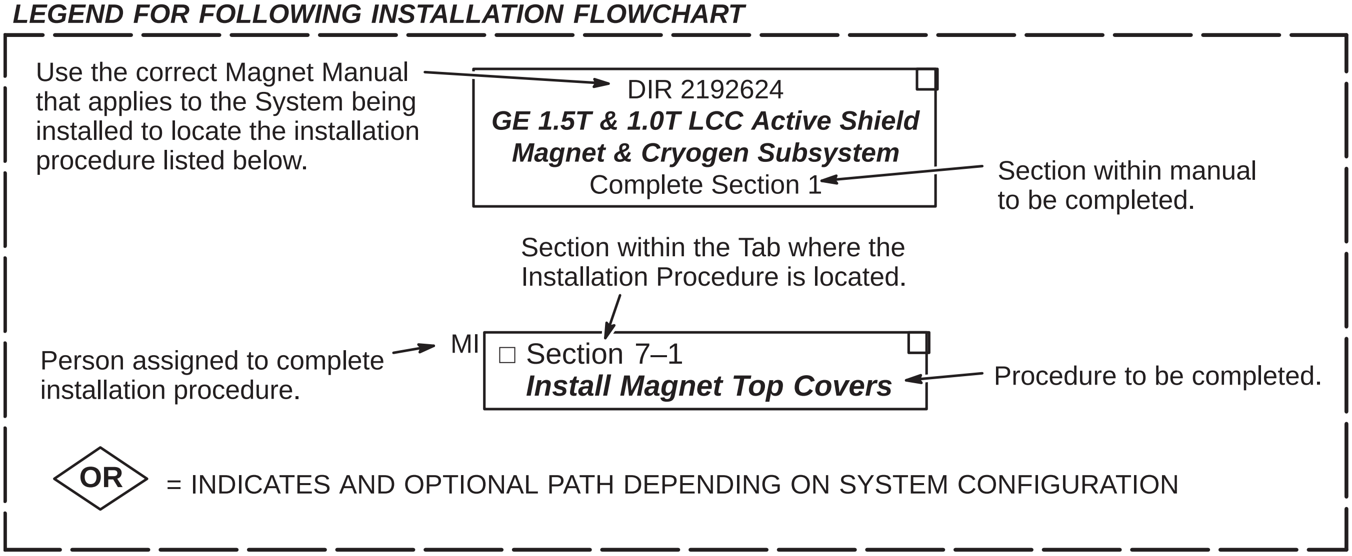
1–12–2 Installation Flowchart Explanation

Shown below are examples of the flow chart Installation Procedures and a definition of the areas within the procedures.

MILESTONE COMPLETION GOAL Refer to Legend on page 1–11. Also, for details of procedures to complete for each session, refer to *INSTALLATION FLOWCHART PROCEDURE DETAILS* table in Section 10–2.



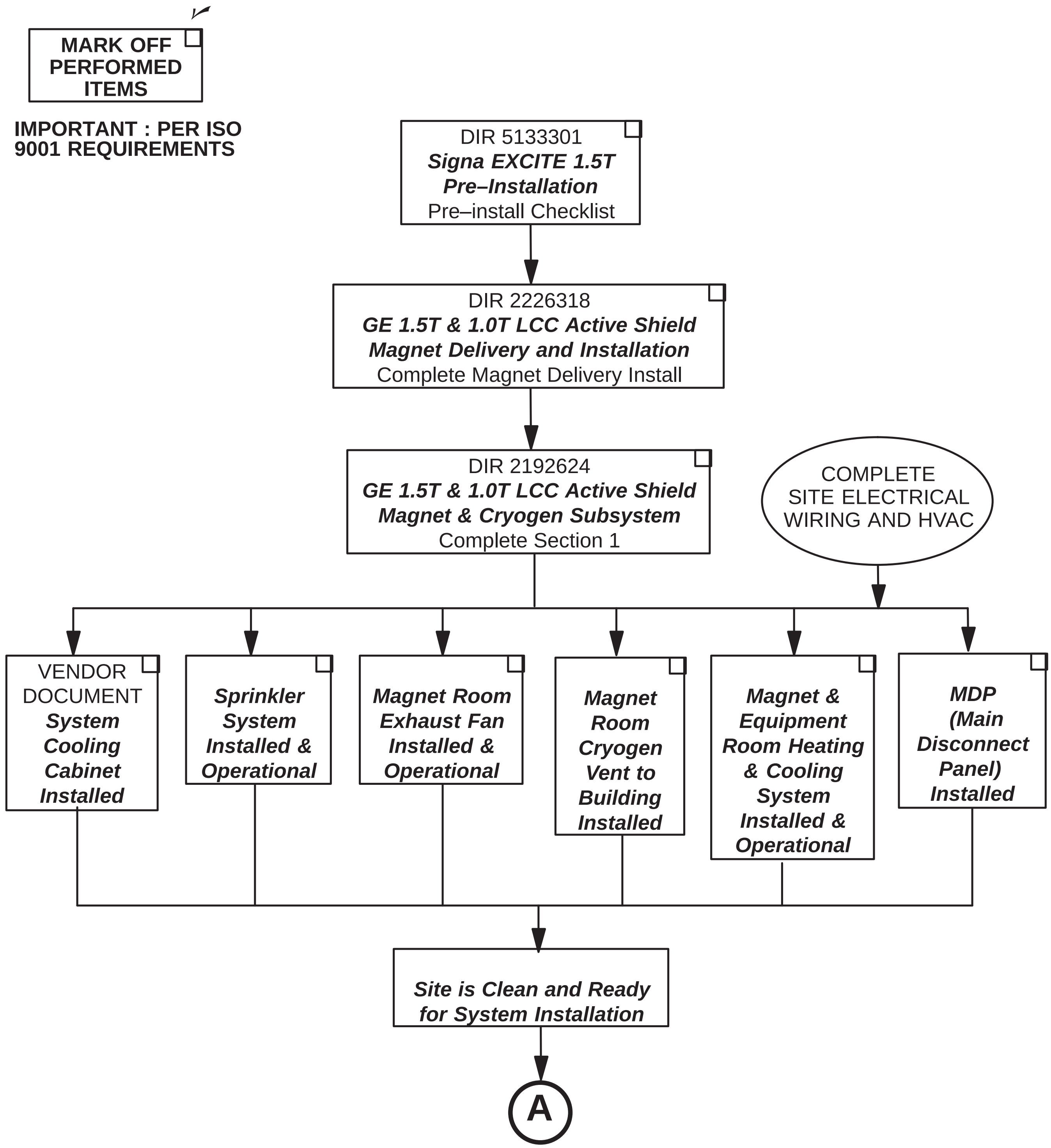
LEGEND FOR FOLLOWING INSTALLATION FLOWCHART



1–13 INSTALLATION FLOWCHART

1–13–1 PREREQUISITES FOR SYSTEM INSTALLATION

Site Installations Required Prior to System Mechanical Installation

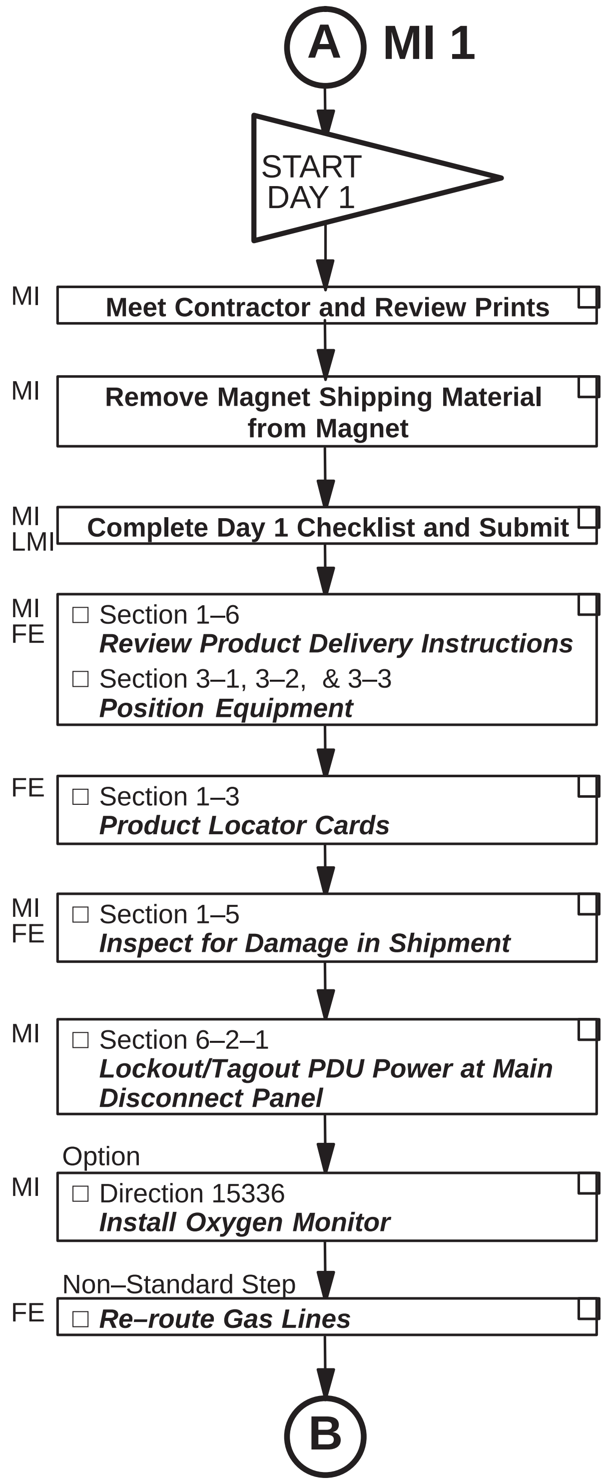


1–13–2 SYSTEM MECHANICAL INSTALLATION

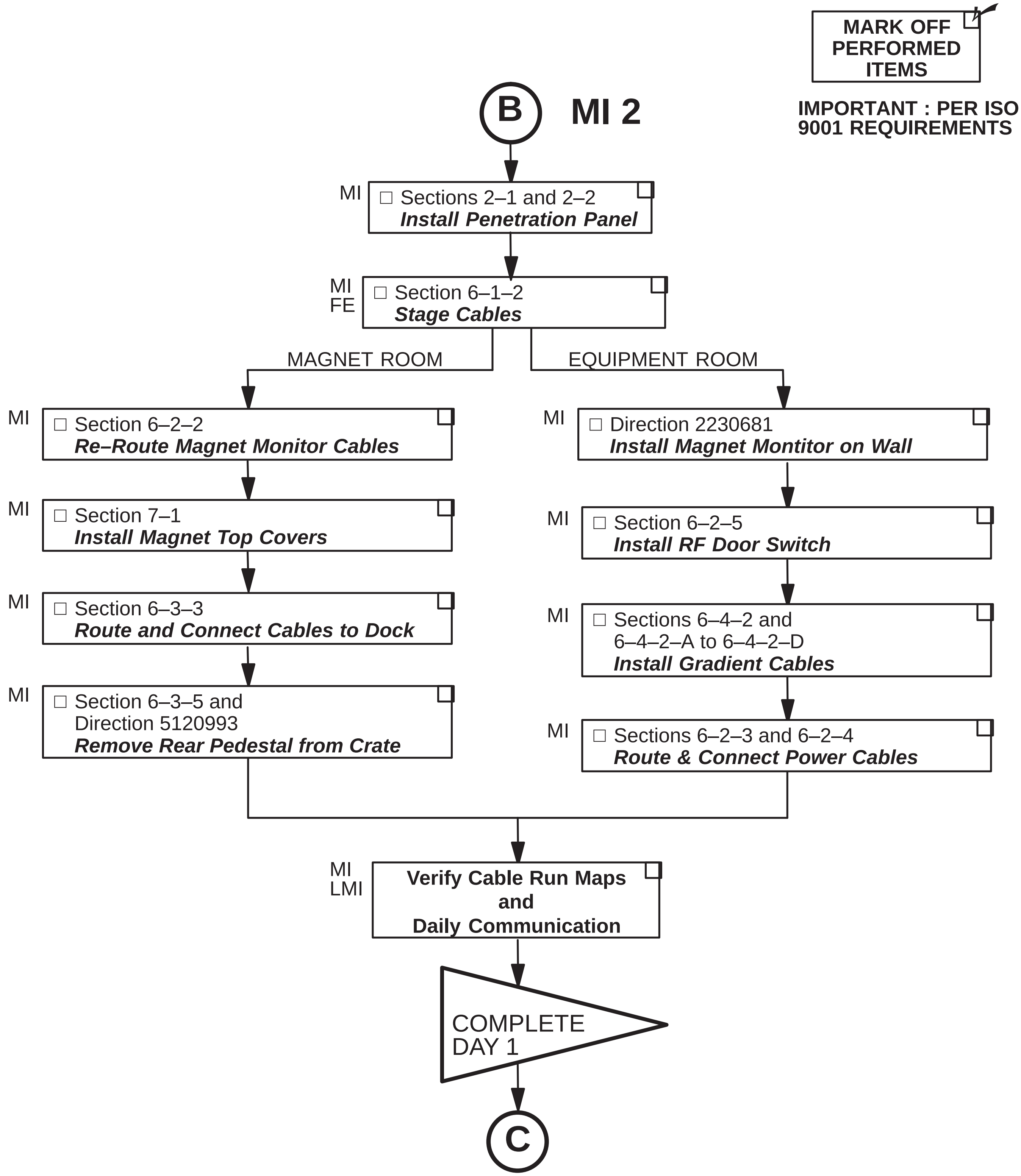
LEGEND:
FE = GE Field Engineer
LMI = Lead Mechanical Installer
MI = Mechanical Installer
MI 1 = Procedures to complete in 1st half of 1st day
MI 2 = Procedures to complete in 2nd half of 1st day
MI 3 = Procedures to complete in 1st half of 2nd day
MI 4 = Procedures to complete in 2nd half of 2nd day
MI 5 = Procedures to complete in 1st half of 3rd day

MARK OFF PERFORMED ITEMS

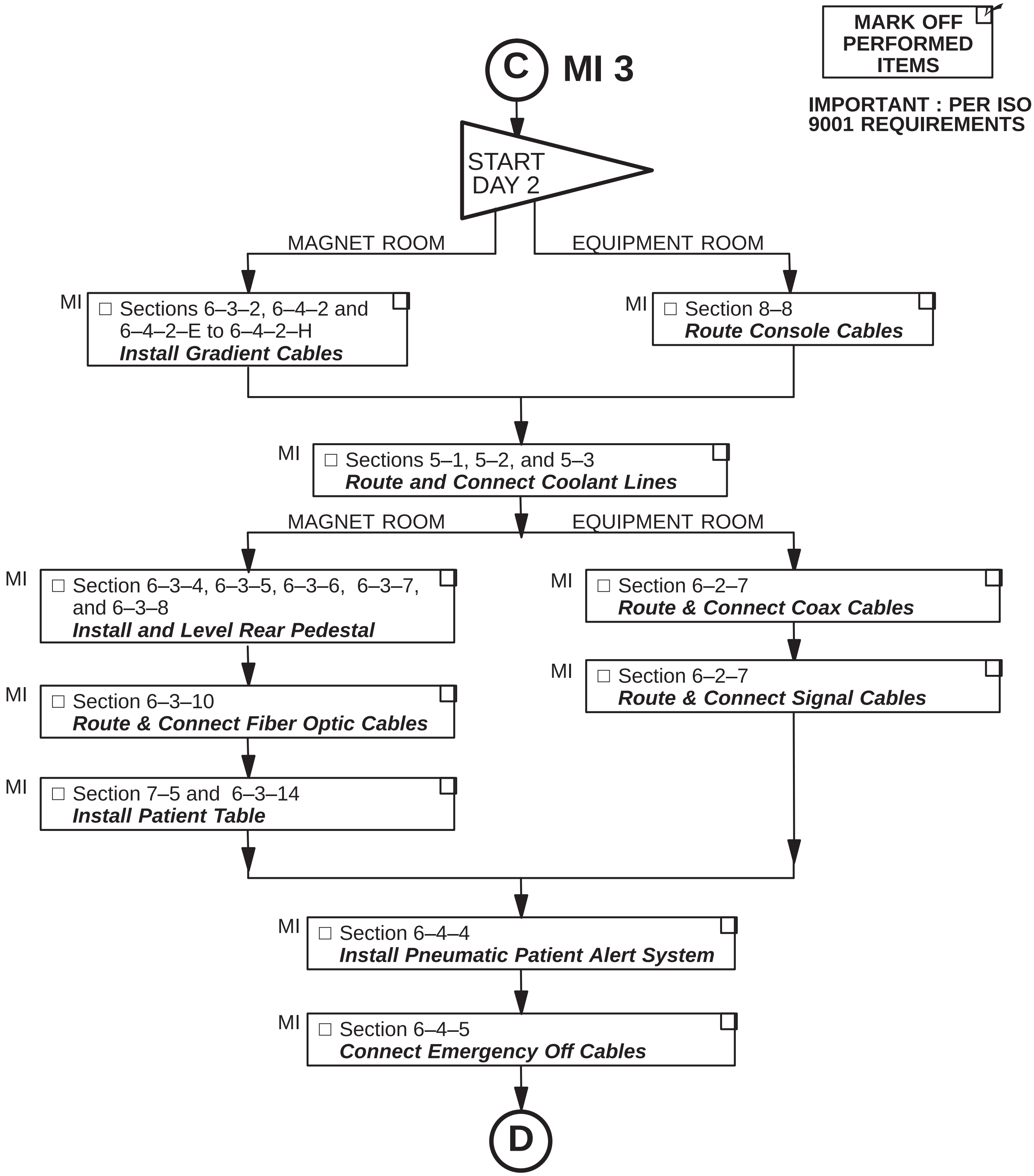
IMPORTANT : PER ISO 9001 REQUIREMENTS



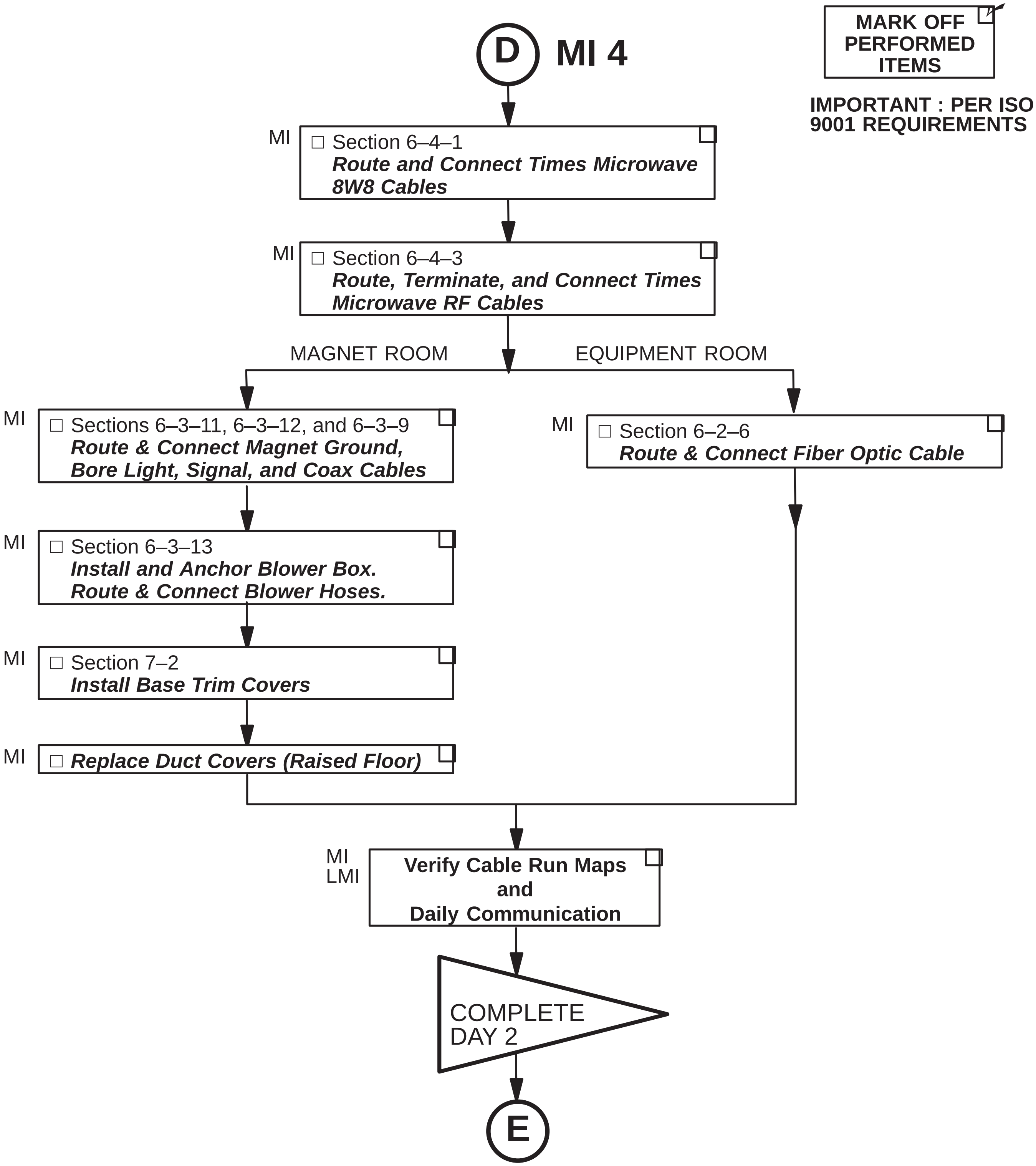
1–13–2 SYSTEM MECHANICAL INSTALLATION (Continued)



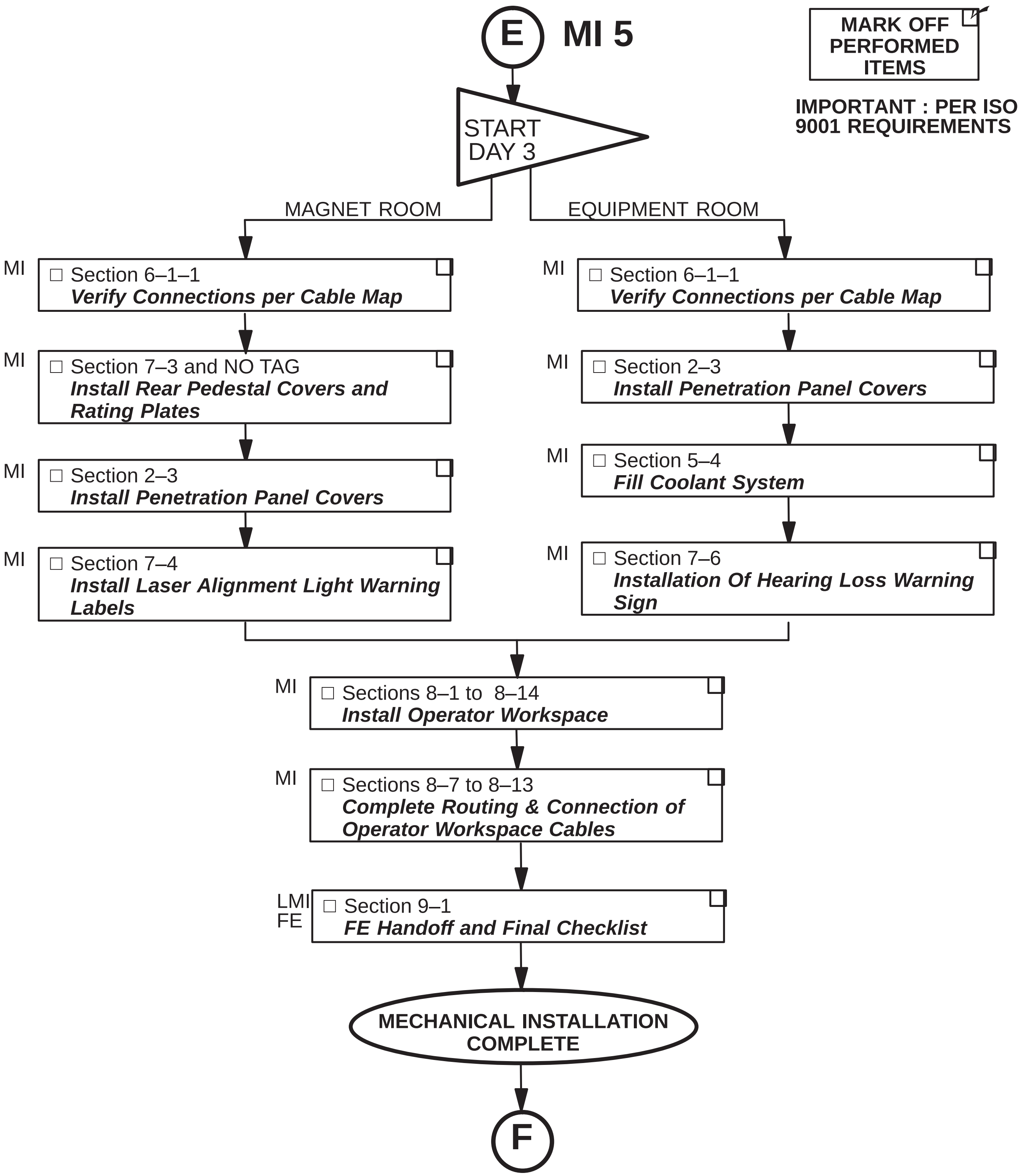
1–13–2 SYSTEM MECHANICAL INSTALLATION (Continued)



1–13–2 SYSTEM MECHANICAL INSTALLATION (Continued)



1–13–2 SYSTEM MECHANICAL INSTALLATION (Continued)



1–13–2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by GE Field Engineer

F

MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS

FE
☐ Section 9–2
GE Field Engineer Responsibilities

FE
MR Service Methods CD–ROM 5133315
☐ Adjustment/Calibrations>System Level Procedures>
Grounding & Leakage Current Checks
1. Ground Resistance Checks

NOTE:
Hard wired signal cables are not to be connected to subsystem cabinets prior to this check. Fiber optic cables may be connected and not impact this check.

FE
☐ Section 8–15
Operator Workspace Power On Information

FE
MR Service Methods CD–ROM 5133315
☐ Adjustments/Calibrations>System Level Procedures>***System Power On***
☐ Adjustments/Calibrations>***Host LCD Monitor Setup and Calibration***

If Host Software is NOT loaded? **OR** If Host Software is loaded?

MR Service Methods CD–ROM 5133315
☐ Software>System Host Software>
Host Operating System Software Load
FE

FE
MR Service Methods CD–ROM 5133315
☐ Installation>Options>
Signa Software Option Install with eLicense
☐ Installation>
EXCITE 1.5T HD Installation Calibration Flows

Applications/Training
☐ **Turn site over to Applications who will instruct users.**

TURN OVER TO CUSTOMER

SECTION 2 – PENETRATION PANEL INSTALLATION

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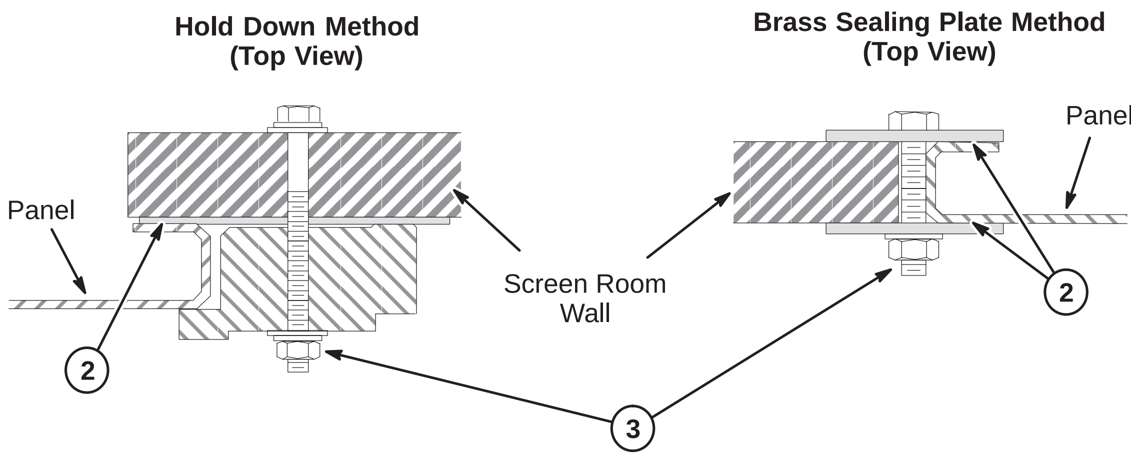
2–1 PREPARE SCREEN ROOM OPENING FOR PENETRATION PANEL

1. If still in place, remove vendor supplied blank panel. Save mounting hardware of either method shown below for installation of Penetration Panel.
2. To make sure of good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**
3. Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).



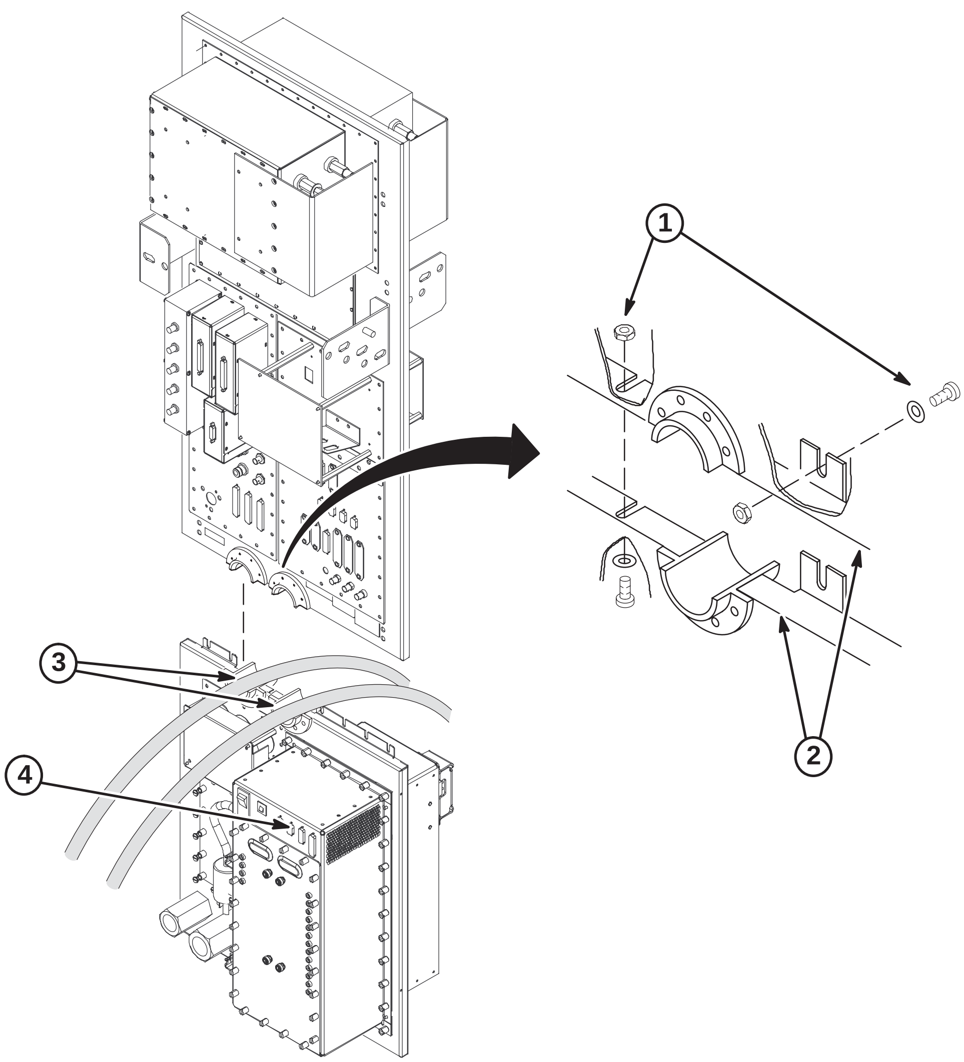
2–2 INSTALL PENETRATION PANEL

Install Penetration Panel using same method and hardware for mounting of blank panel removed in previous procedure.

Note

Install the Penetration Panel Floor plate that is supplied with the cover kit prior to routing and connecting cables to Penetration Panel. Refer to Installation Drawing furnished with the Cover Kit.

1. Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
   (7) #10–32 Nylock Lock Nuts (46–208937P6)
   (7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
   (7) #10–32 x 3/8 Screws (46–208560P82)
2. Seal all joints with copper foil tape with a conductive adhesive (3M #1181 or 46–258218P4, 5, or 6) for RF–tight integrity.
3. Fill inside of J60 and J61 guides with Bronze Wool (46–318068P1) that was supplied with magnet.
4. Install CAN Terminator (2303317) to J155 on PHPS Module.



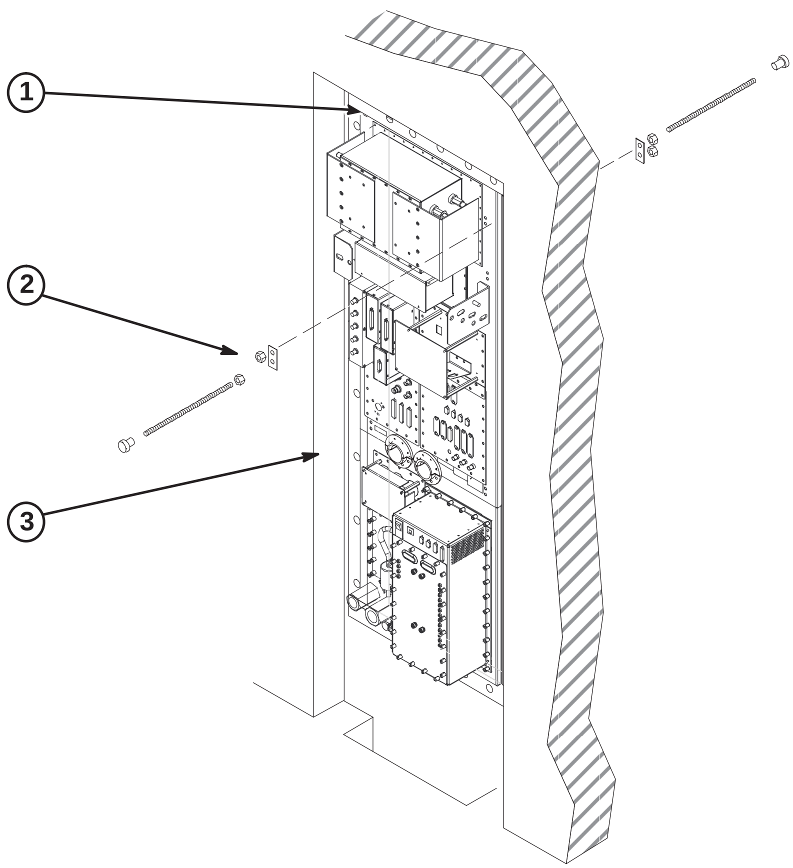
2–3 INSTALL PENETRATION PANEL COVERS

Note

The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

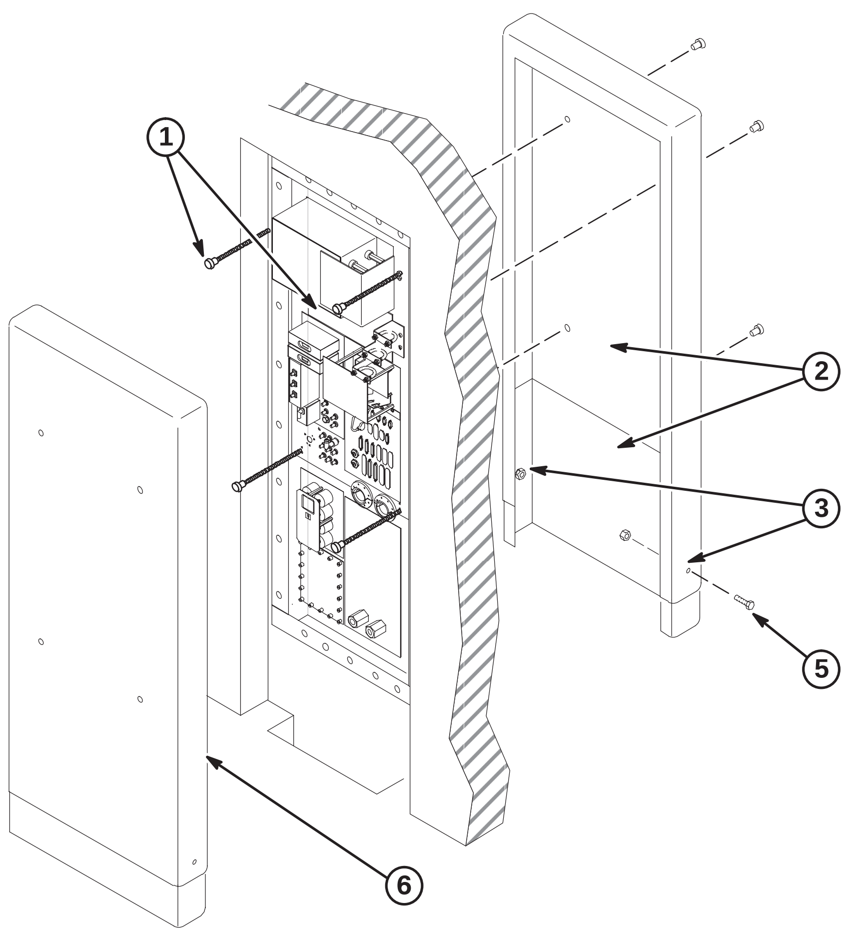
2–3–1 INSTALL PENETRATION PANEL STAND–OFF HARDWARE

1. Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4–1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
2. Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using:
   (16) 5/16–18 x 9/16 Brass Nut (46–208935P9)
   (8) 5/16–18 x 18 inch Threaded Rod (46–301213P2)
   (8) 5/16–18 x 1 inch Barrel Nut (46–301212P1)
3. Measure distance from face of Pen Panel to equipment room wall surface. Add six inches (152mm) to obtain length threaded rod should extend from panel.
4. Repeat step 3 procedure for exam room cover.
5. Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



2–3–2 INSTALL PENETRATION PANEL COVERS (After Cable Installation Complete)

1. Remove all barrel nuts from both sides of Penetration Panel.
2. In the Magnet room, temporarily install Top (46–306936P1) and Bottom (46–306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
3. Adjust position of bottom cover section and mark two lower fastener holes.
4. Remove covers and drill hole in lower cover at each mark.
5. Attach top and bottom covers with furnished 10–32 nuts and screws.
6. Repeat procedure for Equipment room cover.
7. After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.



SECTION 3 – CABINET POSITIONING

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3–1 CABINET INSTALLATION REQUIREMENTS

Positioning of cabinets is critical for installing cables and post–installation system operation. Refer to architectural site layouts for:

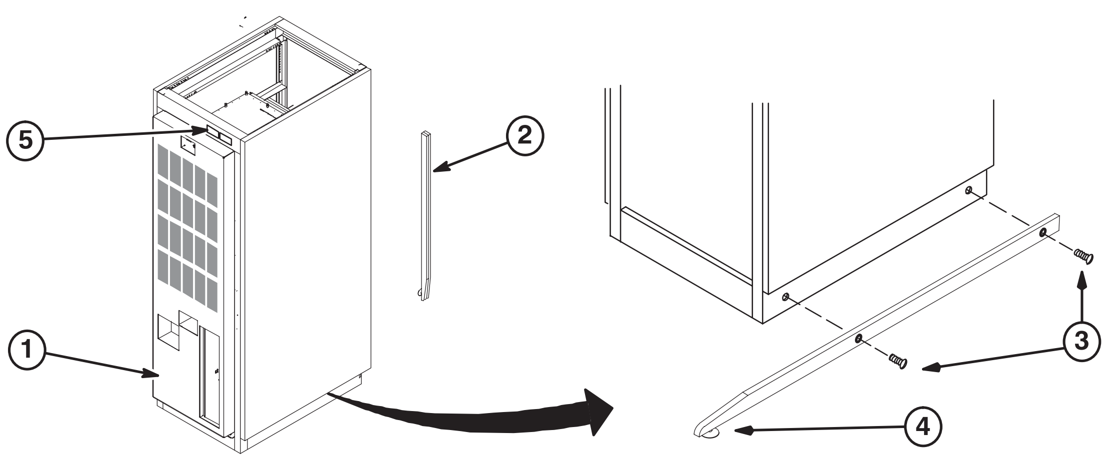
- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

3–2 POSITION HFD CABINET AND ATTACH LABEL

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

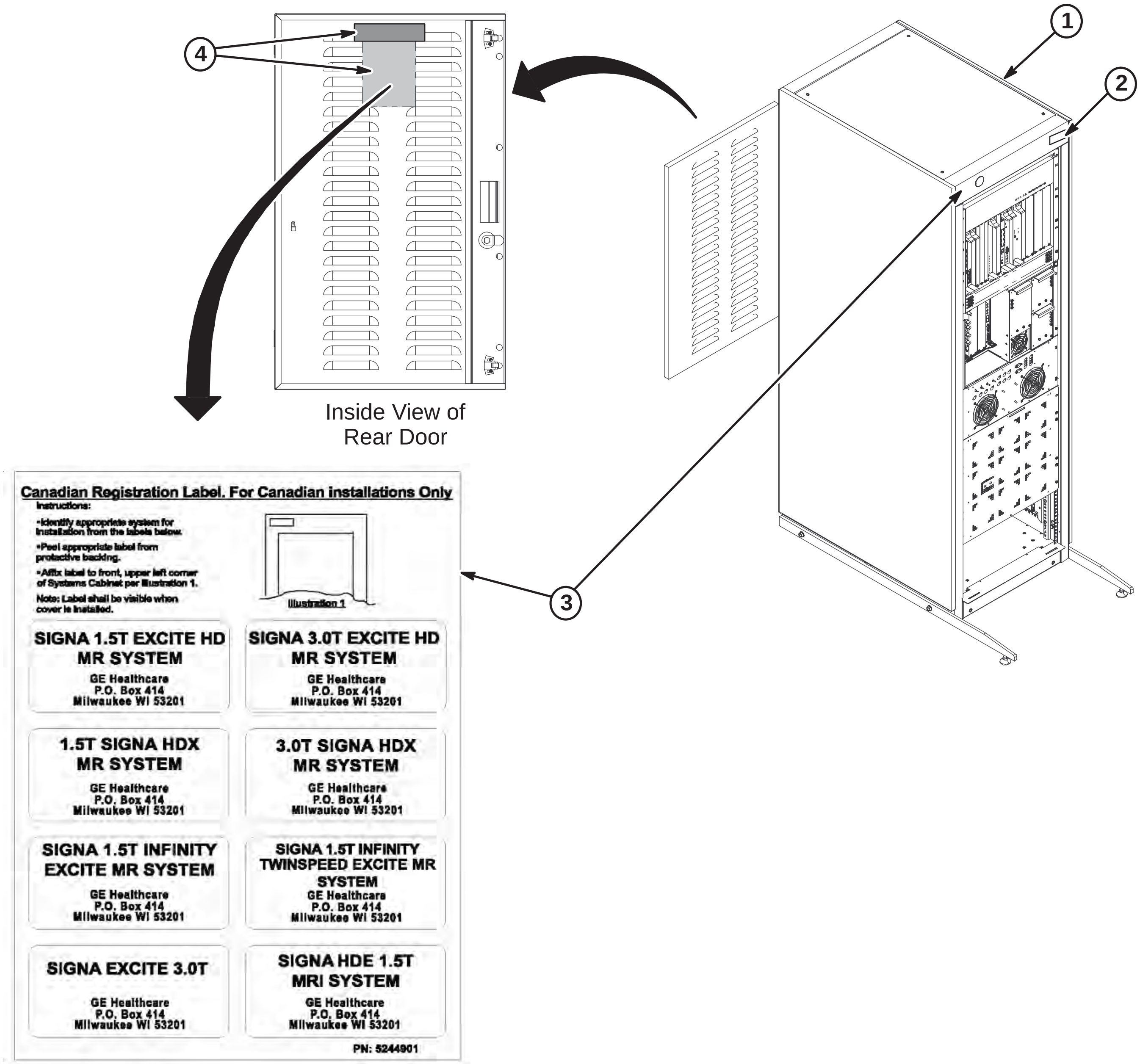
1. Move cabinet to final position as indicated by site layout drawings.
2. Attach the anti–tip bars to cabinet. (Anti–tip bars are shipped separately)
3. Adjust leveling pads to level cabinet.
4. From the Cabinet ID Label Sheet (5126718), locate the “GRADIENT/PDU” cabinet magnet and attach to front of cabinet as shown.



3–3 POSITION RFS CABINET AND ATTACH LABEL(S)

<u>Comply with UL requirements or local codes</u> for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position RF/System cabinet to final position as indicated by site layout drawings.
2. From the Cabinet ID Label Sheet (5126718), locate the ”RF/SYSTEMS CABINET” magnet and attach to front of cabinet as shown.
3. FOR INSTALLATION IN CANADA ONLY: Remove Registration Label sheet (5244901) from plastic sleeve on inside of rear door. Follow instructions on sheet for attaching appropriate label to cabinet.
4. For all non–Canada installations, remove plastic bag with labels and dispose. Label is not required.



SECTION 4 – HFD CABINET SET–UP

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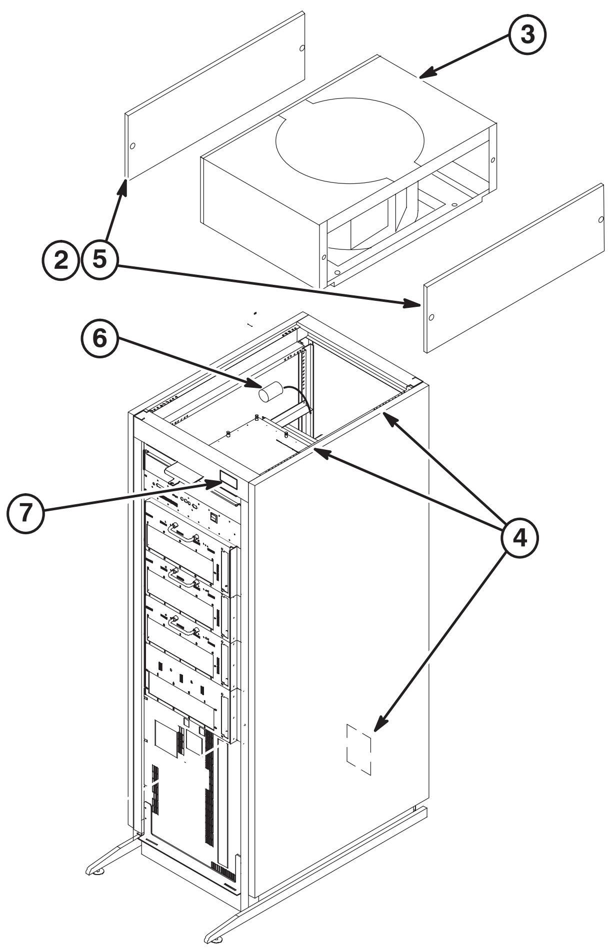
4–1 INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET

1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet. The fasteners are provided in a plastic bag located in PDU module. Rear PDU covers will need to be removed for access to plastic bags with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from SGA Power Supply to the fan connector.
7. Make sure the “GRADIENT/PDU” cabinet magnet, part of the Cabinet ID Label Sheet (5126718), is attached to the front of cabinet as shown.

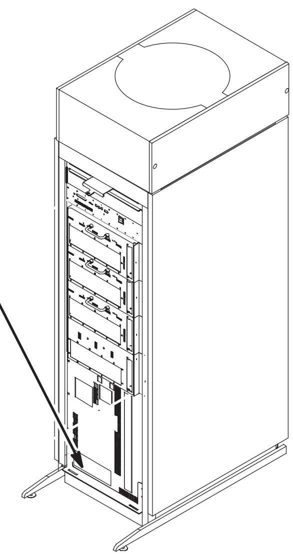


DANGER!!

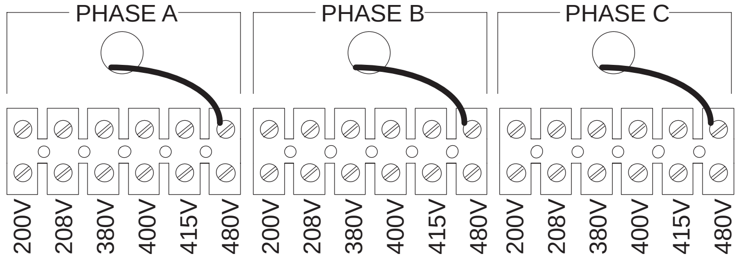
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

4–2 INPUT VOLTAGE SELECTION

1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



(SHOWN WIRED FOR 480V)



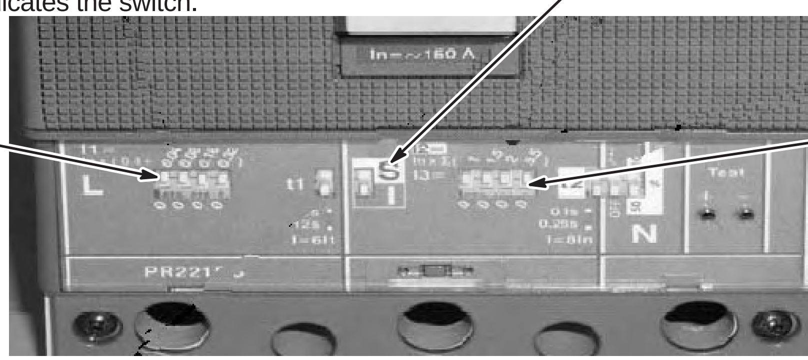
4–3 CIRCUIT BREAKER DIP SWITCH SETTINGS

1. Table below shows the position of the DIP switches for each input voltage selection. Set DIP switches according to input voltage. The white portion indicates the switch.

2. The "S/I" switch must be in the "I", down position

DIP SWITCH "L" SETTING

DIP SWITCH "I" SETTING



INPUT VOLTAGE	200	208	380	400	415	480
"L" SWITCH SETTINGS	□■■■ ■□□□	■□■■ □■□□	■■□□ □□■■	□■□□ ■□■■	□■□□ ■□■■	□□□□ ■■■■
"I" SWITCH SETTINGS	■□□■ □■■□	■□□■ □■■□	■■■□ □□□■	■■■□ □□□■	■■■□ □□□■	■■■□ □□□■

SECTION 5 – COOLING SYSTEM INSTALLATION

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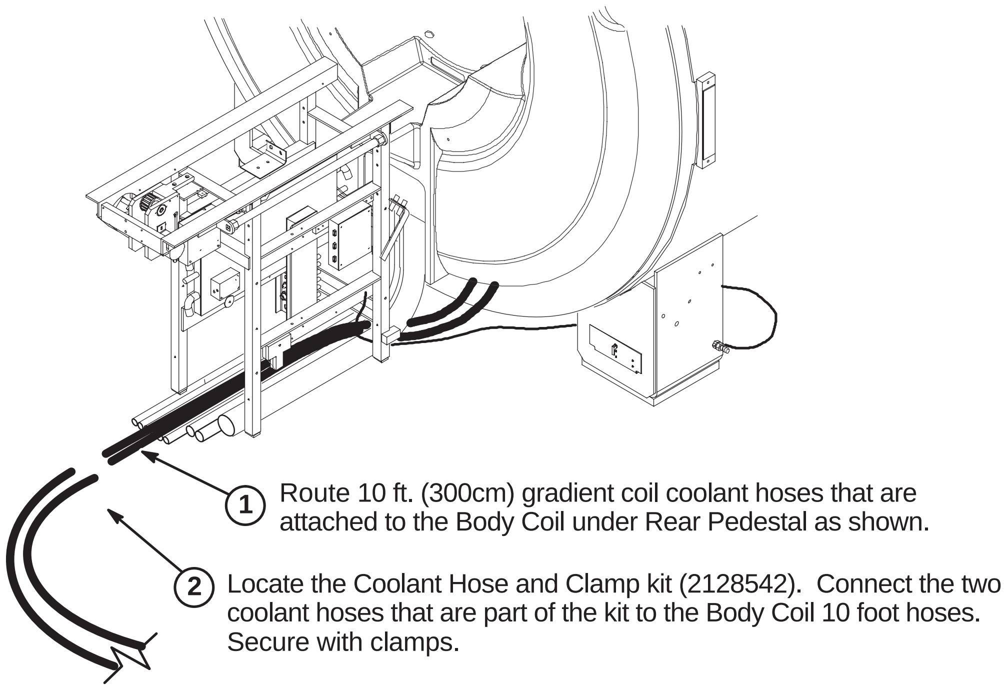
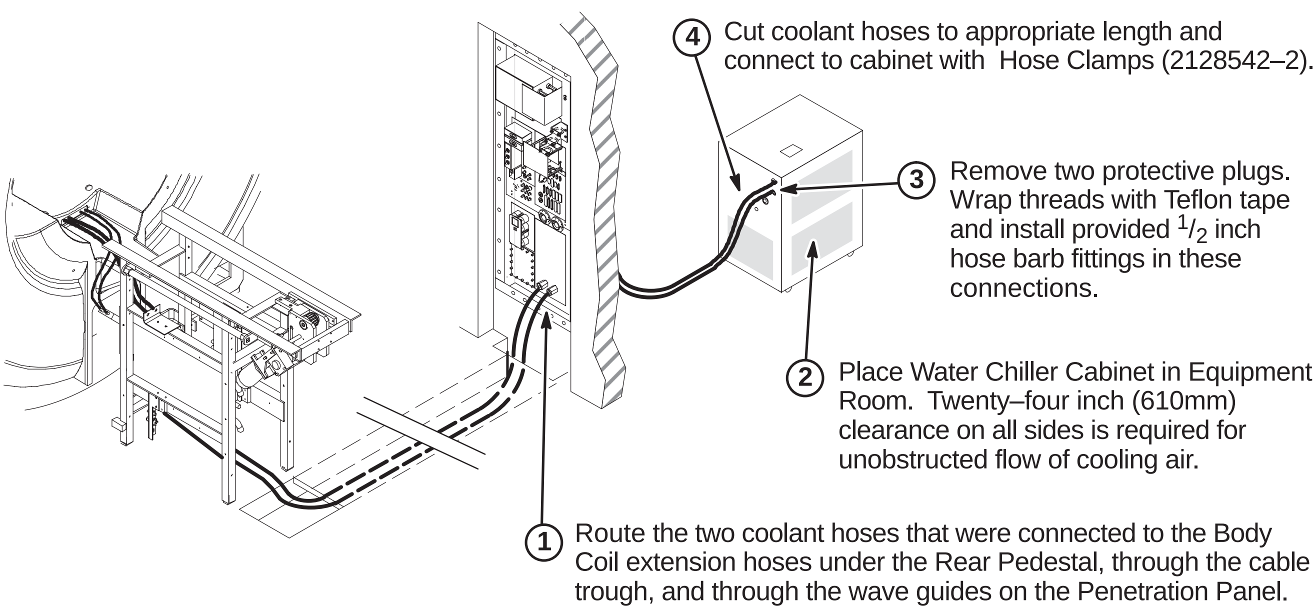
5–1 CONNECT BODY COIL COOLANT HOSES TO WATER CHILLER HOSES



5–2 POSITION WATER CHILLER CABINET AND CONNECT HOSES

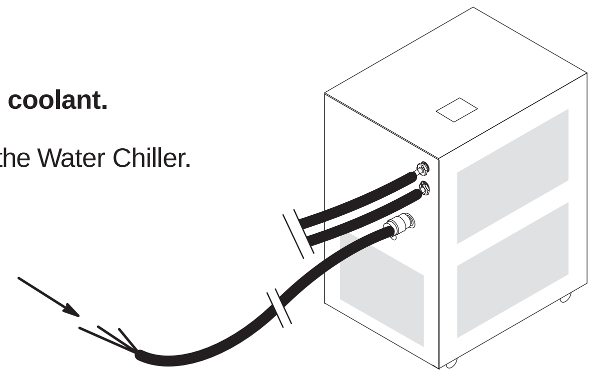


5–3 INSTALL POWER CORD

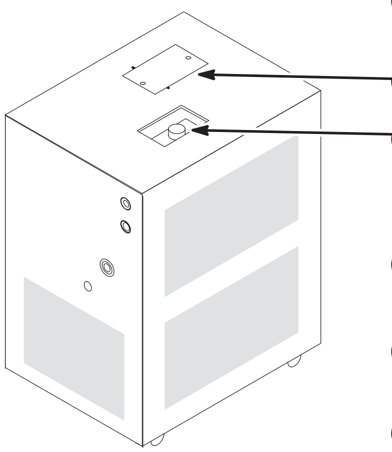
CAUTION

Do not start Water Chiller pump before filling reservoir with coolant.
Consult vendor Water Chiller manual –
Installation, Operation, and Service Manual that is shipped with the Water Chiller.

1. Route and connect power cable Run 970 (2280289) to PDU as marked (TS1–16,17,GND).



5–4 FILLING SYSTEM WITH COOLANT AND START UP



1. Locate the twelve (12) gallons of Body Coil Coolant (2138791) that was shipped with System.
2. Remove cover.
3. Consult vendor Water Chiller manual – *Installation: Filling Requirements*. Initial Filling of Reservoir, and the following steps for the proper procedure on priming and filling the system with coolant.
4. Consult vendor Water Chiller manual – *Operation: Start Up.* Follow instructions in manual for starting water chiller operation.
5. **Verify unit is set to control at 20° C.**
6. If supplied, attach plastic sleeve to side of Water Chiller and insert vendor manual. This will allow easy access to any future service/ maintenance procedures. If plastic sleeve is not present, attach vendor manual to Water Chiller.

SECTION 6 – CABLE & SYSTEM INSTALLATION

TABLE OF CONTENTS

6–1 CABLE INTERCONNECT DOCUMENTATION

The cable maps illustrate the Signa EXCITE HD 1.5T system interconnect for all supplied cables and interface with customer supplied wiring.

- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
  - Instructions for connections for optional laser cameras are supplied with option.
  - Connection procedures for other options are covered in the installation manual shipped with the option.
  - Spectroscopy option cabling can be found on page 6–7.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

6–1–1 Cable Interconnect Maps

For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables when their connections are complete.

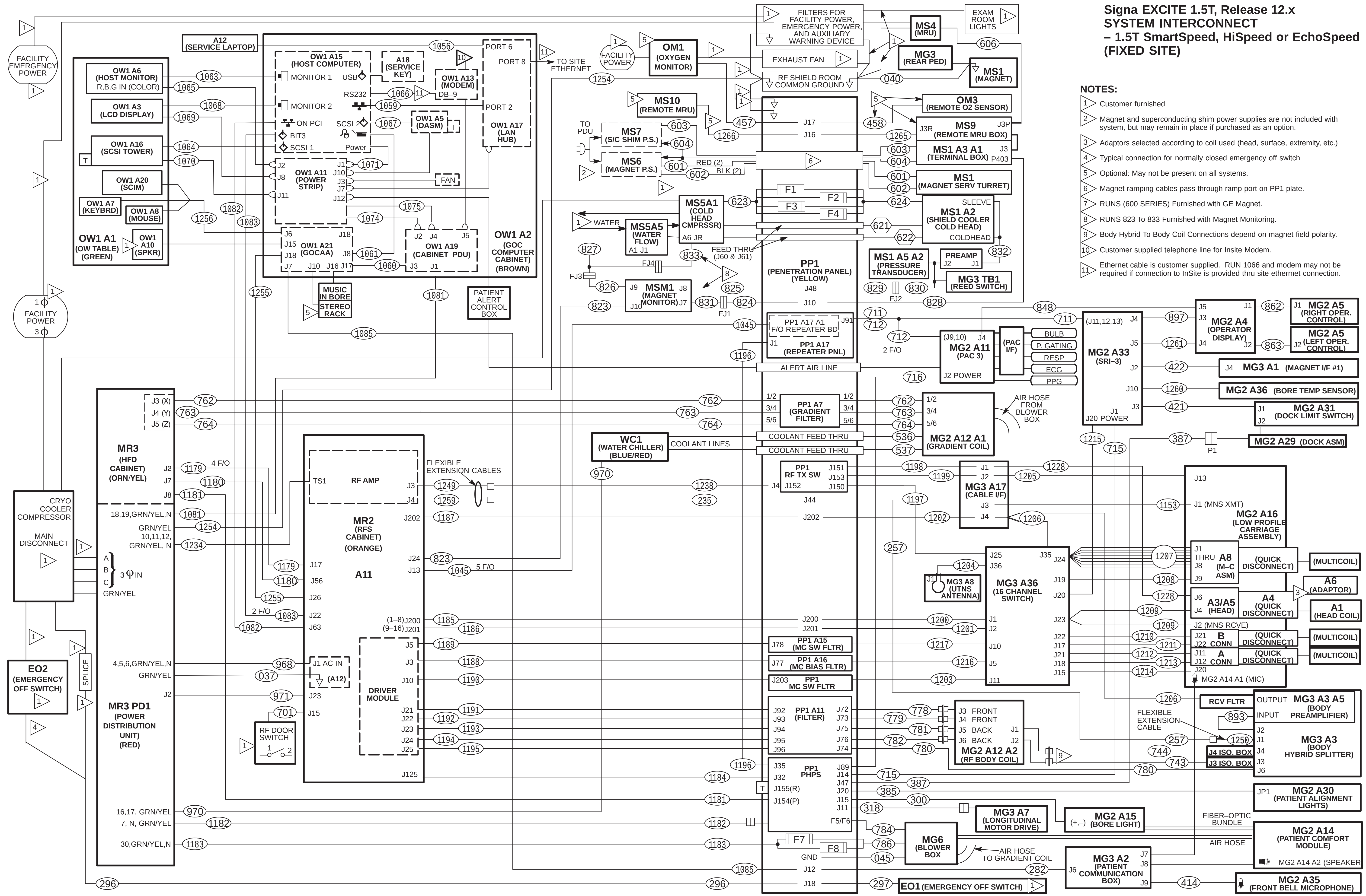
Fold–out EXCITE HD 1.5T System Interconnect Cable Map is provided on the following pages:

Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off routed cables. Additional information on each cable is located in the Appedndix. Upon completion of connecting cables, return fold–out maps to the binder for future reference.

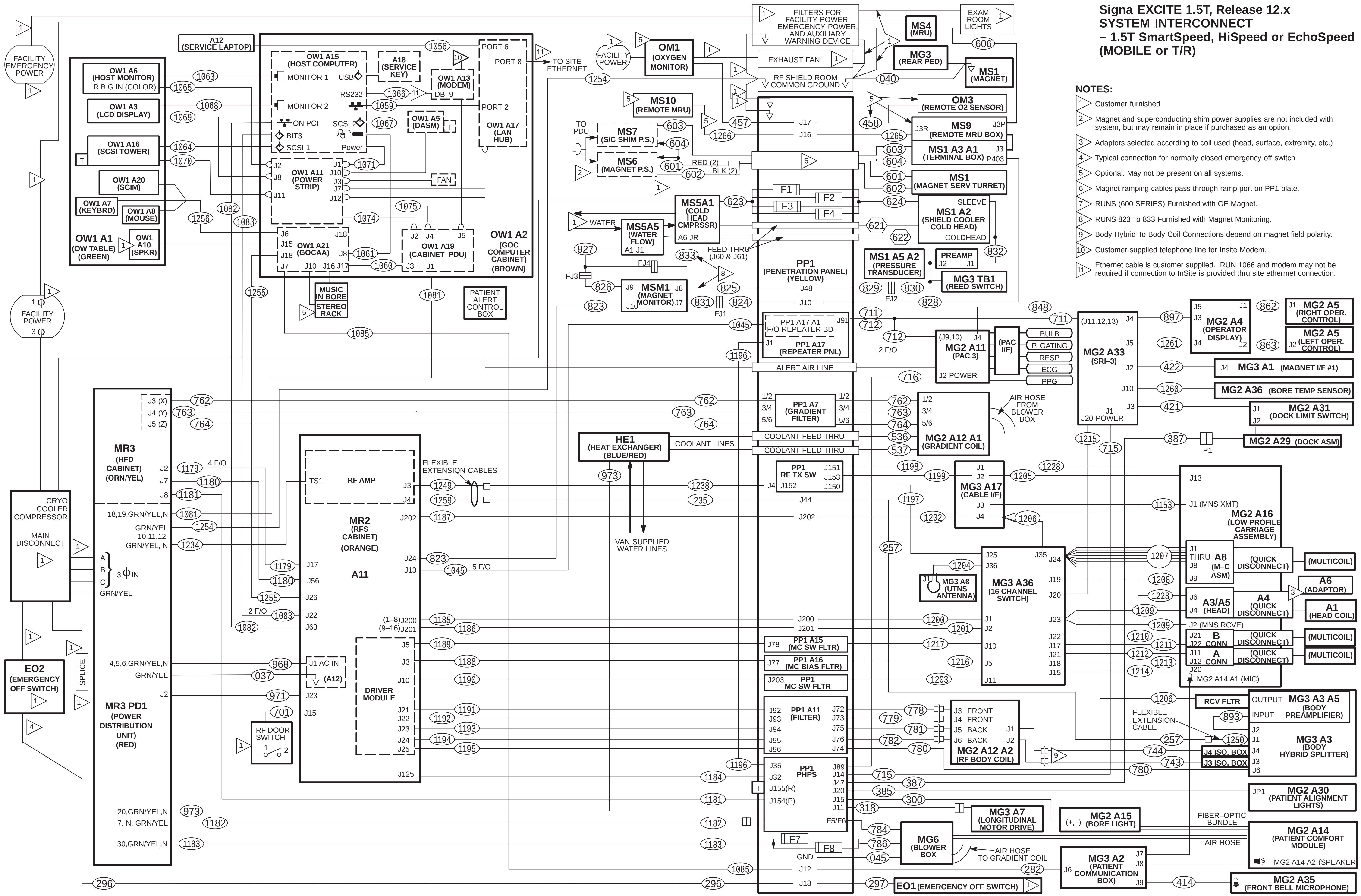
Signa EXCITE 1.5T, Release 12.x
SYSTEM INTERCONNECT
– 1.5T SmartSpeed, HiSpeed or EchoSpeed
(FIXED SITE)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



Signa EXCITE 1.5T, Release 12.x
SYSTEM INTERCONNECT
– 1.5T SmartSpeed, HiSpeed or EchoSpeed
(MOBILE or T/R)



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
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Signa EXCITE 1.5T, Release 12.x
SYSTEM INTERCONNECT
– 1.5T SmartSpeed, HiSpeed or EchoSpeed
(FIXED SITE)
(DUPLICATE OF PAGE 6–3)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
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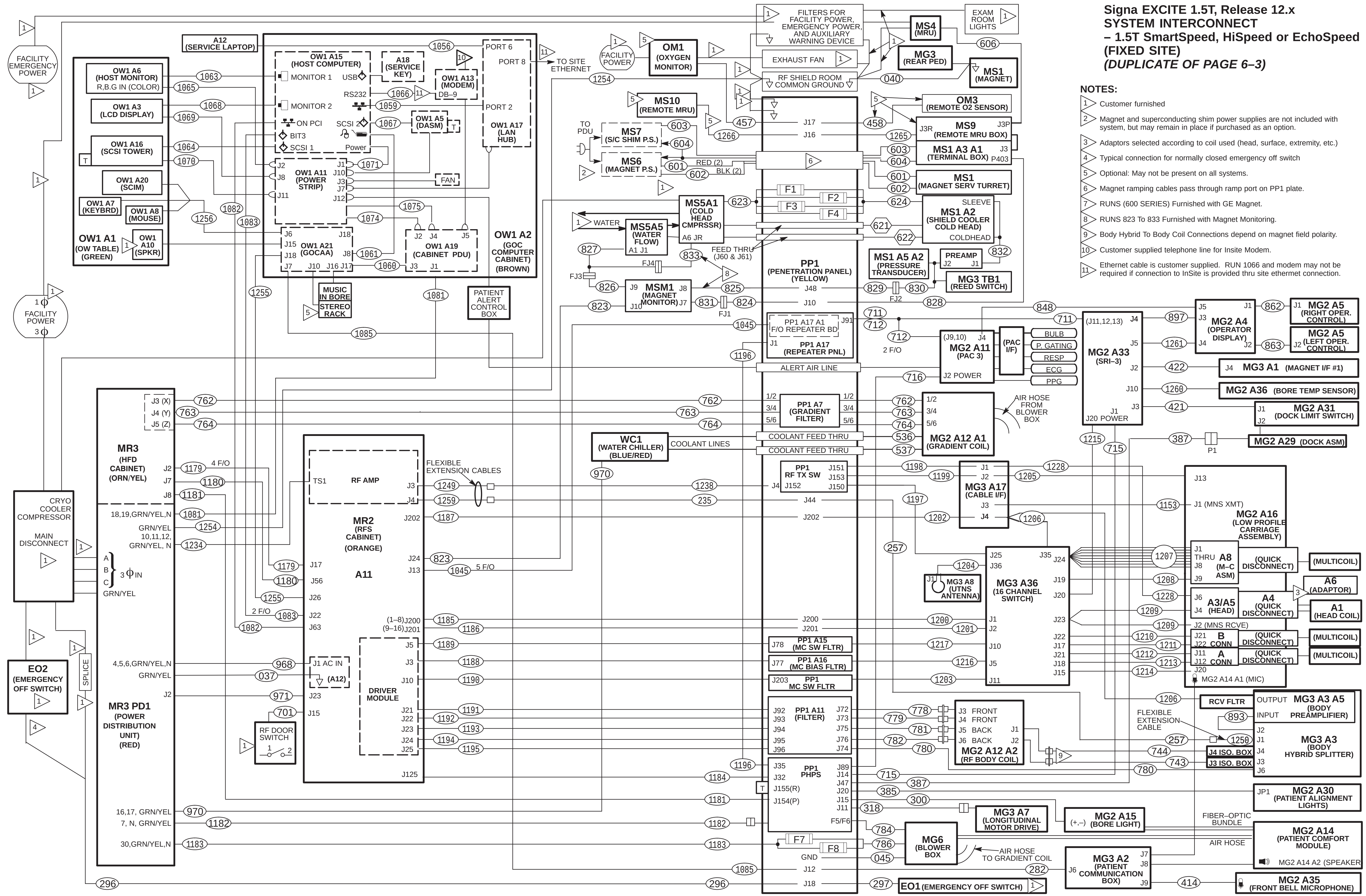

Signa EXCITE 1.5T, Release 12.x
SYSTEM INTERCONNECT
– 1.5T SmartSpeed, HiSpeed or EchoSpeed
(MOBILE or T/R)
(DUPLICATE OF PAGE 6–4)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface, extremity, etc.)
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
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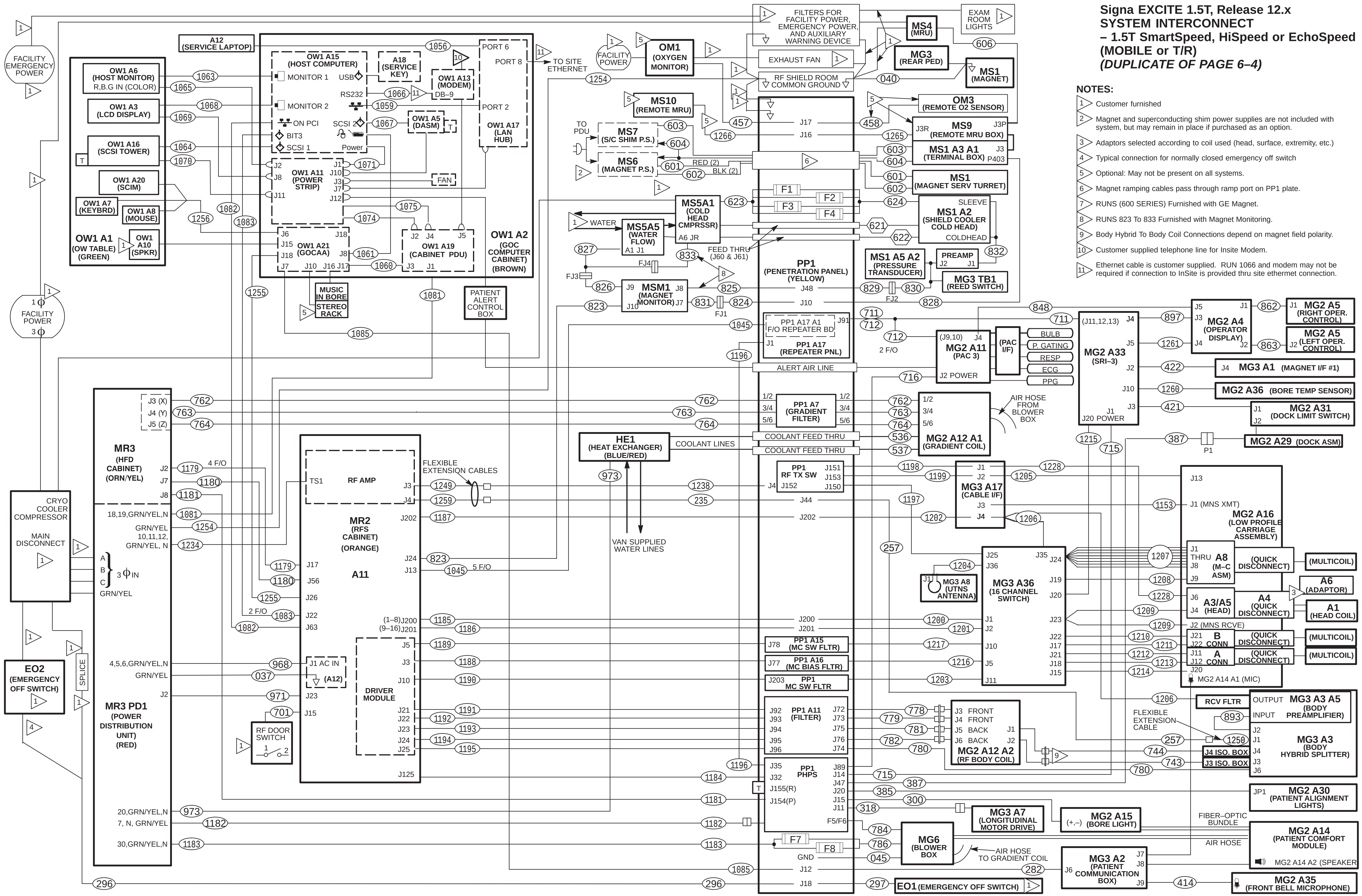
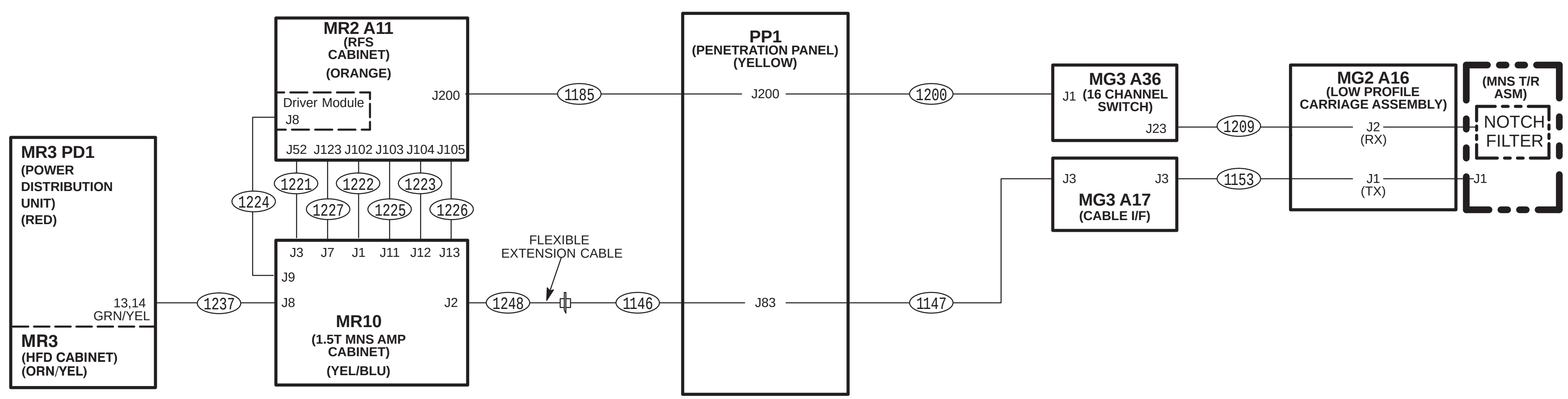


Signa EXCITE 1.5T, Release 12.0: 2KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



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6–1–2 Sorting and Routing Cables

CAUTION

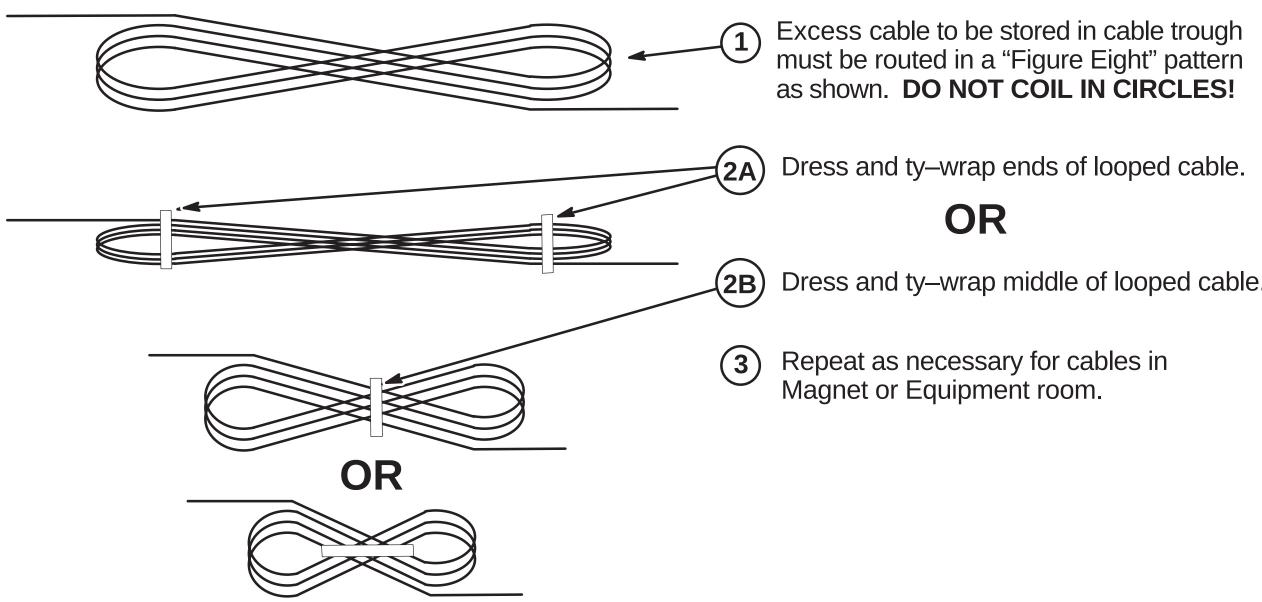
Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2–1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber–optic cables.



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 2–1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

6–2 EQUIPMENT ROOM INSTALLATION

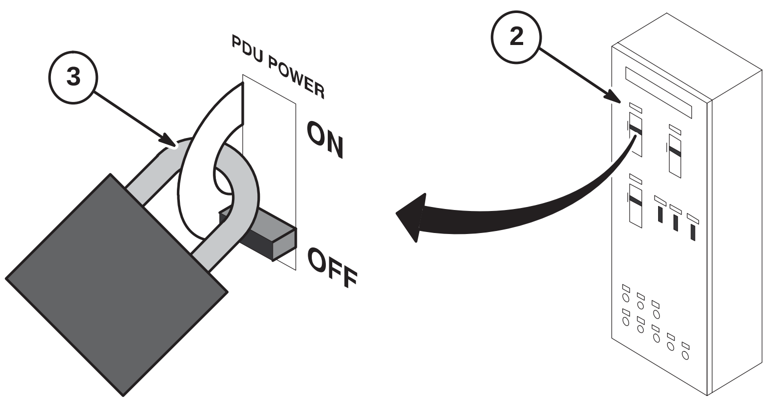
The equipment room cables and cabinets can be installed simultaneously. Cable routing depends on location of cabinets.

6–2–1 LOCKOUT AND TAGOUT PDU AT MAIN DISCONNECT PANEL

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT THE PDU CIRCUIT BREAKER ON THE FACILITY MAIN DISCONNECT PANEL IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

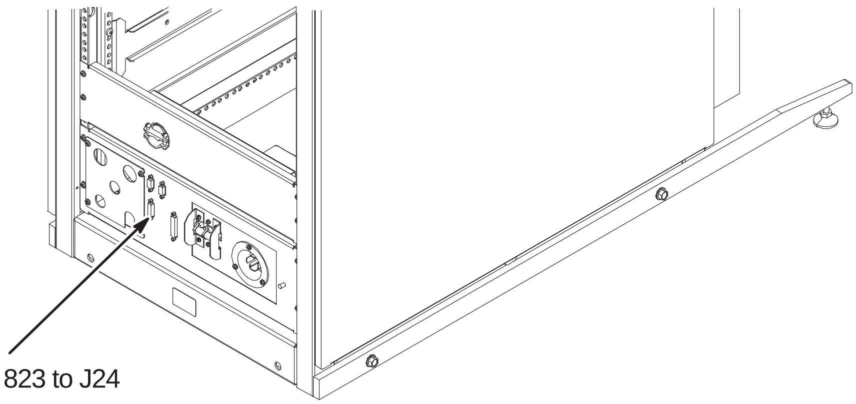
1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
2. Locate and shut off main disconnect supplying power to the PDU.
3. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
4. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring



6–2–2 COMPLETE INSTALLATION OF MAGNET MONITOR SYSTEM

Locate Direction 2230681, *Magnet Monitor Hardware Installation Manual.* This manual should be found in the box containing the components for the Magnet Monitor installation. Refer to instructions in the manual for:

- Installing Magnet Monitor module on wall (if not already completed)
- Routing and connecting Magnet Monitor cables
- Connecting Run 823 to the System Cabinet (see illustration below)



1 Connect Run 823 to J24

6–2–3 HFD/PDU CABINET CABLE CONNECTIONS

DANGER!!

MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

6–2–3–A HFD Cabinet Power and Ground Cable Installation

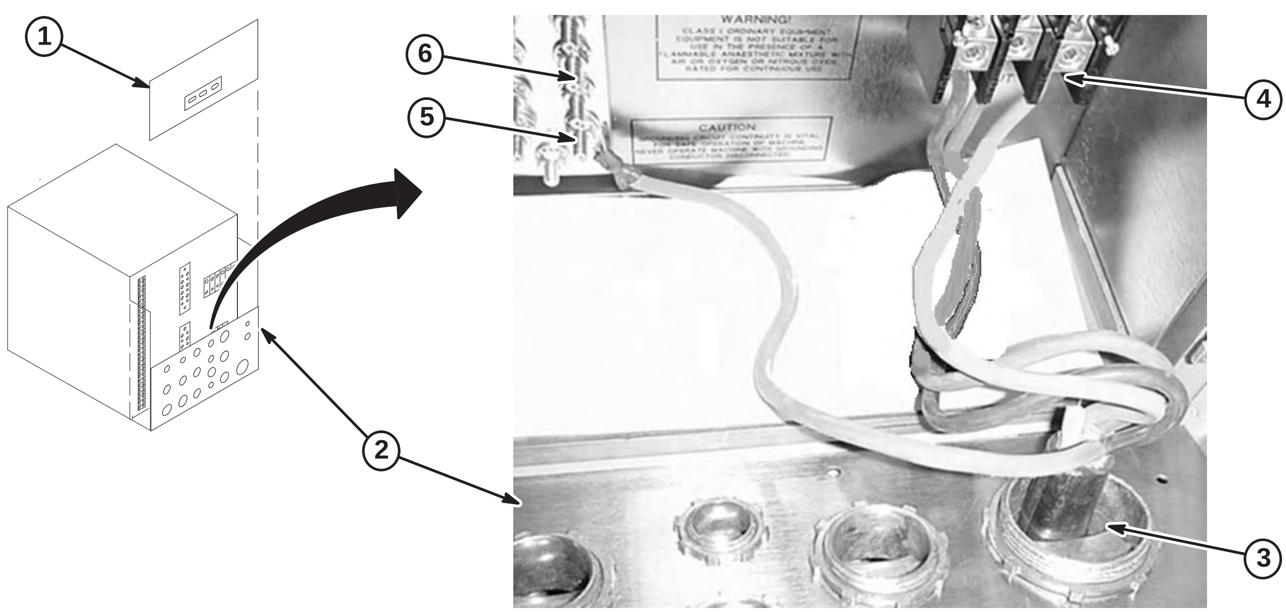
The procedure of routing and connecting system power from the Main Disconnect panel to the PDU Module in the HFD/PDU cabinet must be completed by customer supplied licensed electrician.

Instructions for connecting to the Main Disconnect panel should be provided with the panel. The steps below are for connecting power to the PDU module in the cabinet.

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND <u>NOT</u> BROUGHT TO THE HFD CABINET.

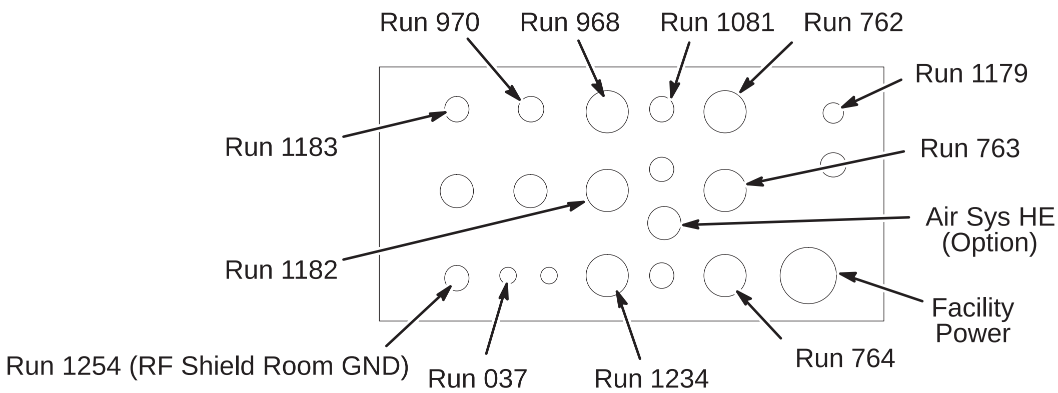
1. Unfasten and remove the top rear PDU module cover and ty–wrap to cabinet frame.
2. Unfasten the lower rear PDU interface panel, loosen cable clamps, and tilt top away from cabinet.
3. Route facility input power cable and ground wire thru interface panel.
4. Connect L1–L3 conductors to terminal block.
5. Connect facility input dedicated <u>ground</u> conductor to Ground (GRN/YEL) Bus Bar.
6. Connect RF screen room ground conductor to Ground Bus Bar.



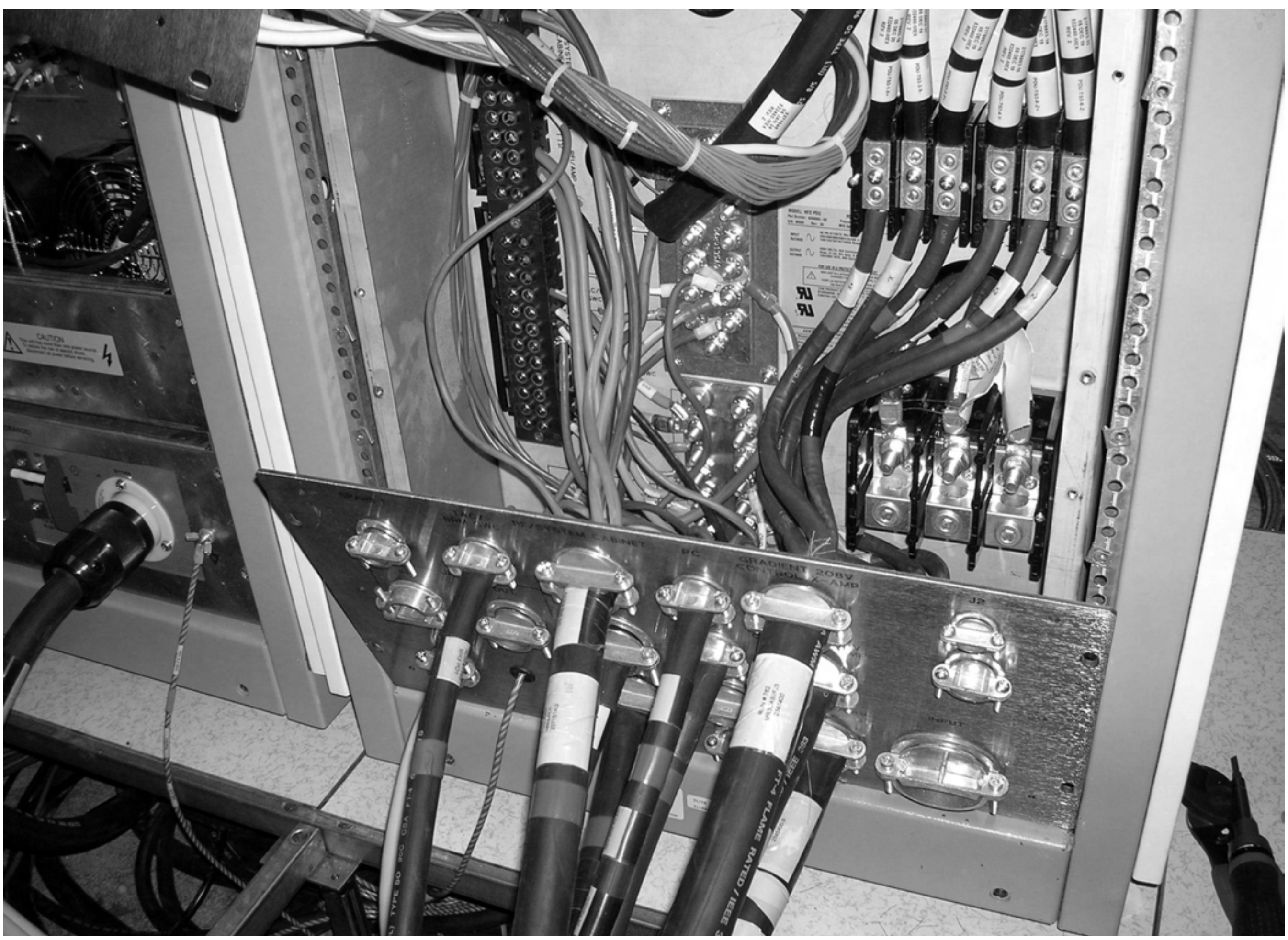
6–2–3–B Route Power Cables Thru Interface Panel

Install cables right to left, bottom to top.

1. Route Runs thru interface panel as marked.
2. Route any option Runs as marked.

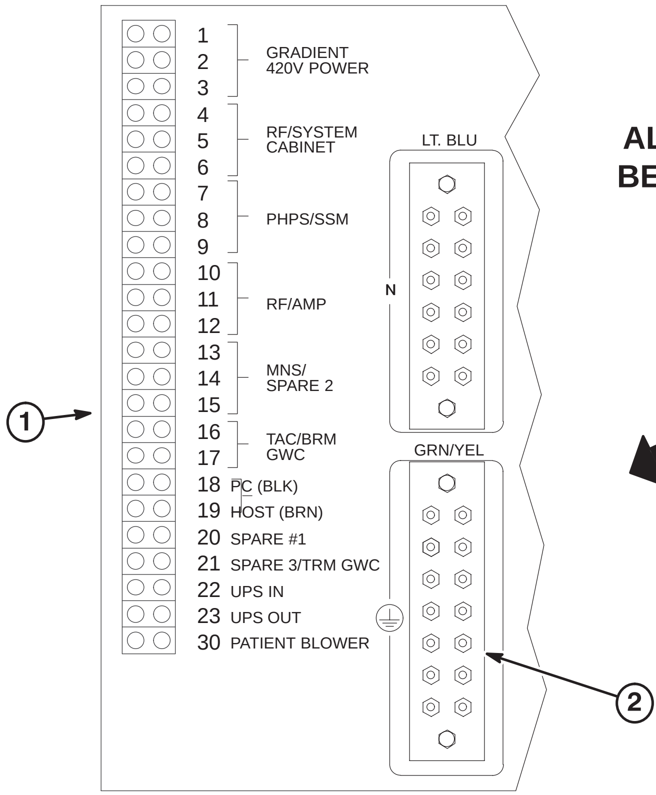


The image below shows the interface panel with cables routed thru and connected to PDU module. Section 6–2–3–C details the connection of cable wires to the terminal strips in the PDU.



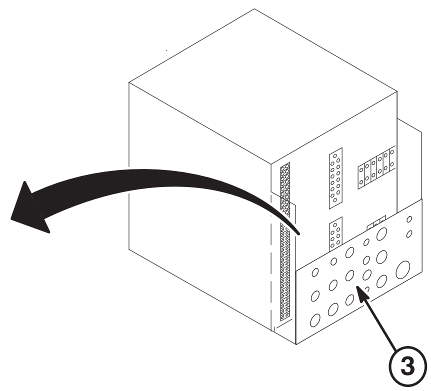
6–2–3–C Connect Power Cables to PDU Module

1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
2. Connect ground wire cables to Ground bus connectors. Trim and reterminate if necessary.
3. Connect neutral wire cables to Neutral bus connectors.
4. Tighten clamps of interface panel to secure cables in place.



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.



CONNECT TO	RUN	PHASE	DESTINATION	
4,5,6	968	ϕ A, B, C	RFS Cabinet	3ϕ, Neutral, GND
7	1182	ϕ A	PHPS Module	Single ϕ Neutral, GND
10,11,12	1234	ϕ A, B, C	RF Amp in RFS Cabinet	3ϕ, Neutral, GND
16,17	970	ϕ A,B	Water Chiller (Fixed Site)	Single ϕ GND
18,19	1081	ϕ C,A	Operator Workspace	Single ϕ, Neutral, GND
20	973	ϕ A	Heat Exchanger (Mobile)	Single ϕ, Neutral, GND
30	1183	ϕ A	Blower Box	Single ϕ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire	
GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	

COLOR CODE: for remainder of cables:
3ϕ = Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
Single ϕ = Brown, Black

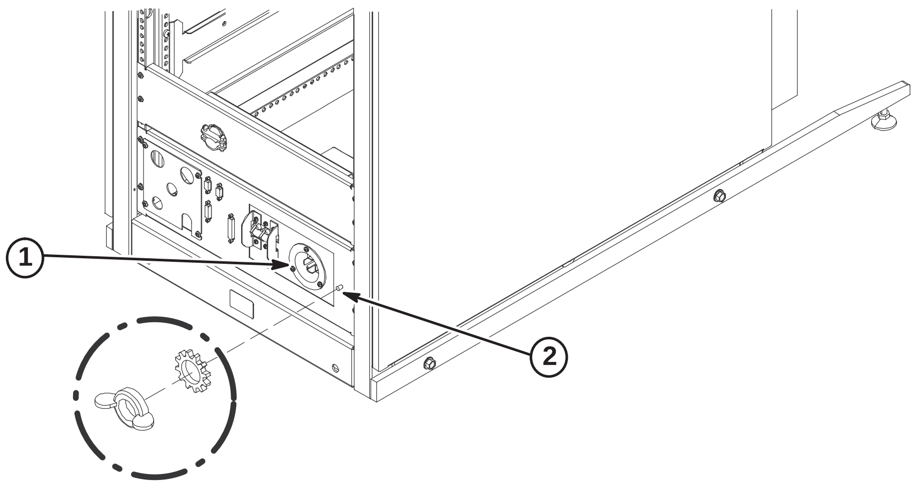
* Connection to Main Disconnect Panel to be completed by customer electrician.

6–2–4 RF/SYSTEM CABINET POWER AND GROUND CABLE CONNECTIONS

WARNING!

CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

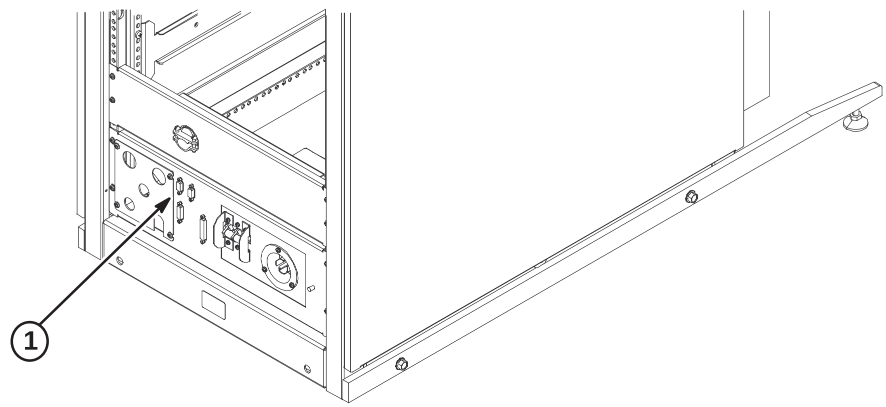
1. Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.



6–2–5 INSTALL RUN 701 – RF DOOR SWITCH

Run 701 is a 100 ft cable with a 9–pin subminature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminature D plug.

1. Connect Run #701 to J15 on System Cabinet and route to RF Door Switch. Refer to site engineering drawings for cable routing path.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

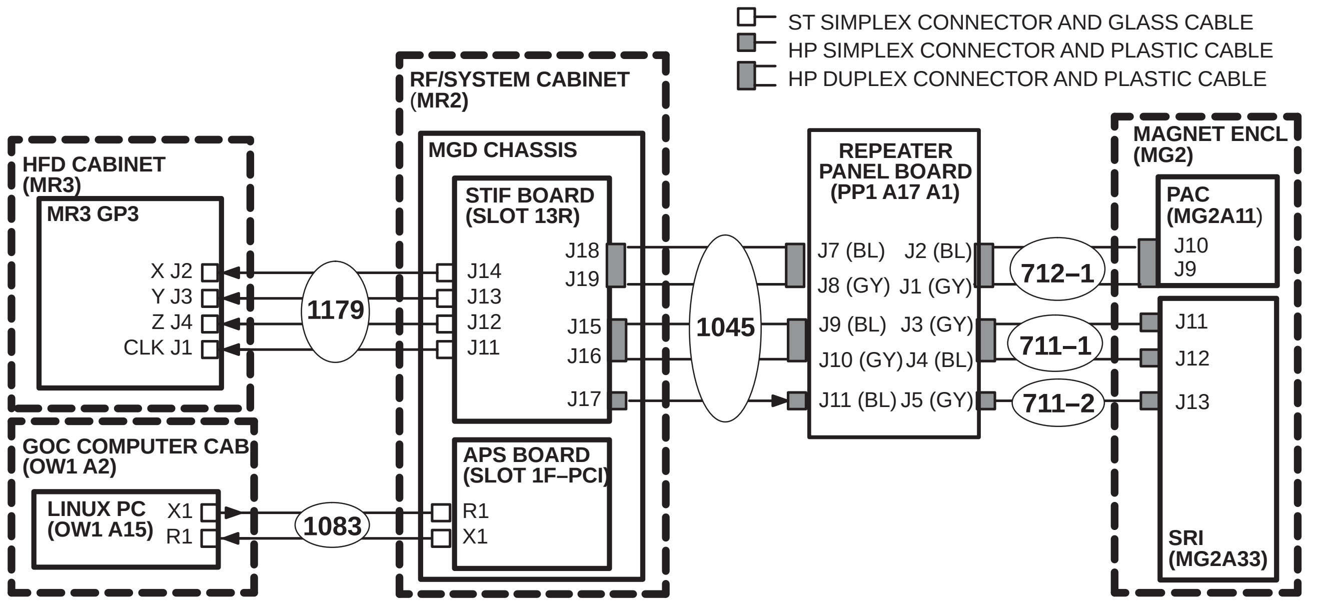


6–2–6 INSTALLATION OF FIBER OPTIC CABLES

Plan for storage of excess Run 1179 conduit of as it cannot be reterminated.

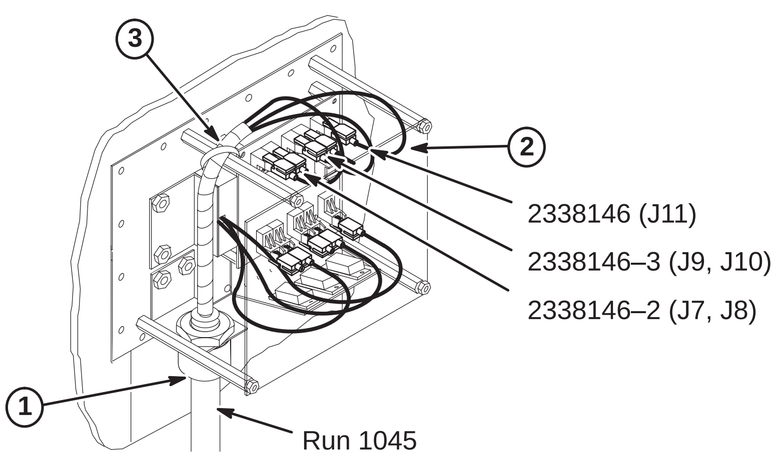
6–2–6–A Routing of Fiber Optic Cables

1. Unroll coiled fiber optic conduits for Runs 711/712, 1045, 1083, and 1179 along intended routes.
2. Connection of Runs is shown in diagram below.



6–2–6–B Connect Run 1045 to Penetration Panel

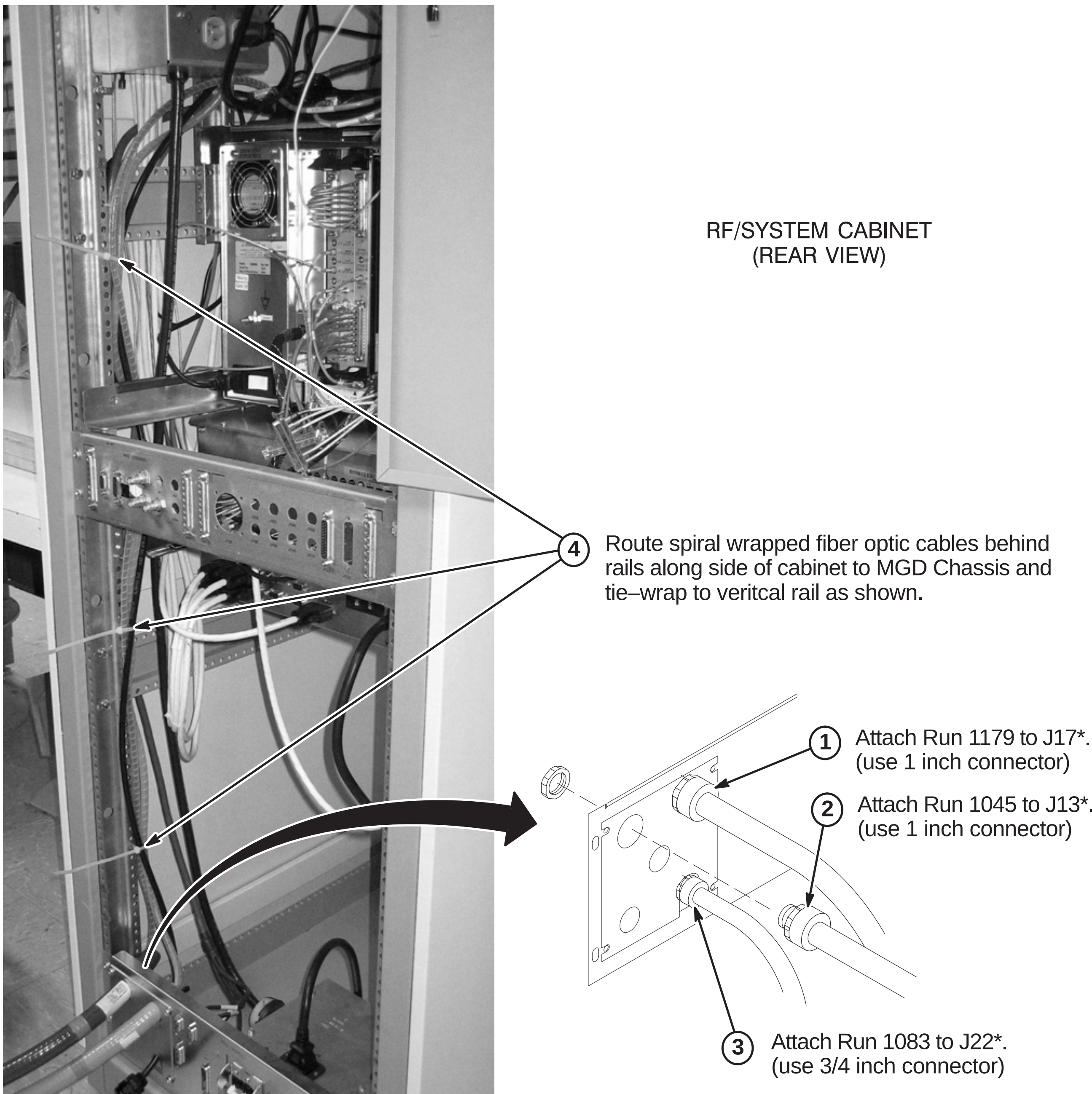
1. Move Run 1045 to the Equipment Room side and attach “PP1–A17” end to access bracket on Penetration Panel. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
2. Adjust route of individual fiber optic runs to maintain large radius and connect Run 1045 as shown.
3. Adjust position of protective spiral wrap and secure with tie–wrap.
4. Route other end of Run 1045 to System Cabinet.



6–2–6–C Route Fiber Optic Runs 1045, 1083, and 1179 in RF/System Cabinet

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



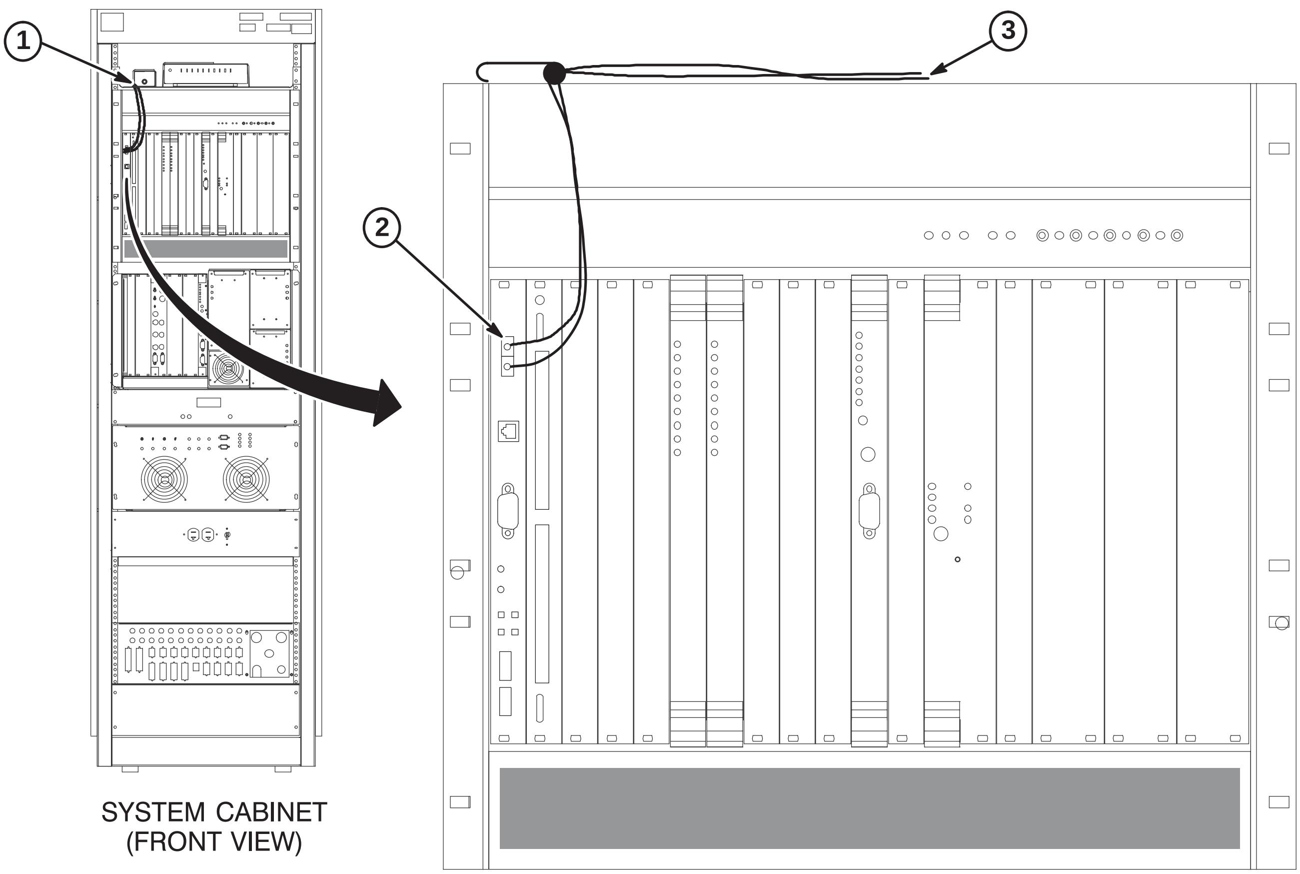
* Instructions provided with connectors.

6–2–6–D Route/Connect Fiber Optic Run 1083 to APS Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route Run 1083 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Connect the Orange and Blue fiber optic cables to the TI–MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



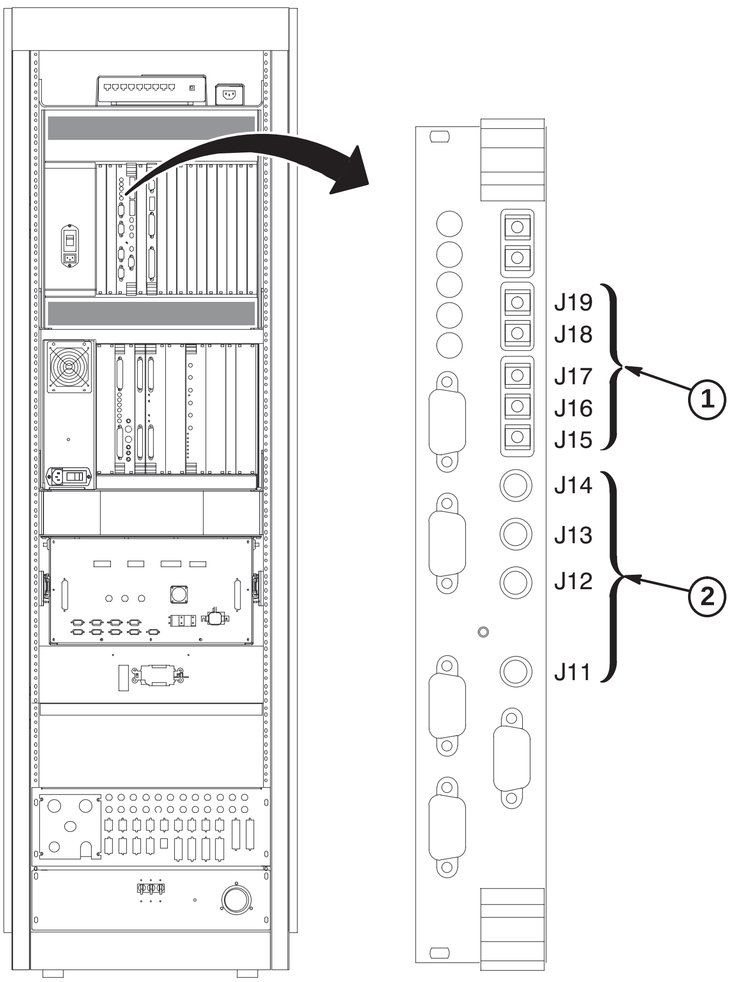
SYSTEM CABINET
(FRONT VIEW)

6–2–6–E Route/Connect Fiber Optic Runs 1079 and 1045 to STIF Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
2. Route/Connect Run 1079 fiber optic cables to J11, J12, J13, and J14 on the STIF Board.



Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

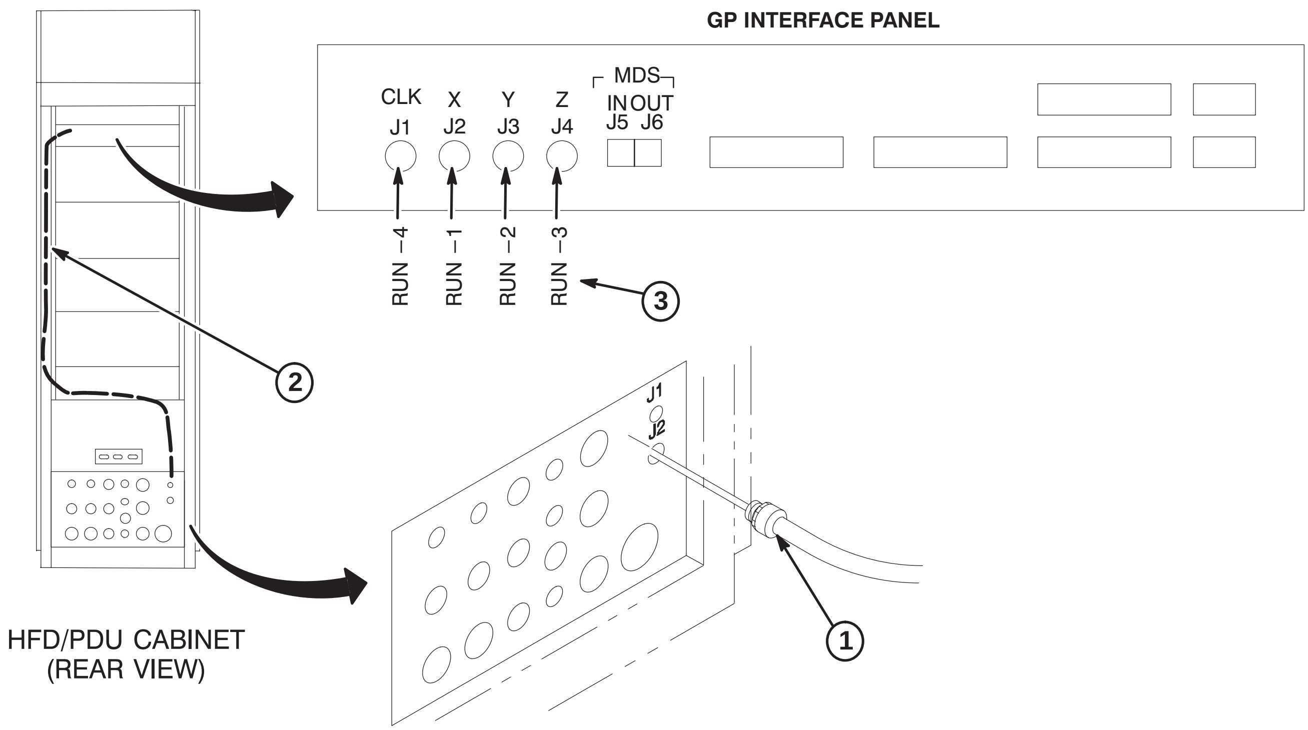
RF/SYSTEM CABINET
(REAR VIEW)

6–2–6–F Route/Connect Fiber Optic Run 1179 to GP3 Module in HFD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.

1. Attach Run 1179 to I/F J2.
2. Route fiber optic cables to GP3 Module. Leave enough slack for extending GP3 module forward.
3. Connect fiber optics to GP3 module according to illustration below and cable map in Section 6–2–6–A.

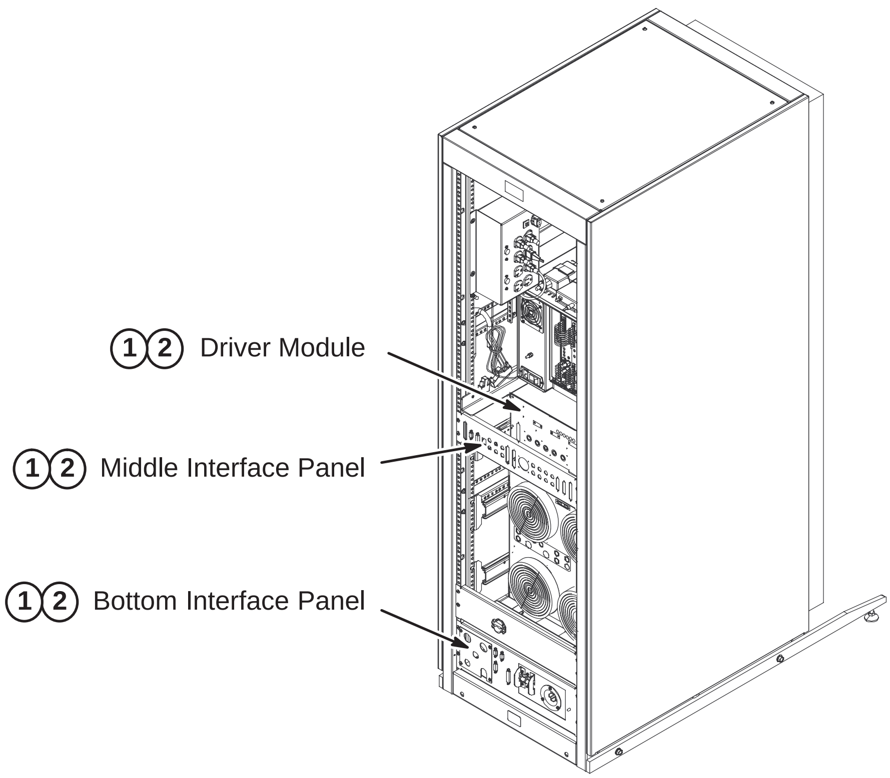


6–2–7 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES

Connect remaining Runs within equipment room. Dress and secure all cables after connections are made. Reference interconnect map on page 6–3 and Cable Table in Appendix for to/from designator information and Run number reference.

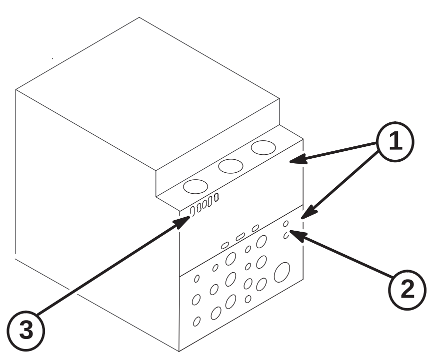
6–2–7–A RF/System Cabinet Connections

1. Connect routed cables to Interface Panels at bottom and middle of cabinet and to the Driver Module as marked. Refer to cable map.
2. Dress and secure cables. Adjust as necessary.



6–2–7–B HFD Cabinet PDU Module Connections

1. Attach rear PDU module covers removed earlier.
2. Connect Run 1179 to J2. Refer to Section 6–2–6–F.
3. Connect Runs 1180 to J7 and 1181 to J8.



6–3 MAGNET ROOM CABLE INSTALLATION

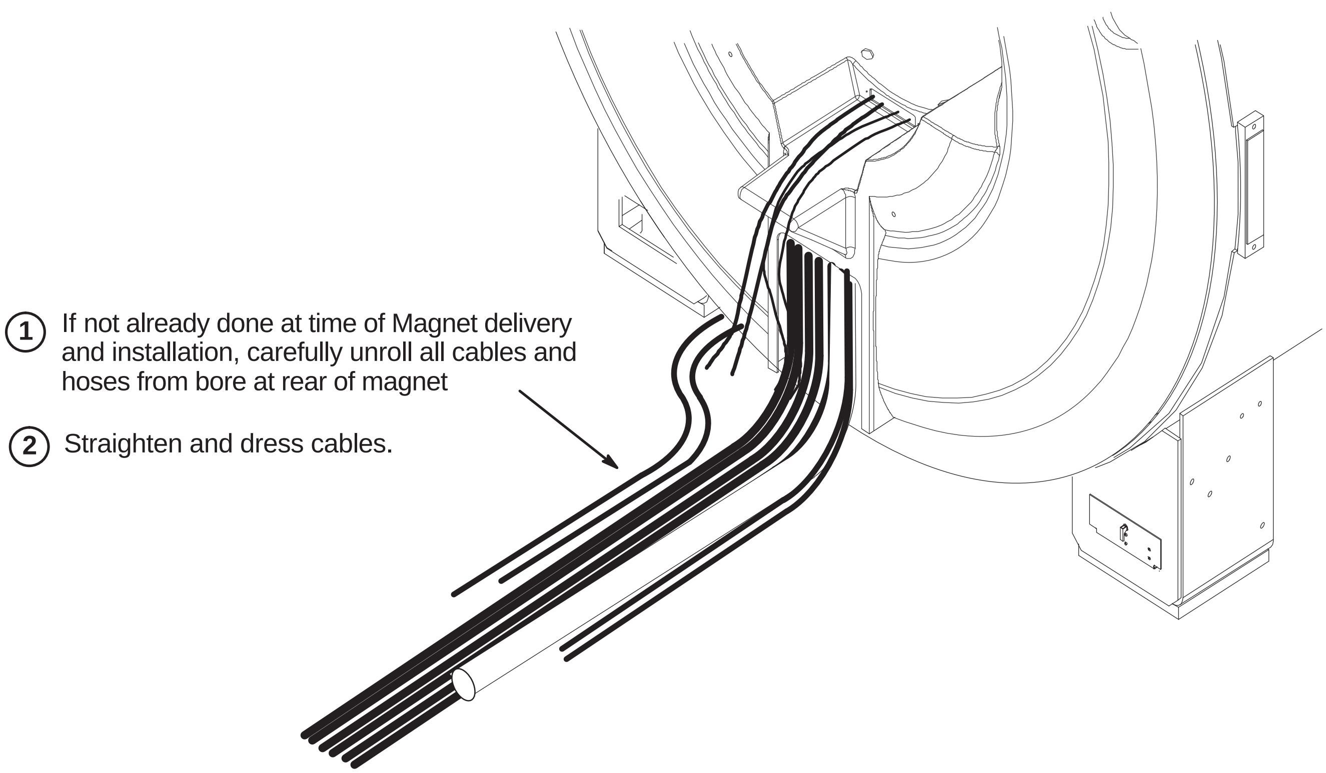
The magnet enclosure and magnet room cables must be installed simultaneously.

Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person’s left and right while facing the patient entrance end of the magnet.

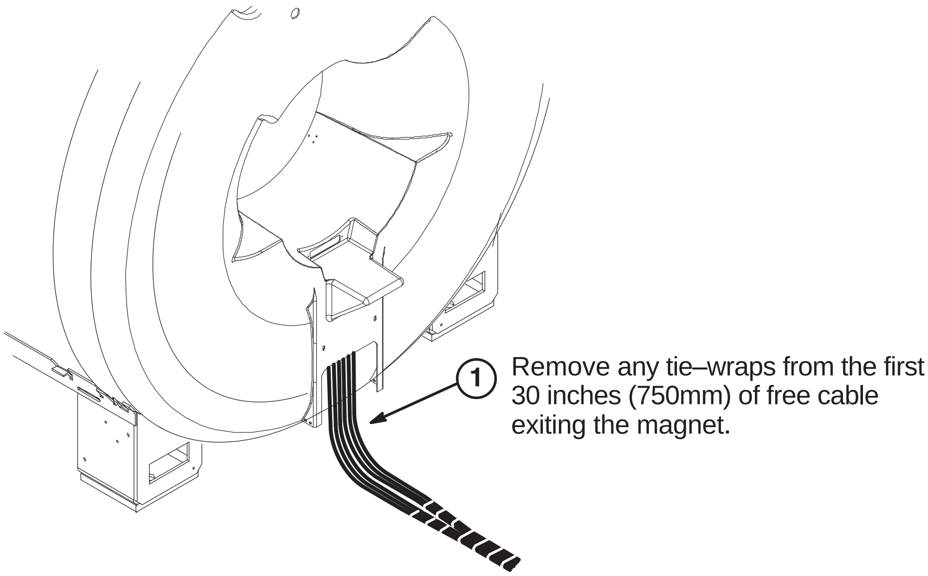
WARNING!

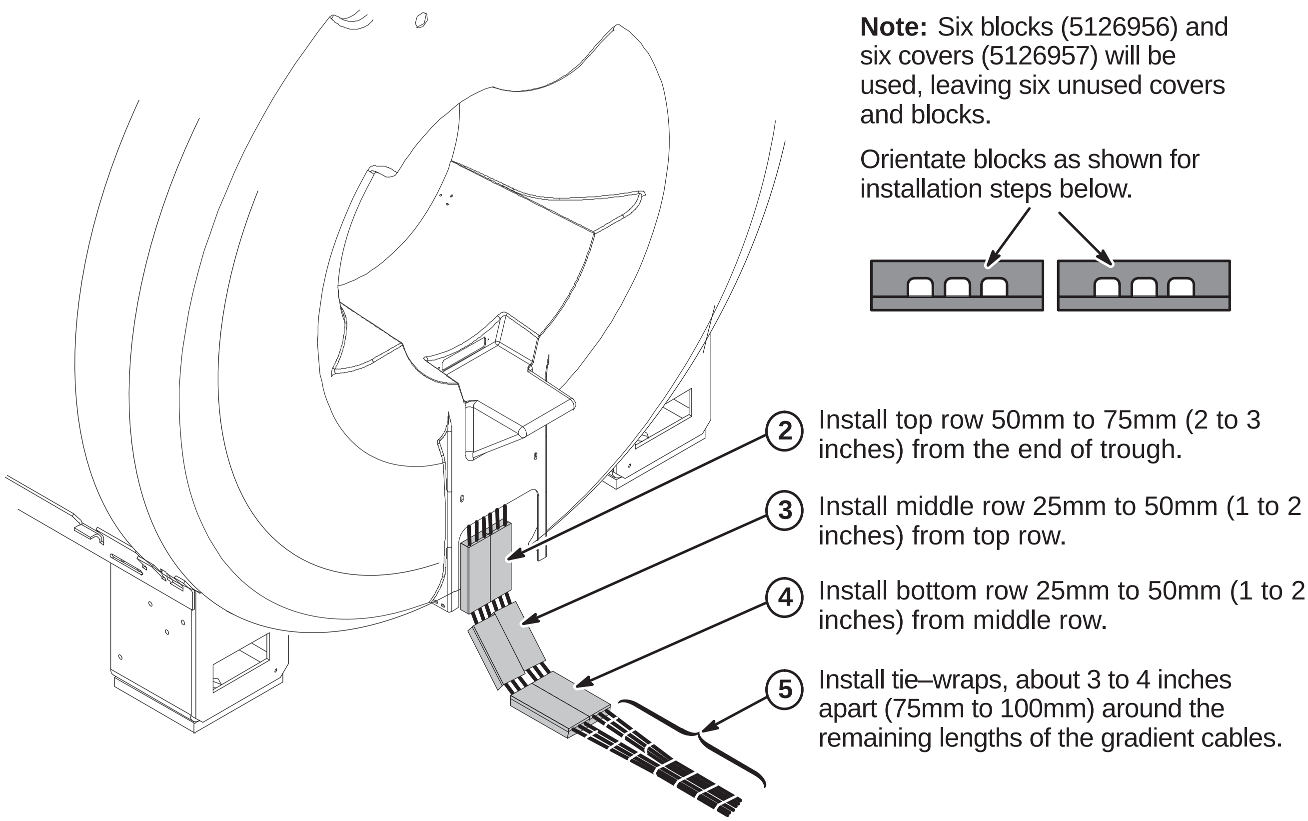
FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

6–3–1 ALIGN CABLES AT REAR OF MAGNET



6–3–2 INSTALL GRADIENT CABLE CLAMP BLOCKS





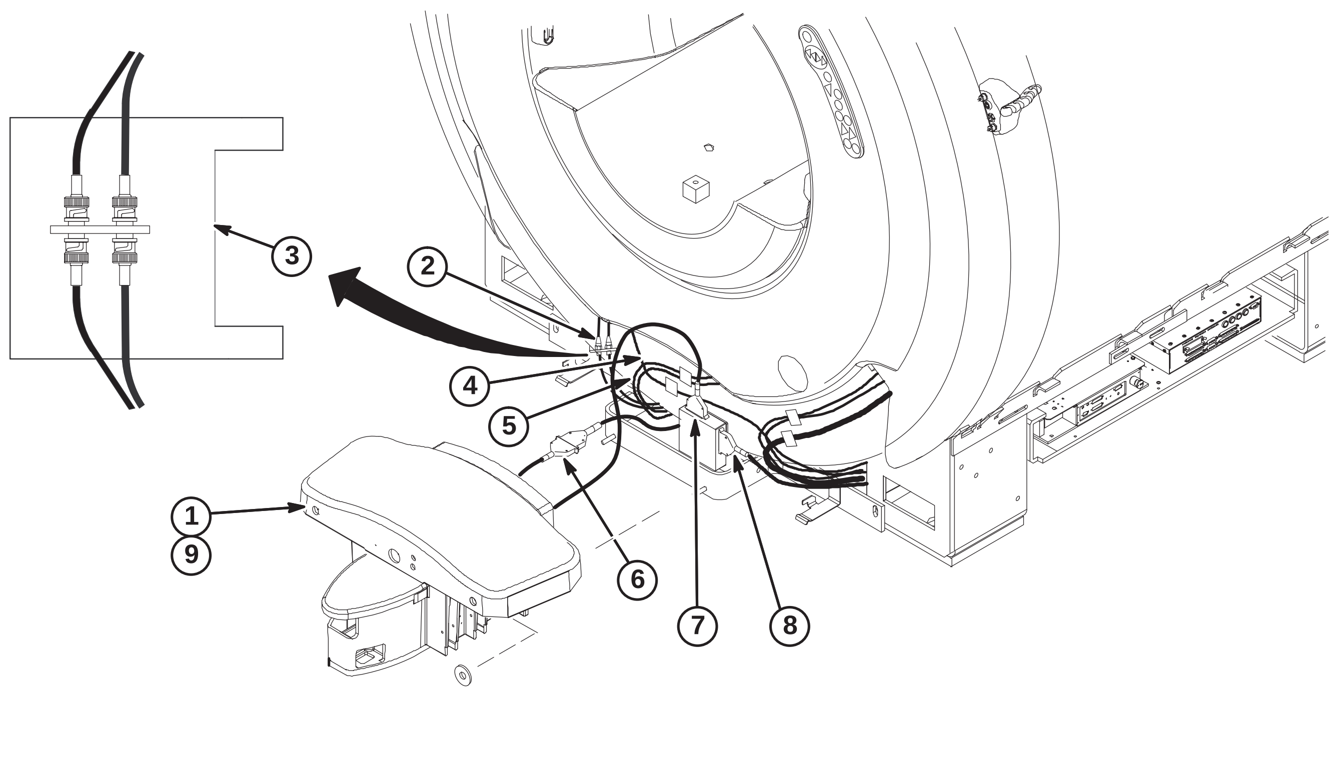
6–3–3 ROUTE AND CONNECT CABLES AT FRONT OF MAGNET ENCLOSURE

CAUTION

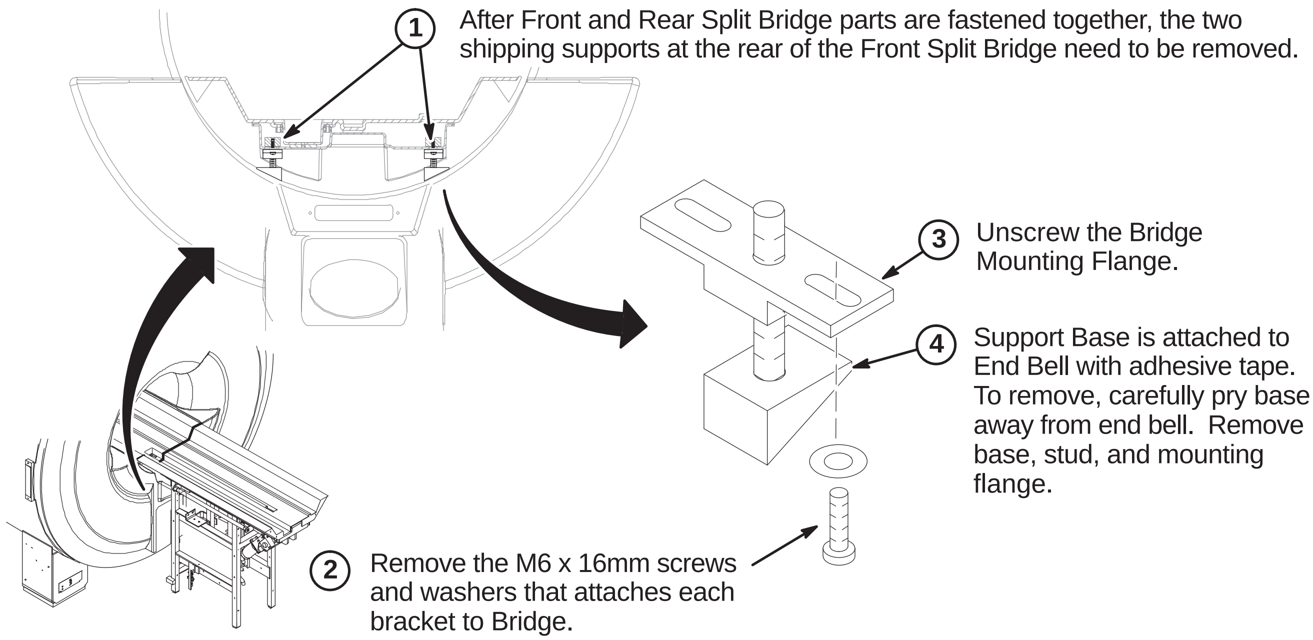
**No metal to metal contact is allowed for connectors to other areas.
i.e. connector touching the magnet.**

For ease in connecting the cables below, Dock should not be attached to mounting bracket.

1. If Dock is fastened to bracket, de–attach and move away from bracket a short distance for ease of making the following cable connections.
2. Route Runs 778 and 779 from J72 and J73 on Penetration Panel and connect to front RF Bias cables J3 & J4.
3. Wrap the RF Bias cable connections with one 177 x 200 Connector Wrap (2236130–2).
4. Route Run 1260, Bore Temperature Sensor, from Front End Bell and connect to J10 on SRI.
5. Route Front End Bell Microphone and Speaker cables underneath magnet to Rear Pedestal and connect to Patient Communication Box as marked.
6. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
7. Connect Dock Cable to J2 on Dock Limit Box
8. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.
9. Re–attach Dock.



6–3–4 REMOVE FRONT SPLIT BRIDGE SHIPPING SUPPORT BRACKETS



6–3–5 INSTALLATION OF REAR PEDESTAL

The Rear Pedestal assembly is delivered to site on a pallet. Refer to instruction manual Direction 5120993 included with Rear Pedestal crate for unpacking and removing from pallet.

6–3–5–A Position Rear Pedestal

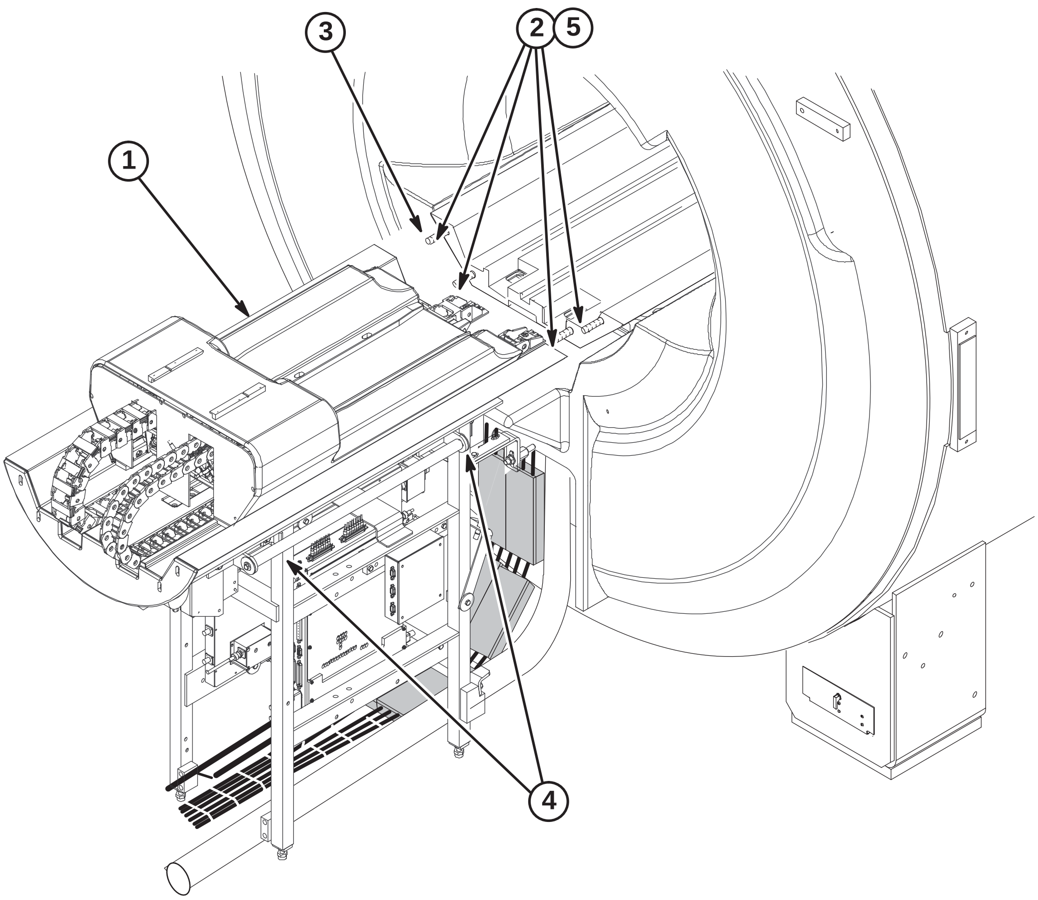
CAUTION

The weight of the Rear Pedestal requires at least THREE people to move Rear Pedestal into position at rear of magnet.

1. Move Rear Pedestal into position at rear on magnet as shown below.

Note: Dress and align all cables and hoses routed thru Rear Pedestal while moving Rear Pedestal closer to magnet.

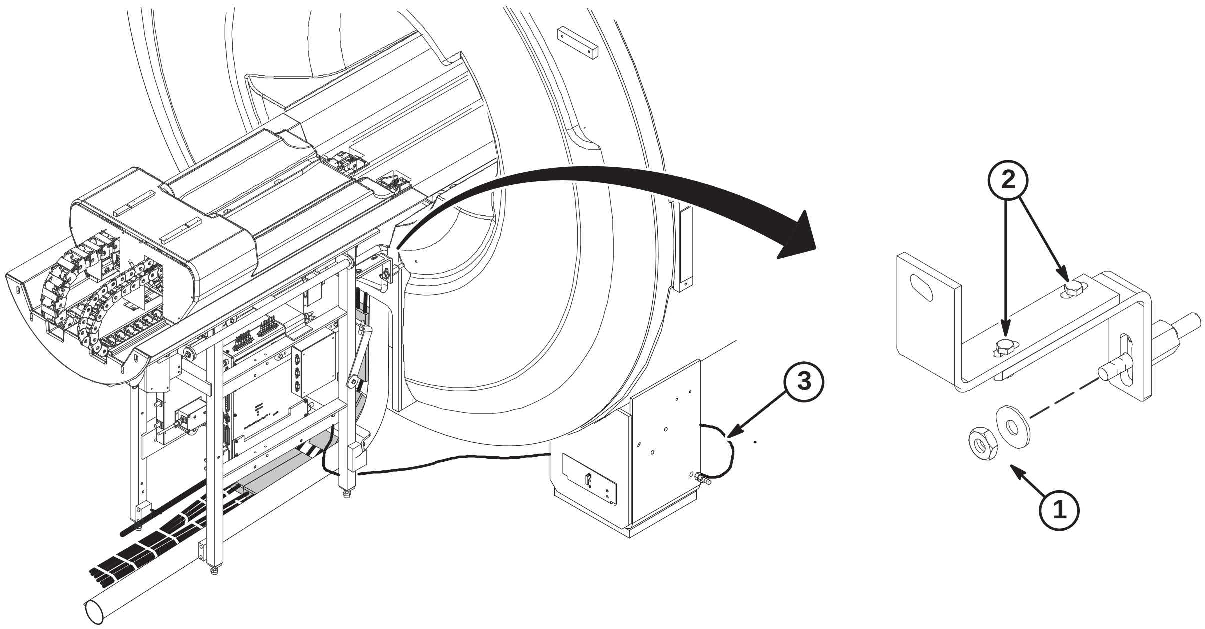
2. Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Split Bridge.
3. Move Rear Pedestal forward and engage front split bridge studs.
4. Adjust the Rear Pedestal on each side to level the Rear Split Bridge with the Front Split Bridge.
5. Fasten Front and Rear Split Bridge with four nuts and washers that were removed from end of Front Split Bridge.



6–3–5–B Connect Front and Rear Split Bridge

Note: Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

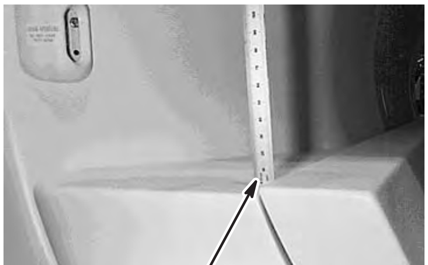
1. Center Rear Pedestal on Rear Endbell and attach to Support Studs using M10 Nut (2109875–4) and Flat Washer (2109875–5) on each stud.
2. Adjustment of Rear Pedestal location can be made at this slotted hole location when conecting the Split Bridge.
3. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.



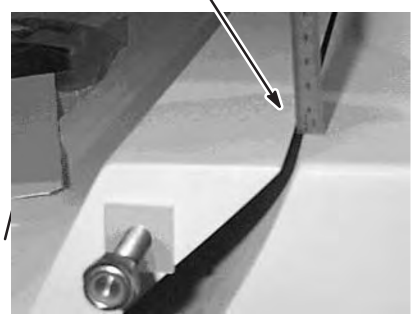
6–3–6 LEVELING OF BRIDGE

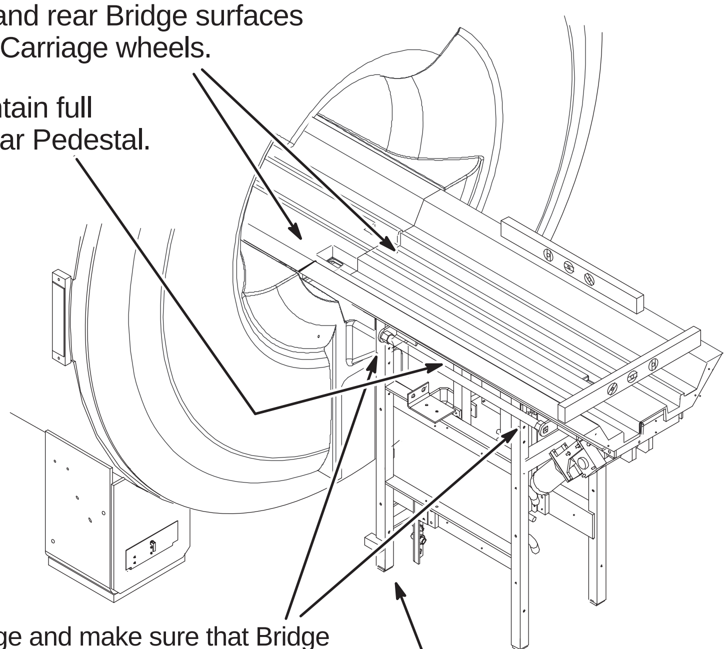
NOTE: Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE: Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.



2mm to 4mm above End Bell surface.

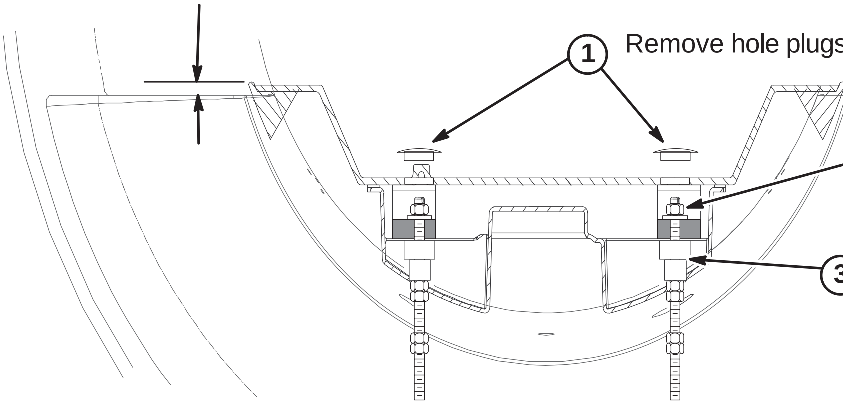




1. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.

6–3–7 HEIGHT ADJUSTMENT AT FRONT OF BRIDGE (IF NEEDED)

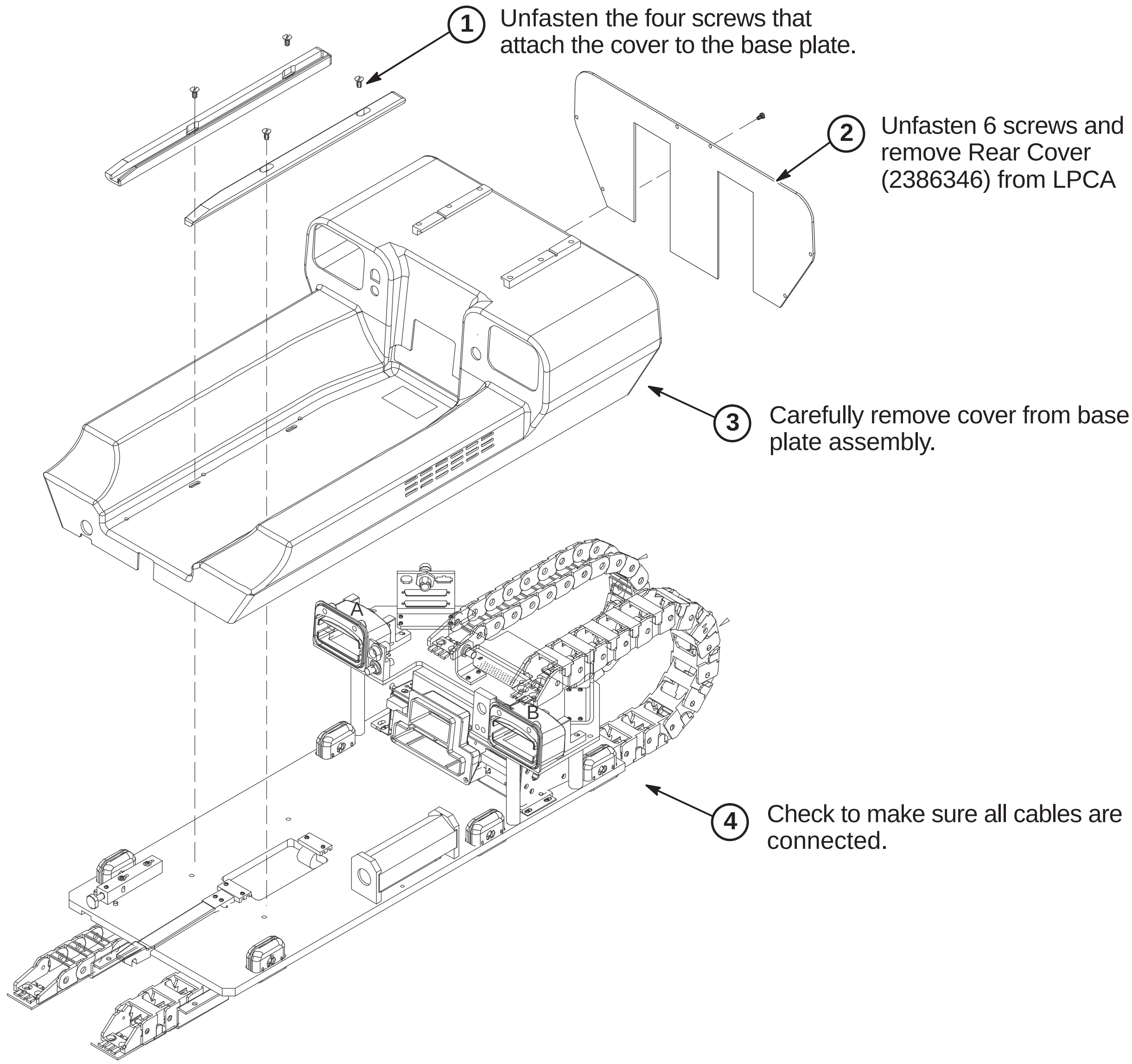
2mm to 4mm above End Bell surface.



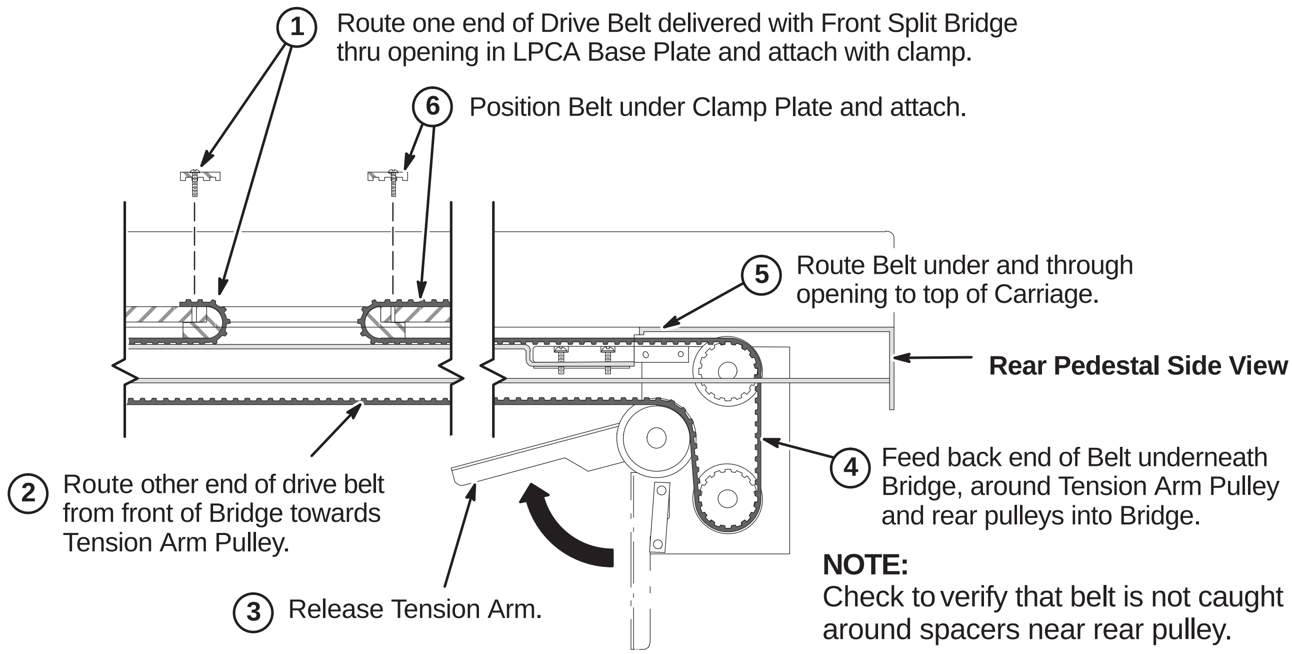
1. Remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Adjust Support Nut (2293381) as needed to achieve correct height.

6–3–8 INSTALLATION OF DRIVE BELT

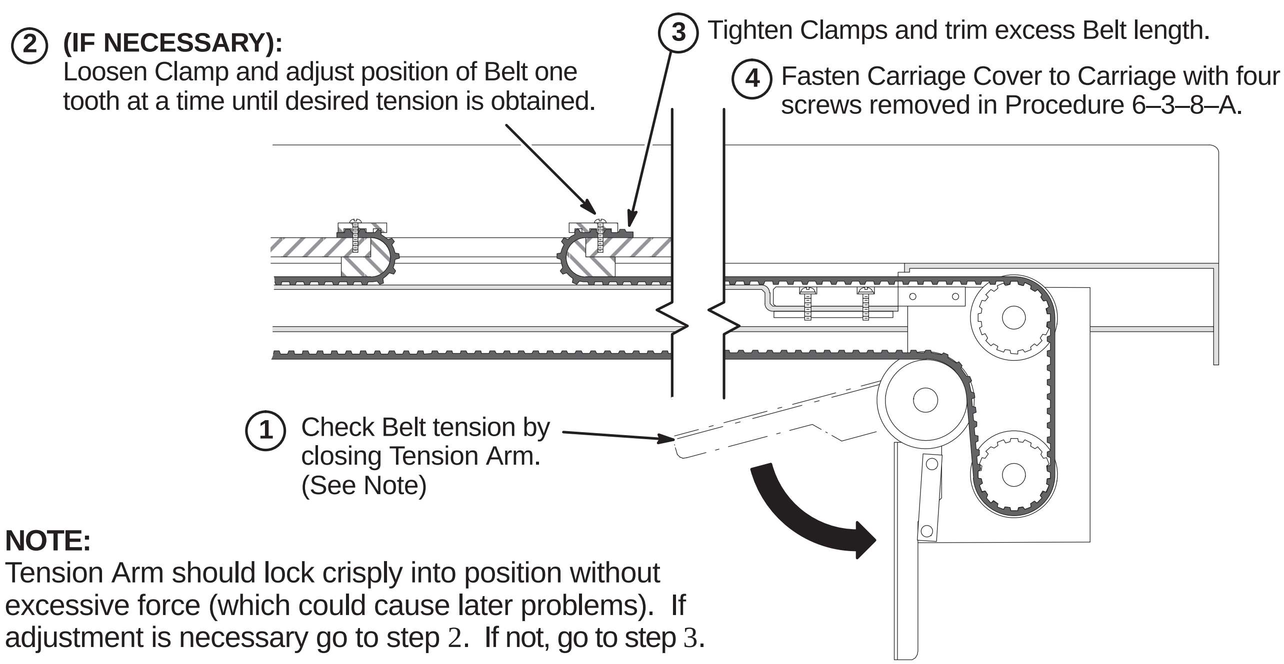
6–3–8–A Remove LPCA Cover



6–3–8–B Route and Attach Drive Belt



6–3–8–C Final Adjustment of Drive Belt



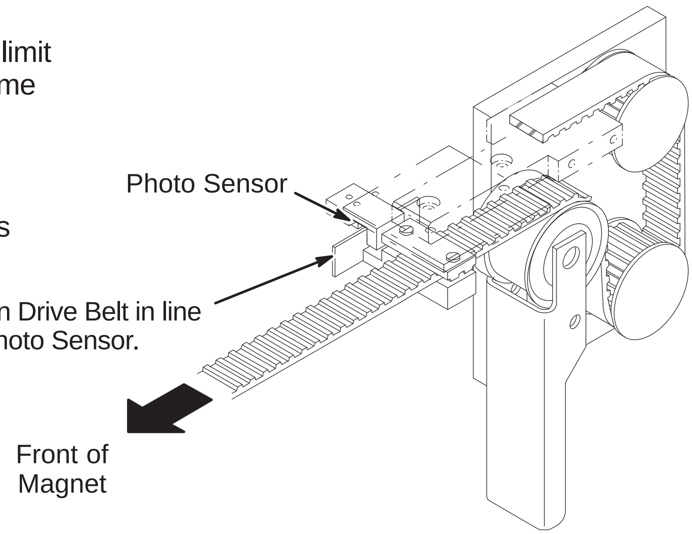
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

6–3–8–D Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1. Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

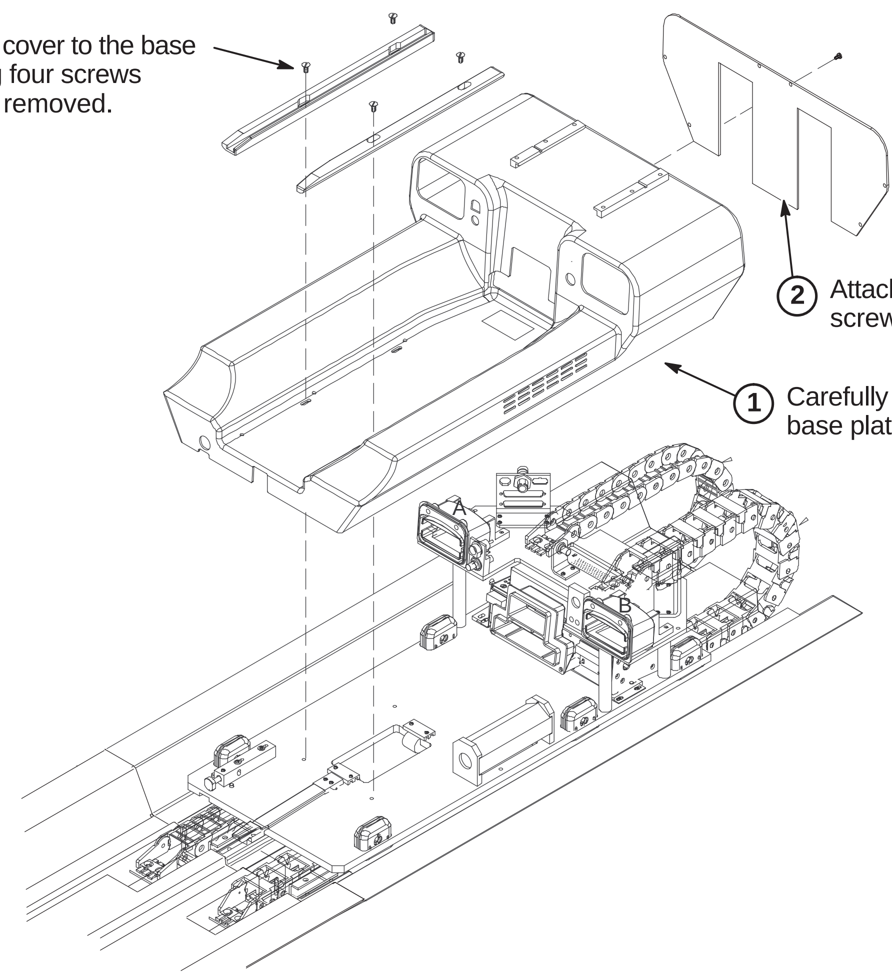


6–3–8–E Installation of LPCA Cover

3. Attach the cover to the base plate using four screws previously removed.

2. Attach Rear Cover using 6 screws previously removed.

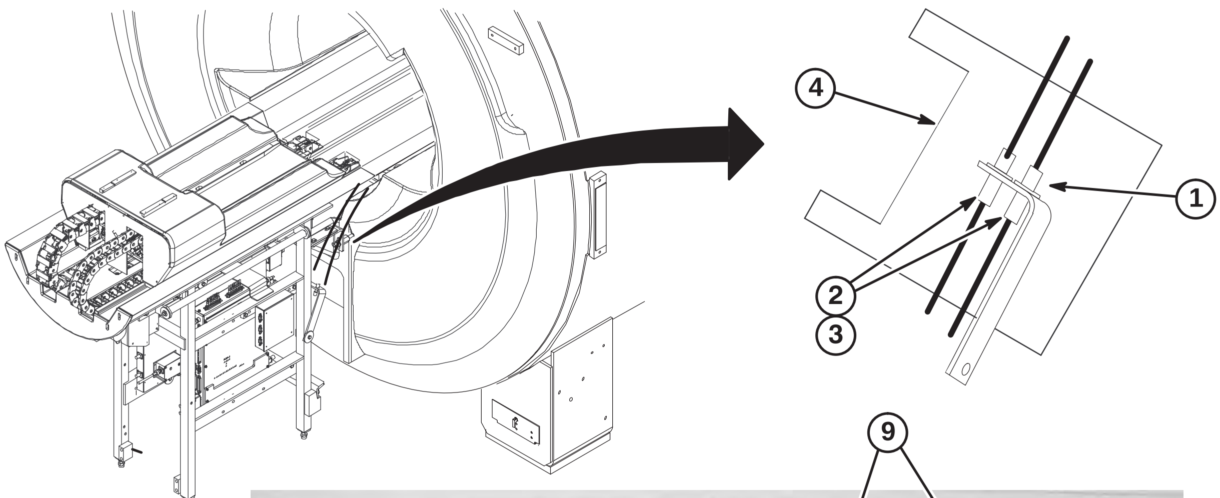
1. Carefully place cover on base plate assembly.

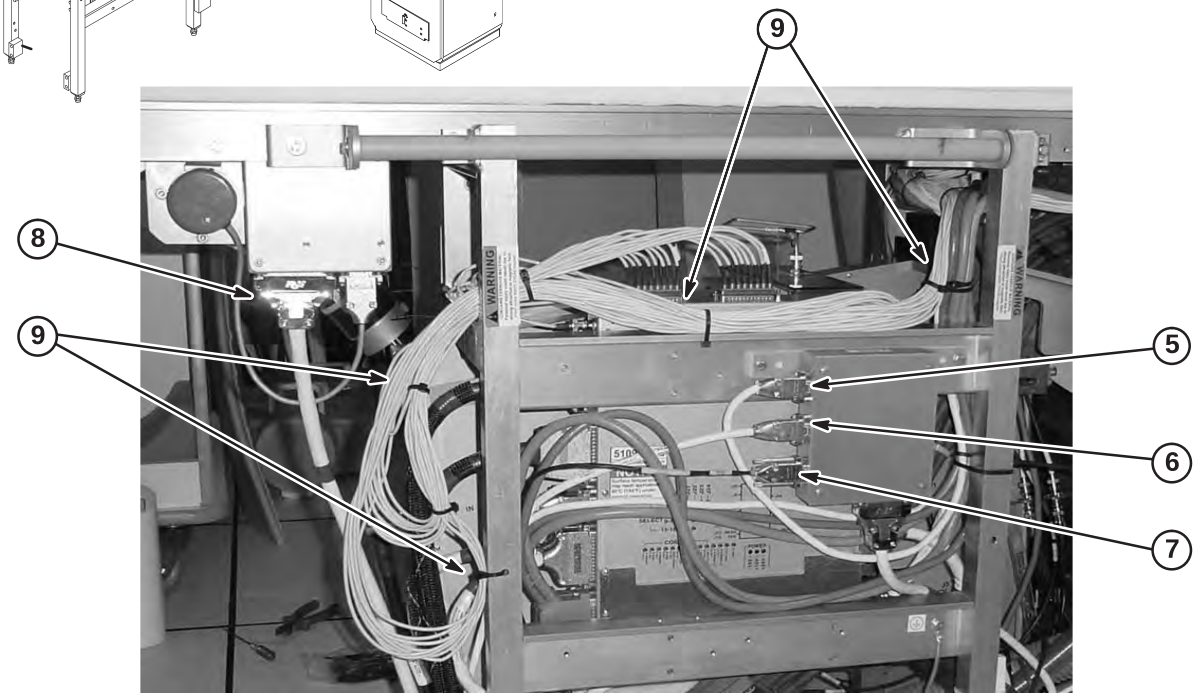


6–3–9 ROUTE AND CONNECT CABLES AT REAR PEDESTAL

6–3–9–A Route/Connect Runs 282, 414, 422, 781, 782, 1057, Multi–coil, & Front End Bell Speaker Cable

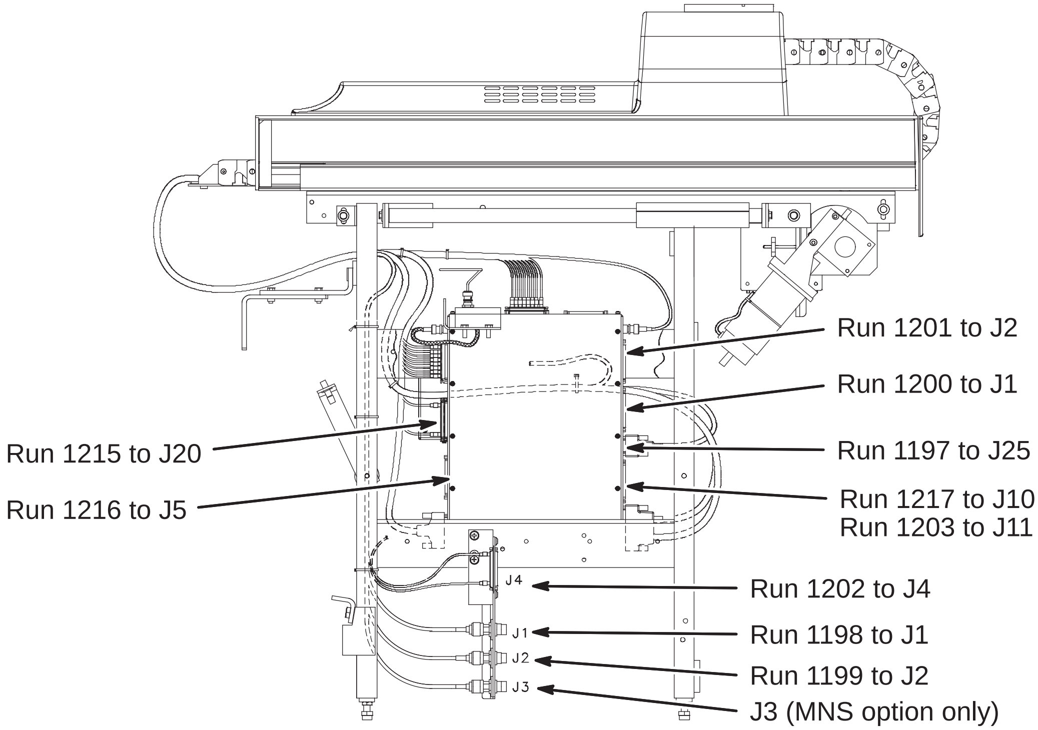
1. Route and connect RF Bias cables from RF Coil to bracket as marked.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. Wrap both Bias connections with 177 x 200 Connector Wrap (2236130–2).
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Patient Comfort Module to J8.
7. Route/connect Run 282 from J12 on Pen Panel to J6.
8. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.
9. Dress and tie–wrap cables as shown. Keep cables away from UTNS antenna, body coil drive cables and home flag on drive belt. There must not be any sharp bends in cable bundle.





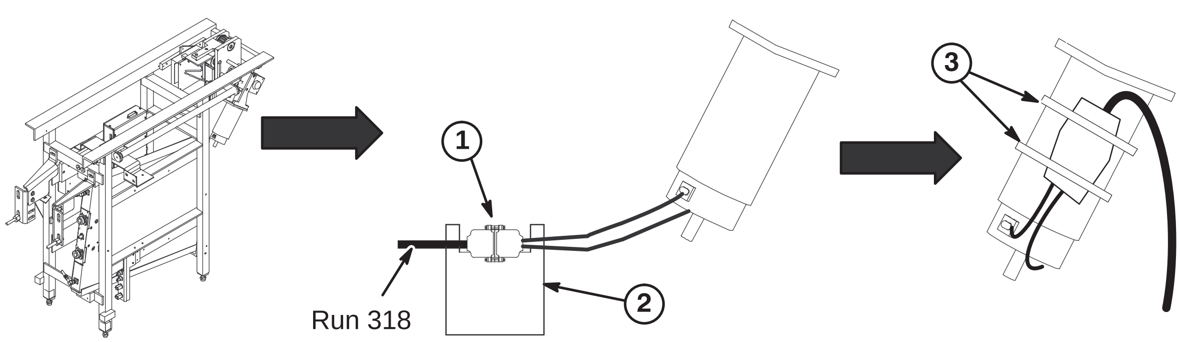
6–3–9–B CONNECT REMAINING CABLES TO REAR PEDESTAL

1. Refer to procedure 6–3–9–A for dressing and tie–wrapping cables from Drive Track. Keep cables away from UTNS antenna, body coil drive cables and home flag on drive belt. There must not be any sharp bends in cable bundle.
2. Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated in illustration below.



6–3–9–C Route/Connect Run 318 to Longitudinal Drive Motor

1. Route Run 318 from J11 on Penetration Panel tand connect to Longitudinal Drive Motor cable.
2. Wrap both connectors with 177 x 200 Connector Wrap (2236130–2).
3. Ty–wrap cable to motor to prevent movement.



6–3–9–D Connect Cables to Hybrid Splitter

CAUTION

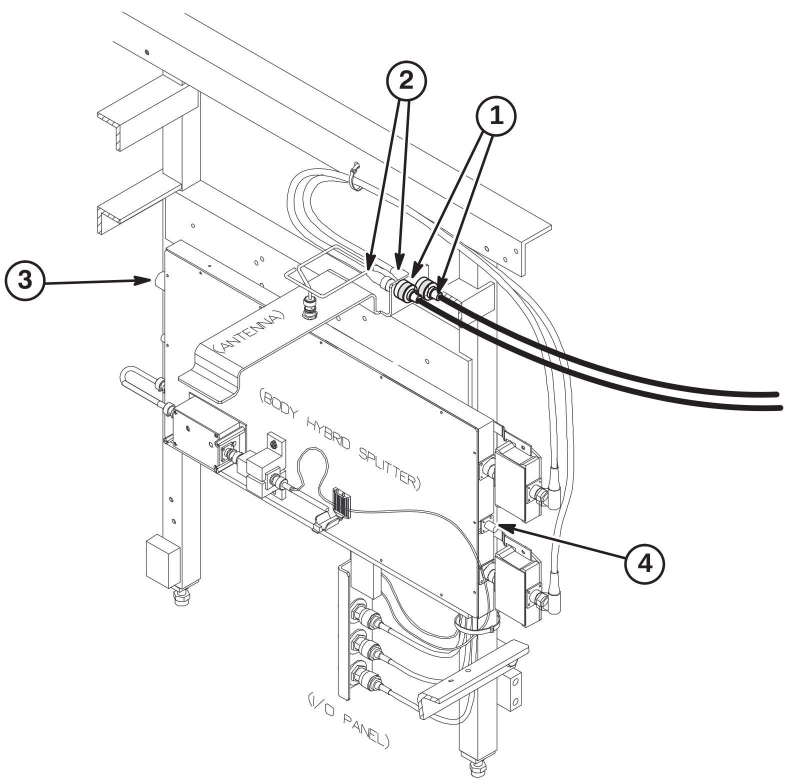
The RF Drive Input cables utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cables to the RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

1. Route and connect I and Q RF Drive input cables from RF Coil and connect as marked to J1 and J2 on bracket.

Note: Runs 743 and 744, Body Coil to Hybrid Splitter cable connections, may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

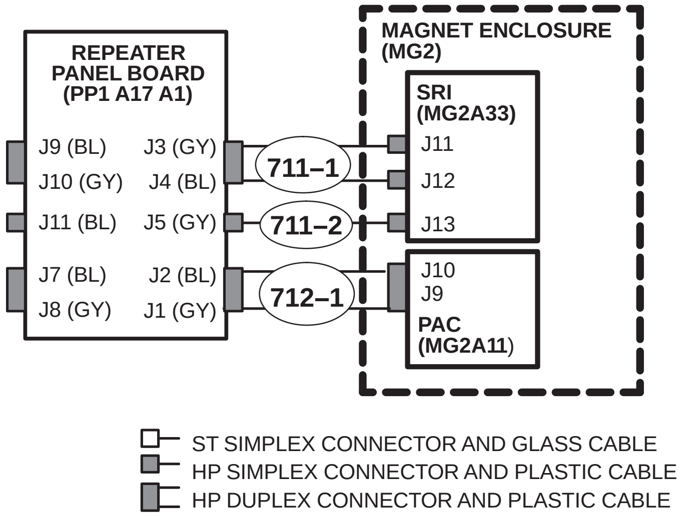
2. Connect Run 743 to J2 and Run 744 to J1 on bracket.
3. Connect RF Flexible cable Run 1250 to J1. Refer to Procedure **6–4–3–B** for instructions on installing the Times Microwave cables. Other end is connected to Run 257.
4. Route and connect Run 780 from J74 on Penetration panel to J6 at front of Hybrid Splitter.



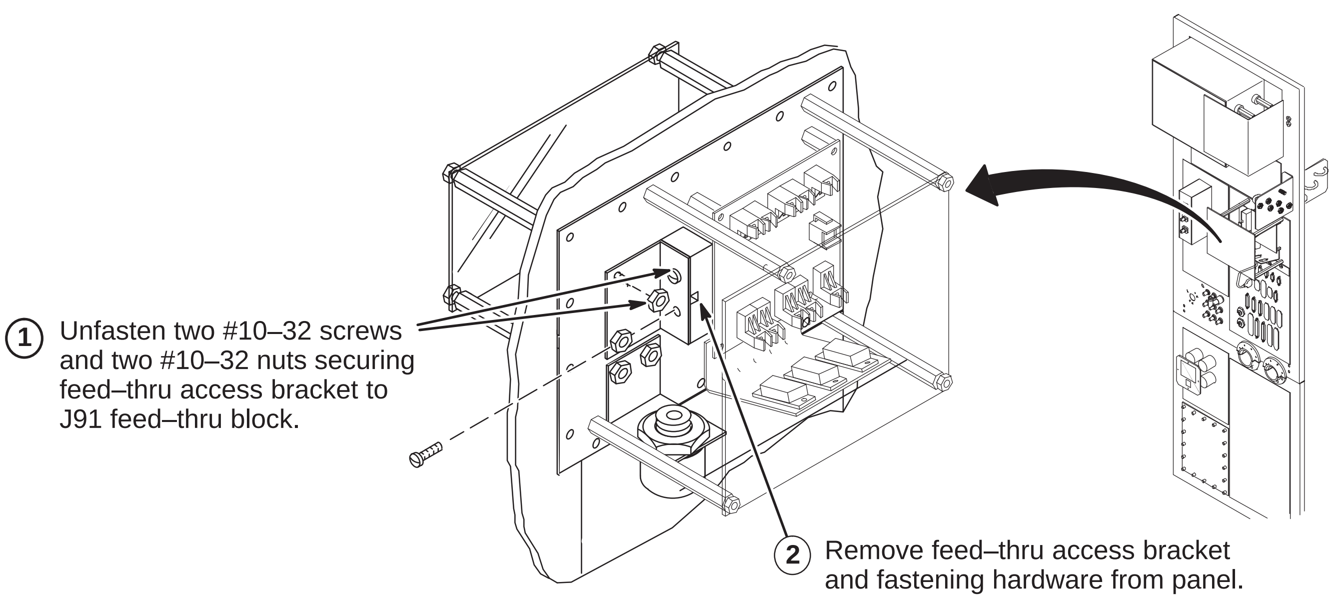
6–3–10 MAGNET ROOM FIBER OPTIC PAC/SRI INSTALLATION

Fiber Optic Run 711/712 is delivered to site connected to the SRI and PAC modules on the platform assembly mounted to the right side of the magnet between the legs. The conduit is stored for shipment in the bore of the magnet. Unroll conduit along designated path in cable trough to Penetration Panel.

Shown below are the connections for the fiber optics cables routed in the magnet room from the Penetration Panel, thru the cable trough to the Rear Pedestal, and under the magnet to the PAC/SRI platform.



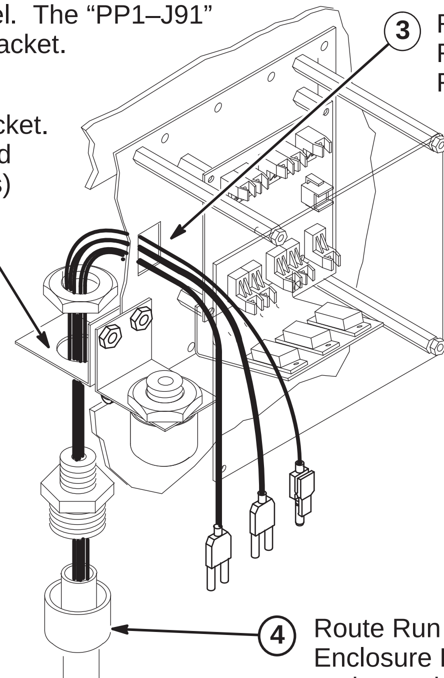
6–3–10–A Repeater Panel Feed–thru Bracket Removal



6–3–10–B Routing Run 711/712 in Magnet Room

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cut and ends terminated with the Fiber Optic Connector Repair Kit. Refer to Appendix for instructions. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

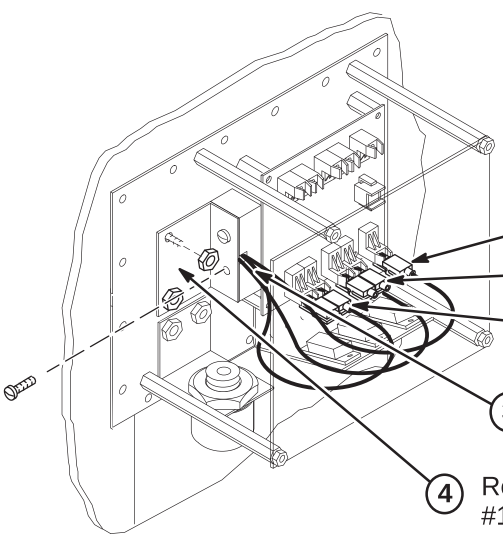
1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The “PP1–J91” end will be attached to interface bracket.
2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)



4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

6–3–10–C Connecting Fiber Optic Run 711/712 to Repeater Panel

Note: Take care to maintain the minimum 2 inch bend radius.



1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.

Run 711–2 (J5)

Run 711–1 (J3, J4)

Run 712–1 (J1, J2)

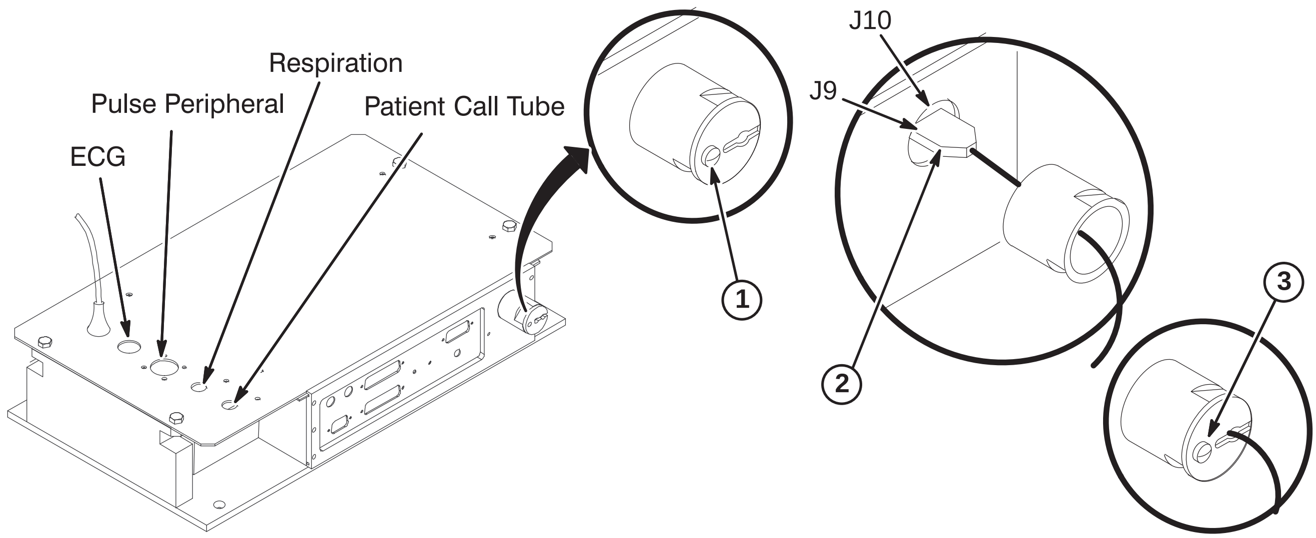
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

6–3–10–D Connecting Fiber Optic Run 712 to PAC Module

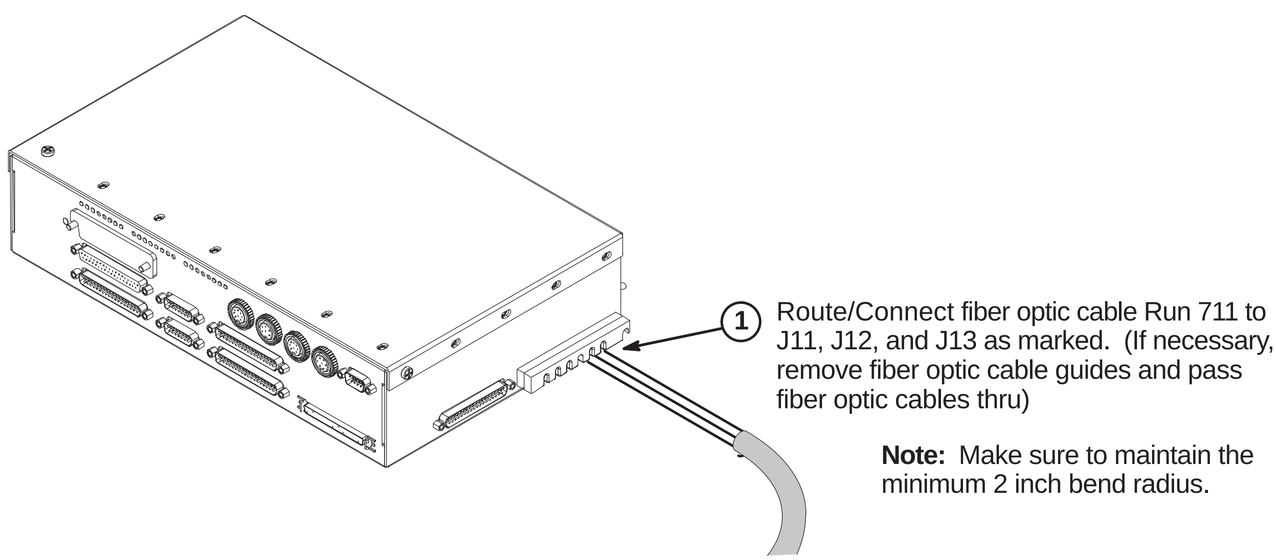
1. Remove shield cover by removing one screw and unscrew shield from module.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.

2. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.
3. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

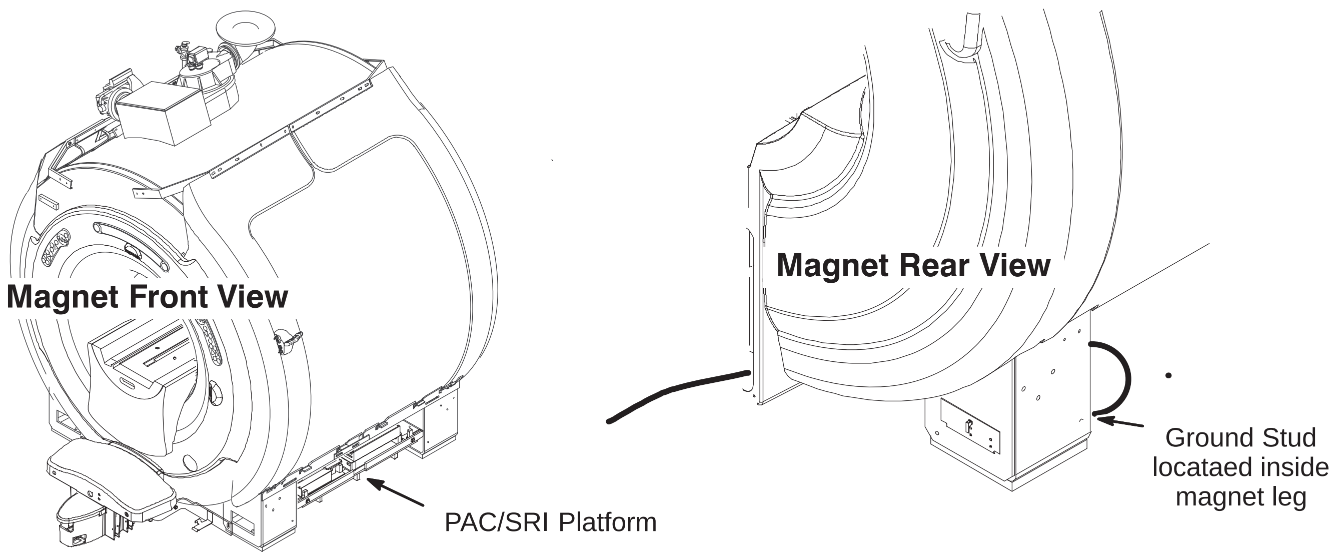


6–3–10–E Connecting Fiber Optic Run 711 to SRI Module



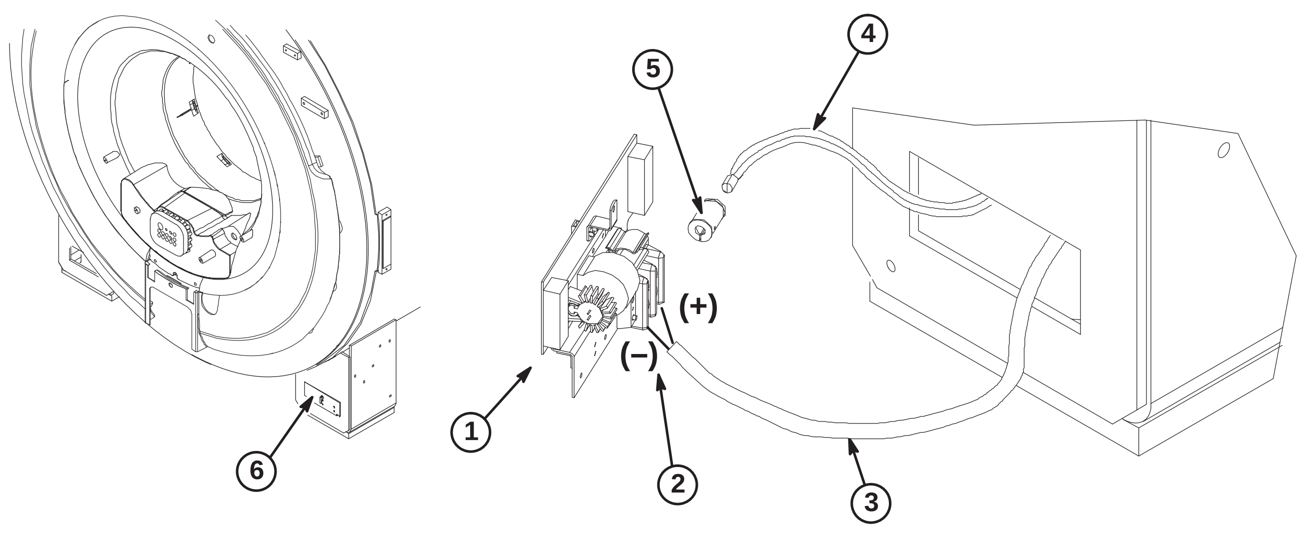
6–3–11 CONNECTION OF MAGNET GROUND, RUN 040

The PAC/SRI Platform is located on right side of magnet. The Magnet ground cable , RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.



6–3–12 INSTALLATION OF BORE LIGHT ASSEMBLY

1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off protective paper on velcro and install assembly in opening in rear leg.



6–3–13 INSTALLATION OF BLOWER BOX AND AIR HOSE FROM REAR OF MAGNET TO BLOWER BOX

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

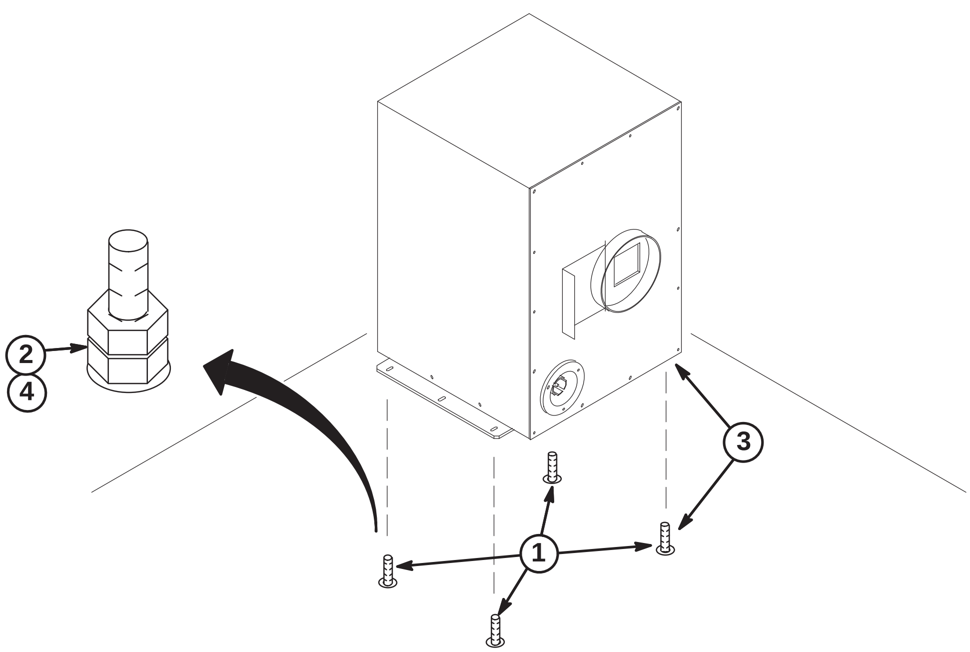
CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions.

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

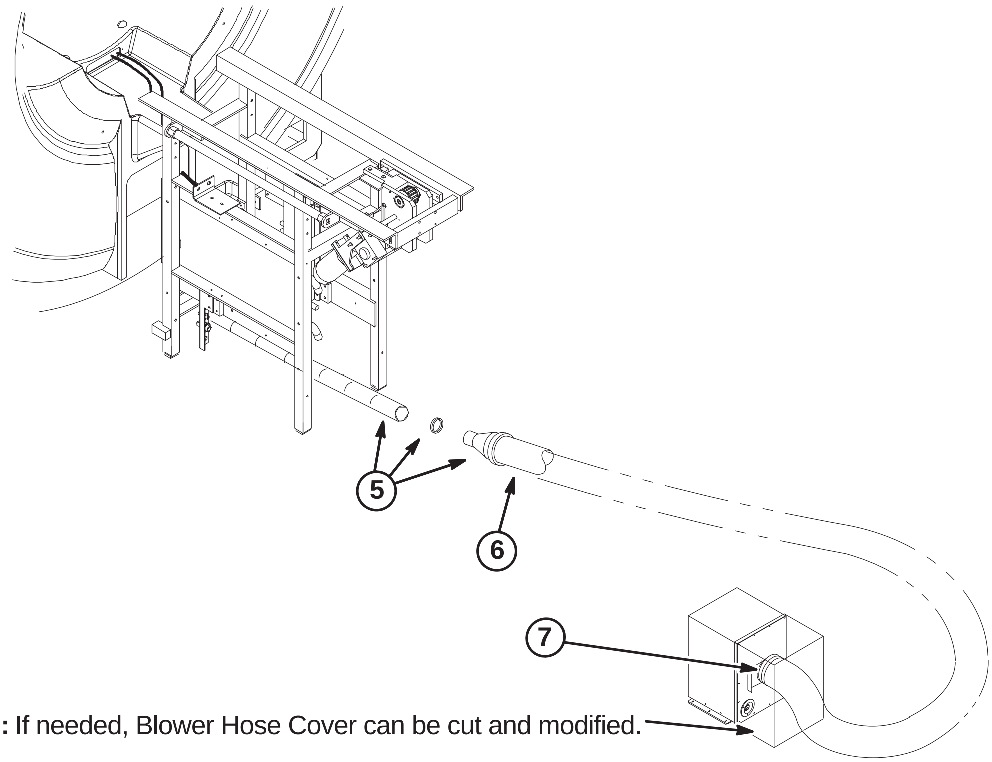
1. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
2. Remove any mounting hardware from the anchors.
3. Position the blower box on top of anchors.
4. Secure the blower box using the hardware provided by the RF shield room vendor to attach.



6–3–13 INSTALLATION OF BLOWER BOX AND AIR HOSE FROM REAR OF MAGNET TO BLOWER BOX (Continued)

The following items are part of Blower Hose Kit, 2129412–5

5. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). **Do not cut or reduce the length of the 2 inch hose.**
6. Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**
7. Route 4 inch hose to Blower Cabinet. Trim to appropriate length and attach to vent opening with 4 inch Hose Clamp (46–208765P34).



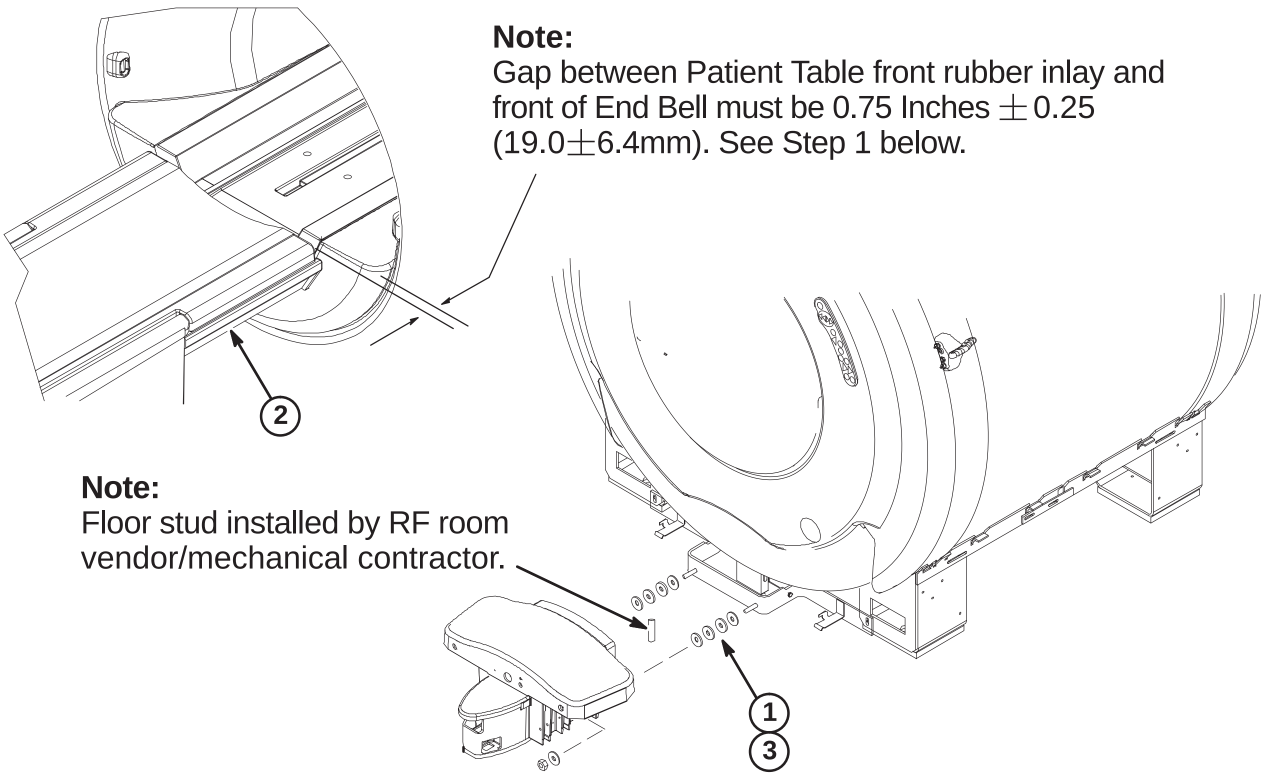
NOTE: If needed, Blower Hose Cover can be cut and modified.

6–3–14 DOCK ADJUSTMENT FOR PATIENT TRANSPORT TABLE GAP

WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION.

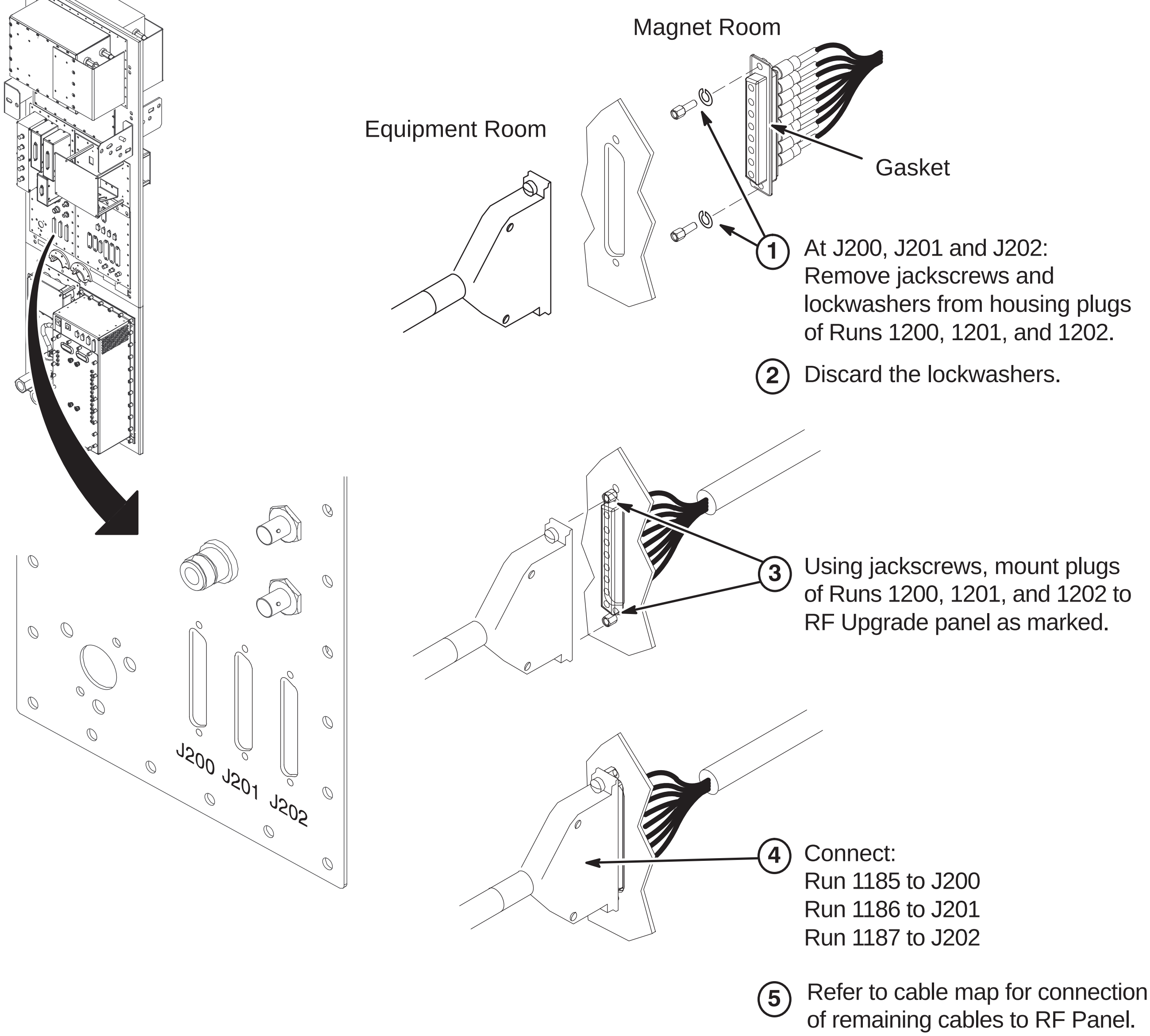
1. Before connecting Patient Table to Dock, make sure there are two (2) 11mm washers on each side between Dock and Dock Support.
2. Move Patient Table into position at front of magnet and connect to Dock. Refer to Note below for proper spacing between Patient Table and End Bell.
3. If adjustment is required, add or remove washers until proper gap is acheived.



6–4 INSTALLATION OF CABLES THAT RUN BETWEEN EQUIPMENT ROOM AND MAGNET ROOM

6–4–1 INSTALLATION OF TIMES MICROWAVE CABLES AT PENETRATION PANEL J200, J201, and J202

Note: Two people are required, one on each side of the Penetration panel, to install the cables in Step 1.



6–4–2 GRADIENT CABLE INSTALLATION

Three cables have been delivered to site. All have three lead wires with terminals at each end. The leads on this cable end are connected to the Equipment room and Magnet room sides of the Gradient Filter on the Penetration Panel. This procedure cuts the cables to length and connects them to the HFD cabinet and the cable leads from the Gradent Coil.

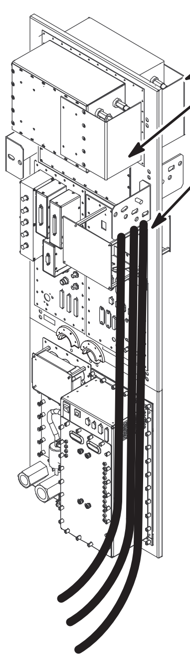
Verify that 100 ft. (30.5 M) length of supplied Run 762, 763, and 764 cable is sufficient to route from HFD Cabinet to Penetration Panel and Penetration Panel to Body Coil Gradient Cables at rear of magnet. **Each cable has identical terminations at each end for connection to <u>both sides of the Penetration Panel Gradient Filter</u>**.

WARNING!

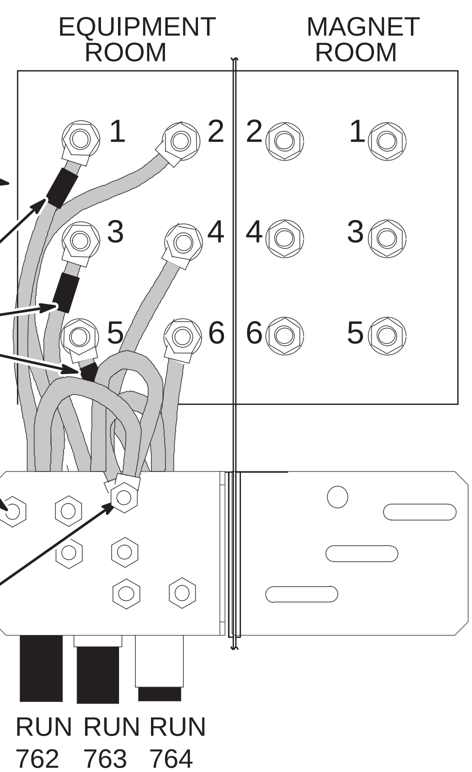
FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

6–4–2–A Connect Gradient Cables to Equipment Room Side of Penetration Panel

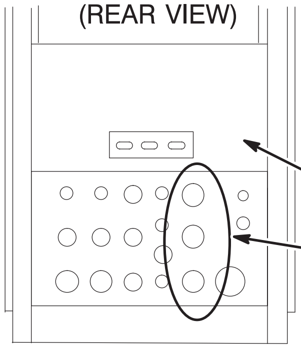


1. Remove gradient filter terminal covers.
2. Route one end of Runs 762, 763, and 764 to gradient filter on Penetration Panel.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.
   Note: Red tape indicates positive (+) wires #1, #3, and #5.
3. Secure each cable Run to strain–relief bracket with U–bolts.
   Note: Do not over–torque the strain relief. Cables may be damaged. Tighten enough to apply friction to cable, but not enough to "crimp" the insulation.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Replace gradient filter terminal covers.



6–4–2–B Route Runs 762, 763, and 764 to HFD Cabinet

HFD CABINET
(REAR VIEW)

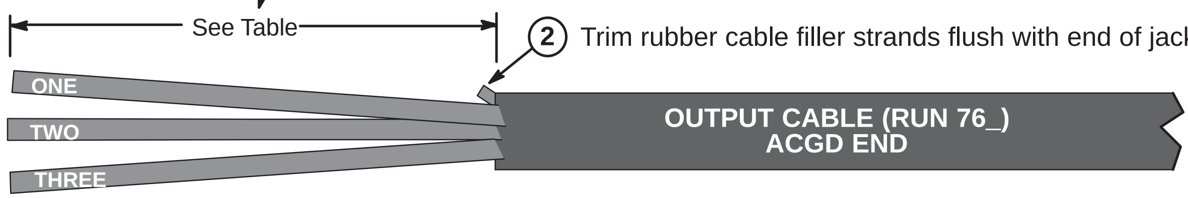


1. Unroll and route each cable from equipment room side of Penetration Panel to HFD Cabinet. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet.
2. Mark location for cutting.
3. Cut cables to length.
4. Remove Rear Cover Plate by removing seven screws that hold it in place.
5. Loosen screws on clamps for gradient output cables. After termination of cables in Procedure **6–4–2–C**, insert ends of cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

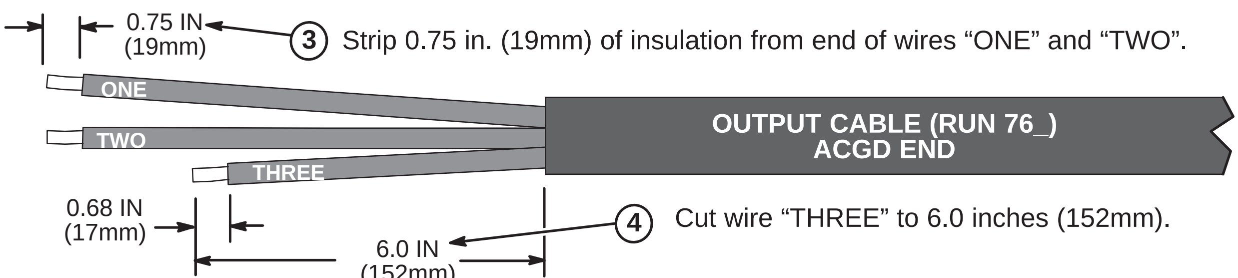
6–4–2–C Terminate Runs 762, 763, and 764 for HFD Cabinet

1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 762 (X), 763 (Y), and 764 (Z).

	Run 762	Run 763	Run 764
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)



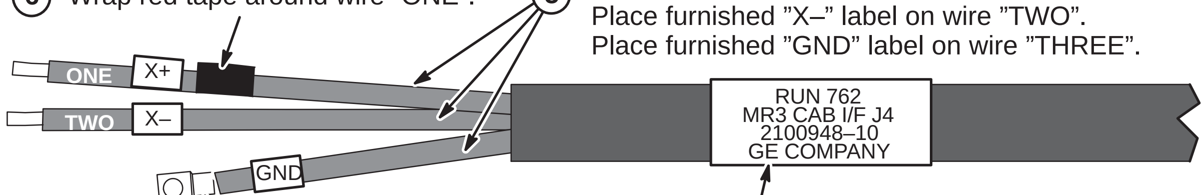
2. Trim rubber cable filler strands flush with end of jacket.



3. Strip 0.75 in. (19mm) of insulation from end of wires “ONE” and “TWO”.
4. Cut wire “THREE” to 6.0 inches (152mm).

Note:
Steps below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y–) and 764 (Z+, Z–).
Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**

6. Wrap red tape around wire ”ONE”.
5. Place furnished ”X+” label on wire ”ONE”. Place furnished ”X–” label on wire ”TWO”. Place furnished ”GND” label on wire ”THREE”.



7. Crimp a furnished T&B #G2–14 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished ”RUN 762” label on cable.

6–4–2–D Connect Runs 762, 763, and 764 at HFD Cabinet

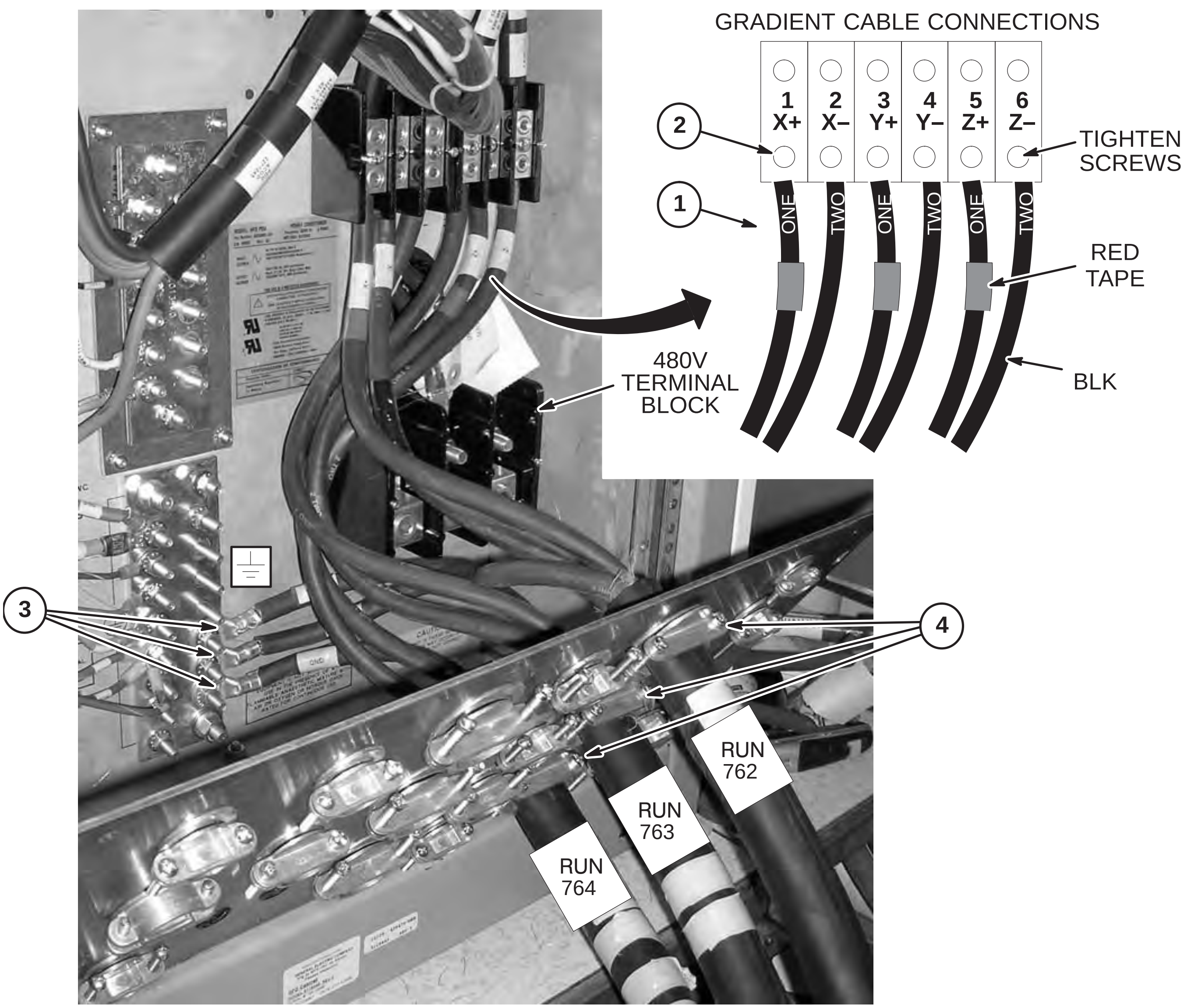
1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

   NOTE: MAKE SURE RUNS 762, 763, AND 764 DO NOT COVER 480V TERMINAL BLOCK

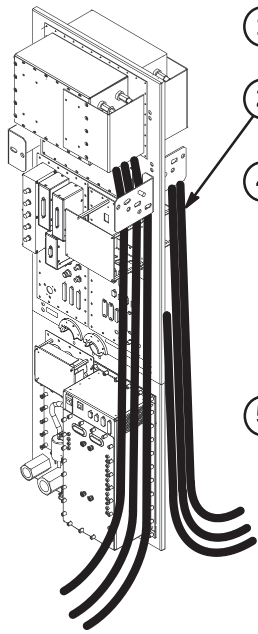
2. Tighten screws to 52–65 inch–lbs (5,88–7,34 N–M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.

   NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.

3. Route the GND leads of each run to the Ground Bus Bar studs. Connect from bottom to top.
4. Tighten clamps on Interface Panel to secure gradient output cables.



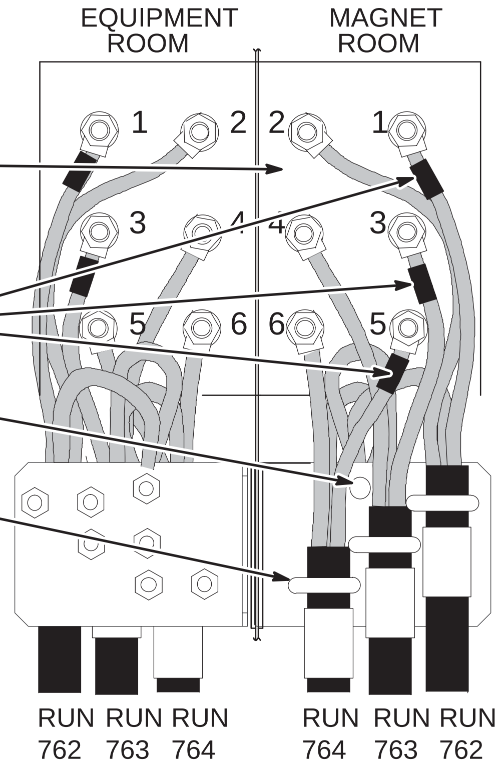
6–4–2–E Connect Gradient Cables to Magnet Room Side of Penetration Panel



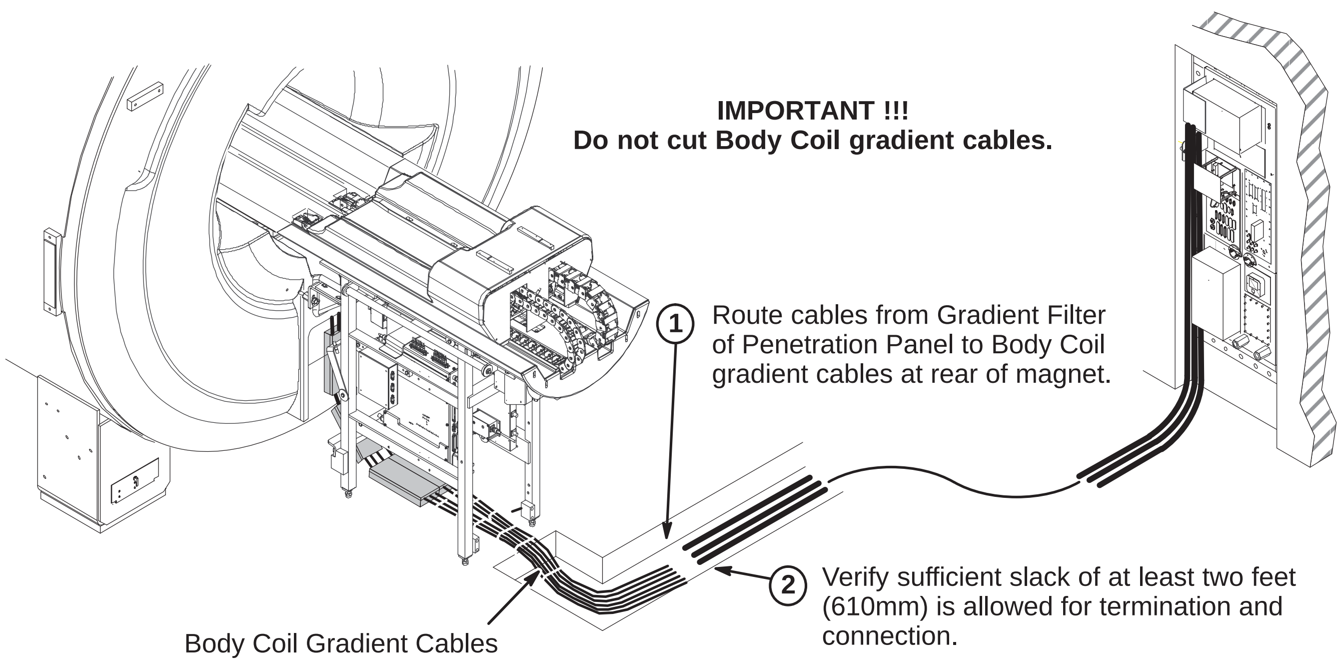
1. Move remainder of gradient cables to Magnet Room.
2. Route Run 762, 763, and 764 to gradient filter on Penetration Panel.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.

   Note: Red tape indicates positive (+) wires #1, #3, and #5.
5. Secure GND lead wire on Ground Stud for each cable Run.
3. Secure each cable Run to strain–relief bracket with U–bolts.

   Note: Do not over–torque the strain relief. Cables may be damaged. Tighten enough to apply friction to cable, but not enough to "crimp" the insulation.
6. Replace gradient filter terminal covers.



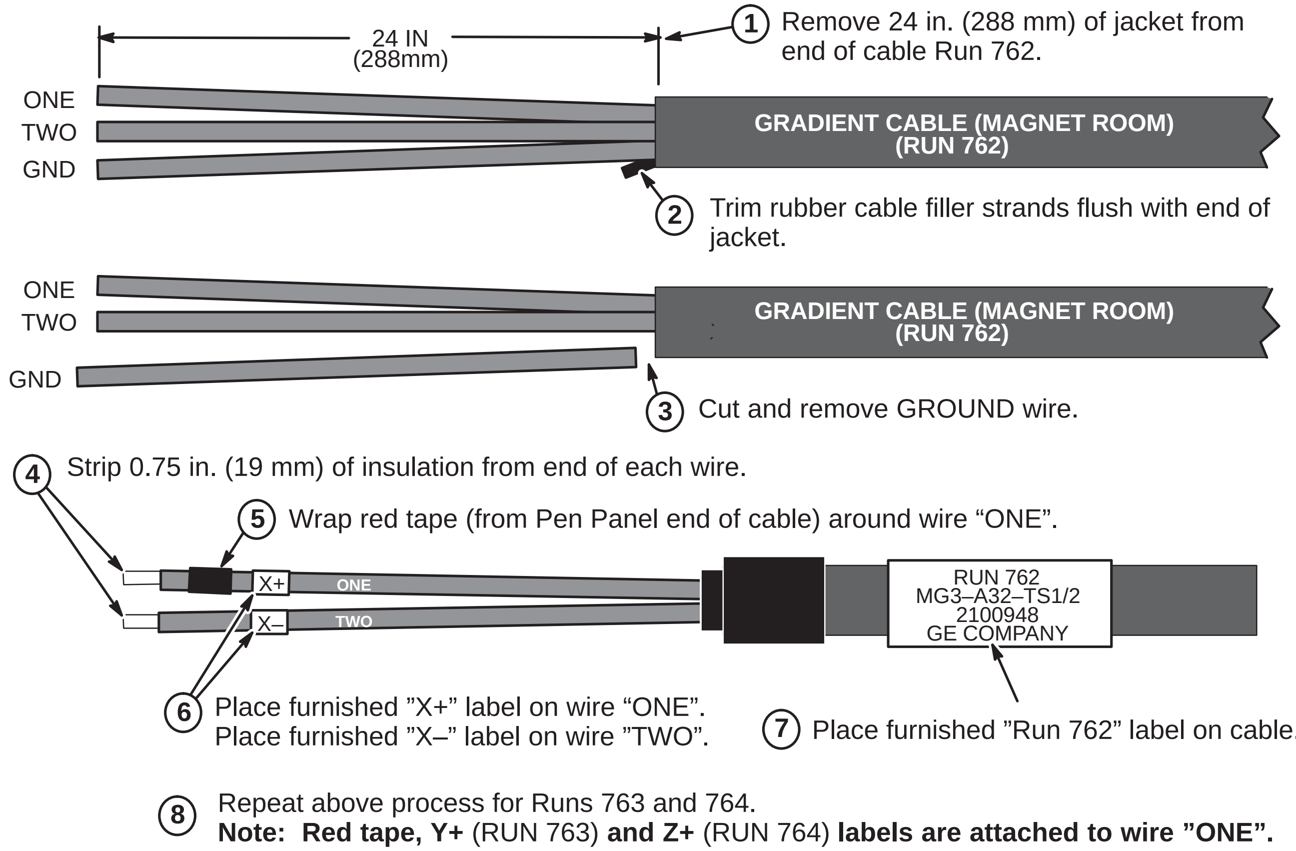
6–4–2–F Route Runs 762, 763, and 764 to Rear Pedestal Area



IMPORTANT !!!
Do not cut Body Coil gradient cables.

1. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
2. Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection.
3. Mark location for cutting and cut cables to length.
4. Label cut ends with applicable run number.

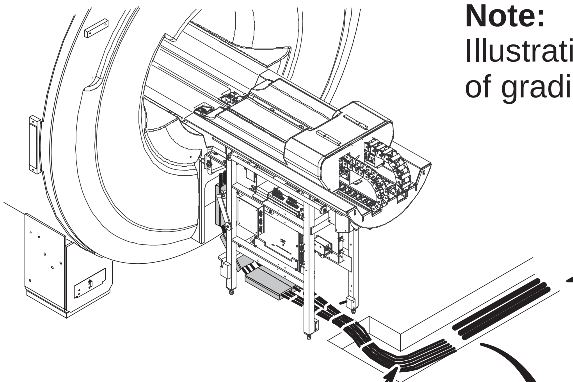
6–4–2–G Terminate Runs 762, 763, and 764 for Magnet Room



6–4–2–H Connection of Body Coil Gradient Cables to runs 762, 763, and 764

Note:
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

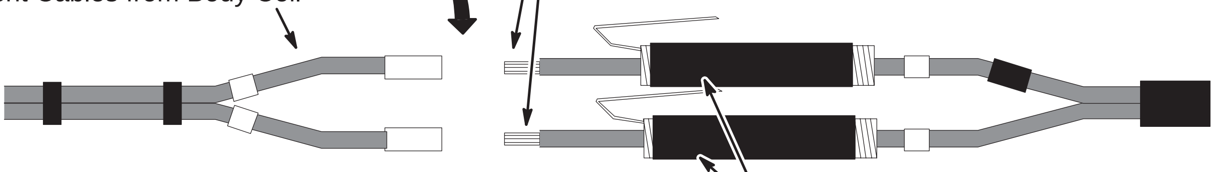
Note:
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet



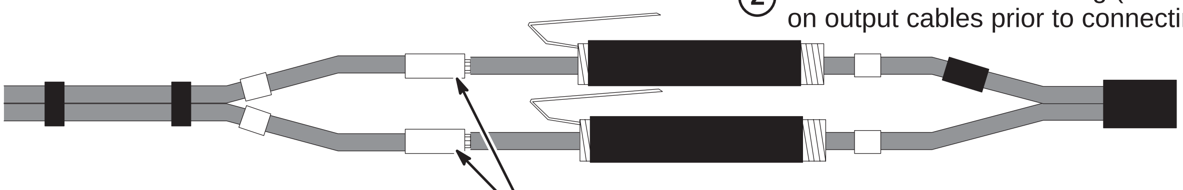
Gradient Output Cables from Penetration Panel

1. Cable connection process shown below is for Run 762. Repeat process for Runs 763 and 764. **Note: Red tape, X+** (RUN 762), **Y+** (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**

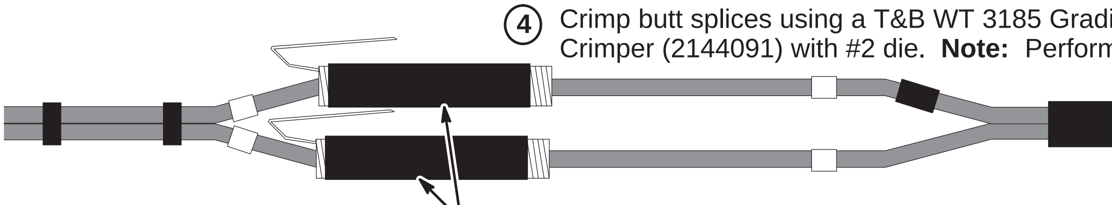
Gradient Cables from Body Coil



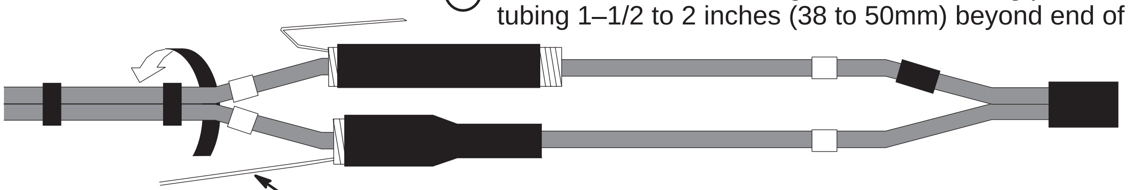
2. Position Cold Shrink tubing (2218148) on output cables prior to connecting.



3. Insert ends of Run 762 into correct butt splice (2218149) on Body Coil gradient cable.
4. Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die. **Note:** Perform tug test.



5. Locate Cold Shrink tubing at desired starting point. Position tubing 1–1/2 to 2 inches (38 to 50mm) beyond end of connector.



6. Pull loose tab gently while unwinding in a counter–clockwise direction. Repeat process for other cable.

7. Connection of cables and installation of Cold Shrink tubing is now complete.

6–4–3 TIMES MICROWAVE RF CABLES INSTALLATION

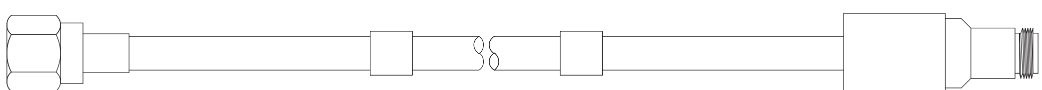
6–4–3–A Cable Information

Five cables, three extension cables, and other associated parts to be installed are provided for this procedure. Also provided are two Times Microwave stripping tools for LMR 900 (2384858) and the LMR 600 (2352193), and the Times Microwave cutting tool (5111565).

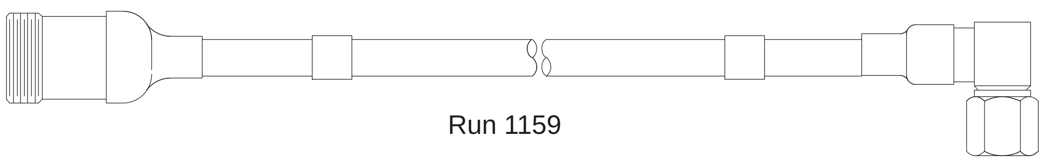
CAUTION

Do not subject LMR–600FR cable (Head) to a bend radius less than 3 inches (75mm). Do not subject LMR–1200FR cable (Body) to a bend radius less than 6.5 inches (165mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Two extension cables, 5116213–2, Run 1249 (Head) and 5121269, Run 1159 (Body) are connected to the RF Amplifier module in the RF/System Cabinet. The other extension cable, Run 1250, 5121245, is connected to the Body Hybrid Splitter in the Rear Pedestal.

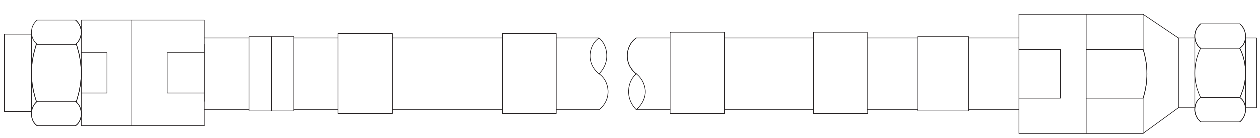


Run 1249 and Run1250



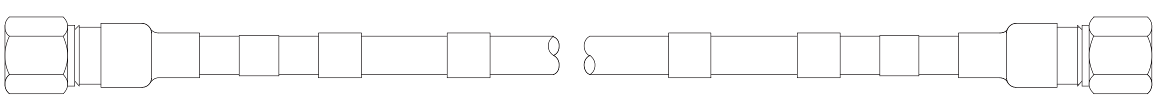
Run 1159

Cable 2316584–2 (Body, Runs 235 & 257), will be cut to length to make two cables and terminated for routing in the equipment and scan rooms.



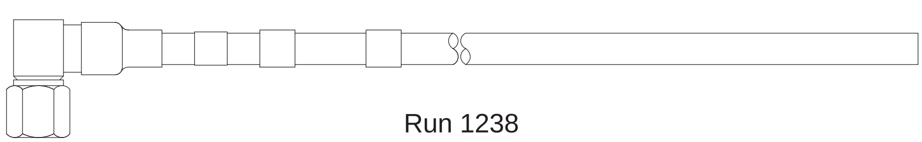
Run 235 and Run 257

Cable 5115915 (Head, Runs 1198 & 1199), will be cut to length to make two cables and terminated for routing in the scan room.



Run 1198 and Run 1199

Cable 5115916 (Head, Run 1238), will be cut to length and terminated for routing in the equipment room.



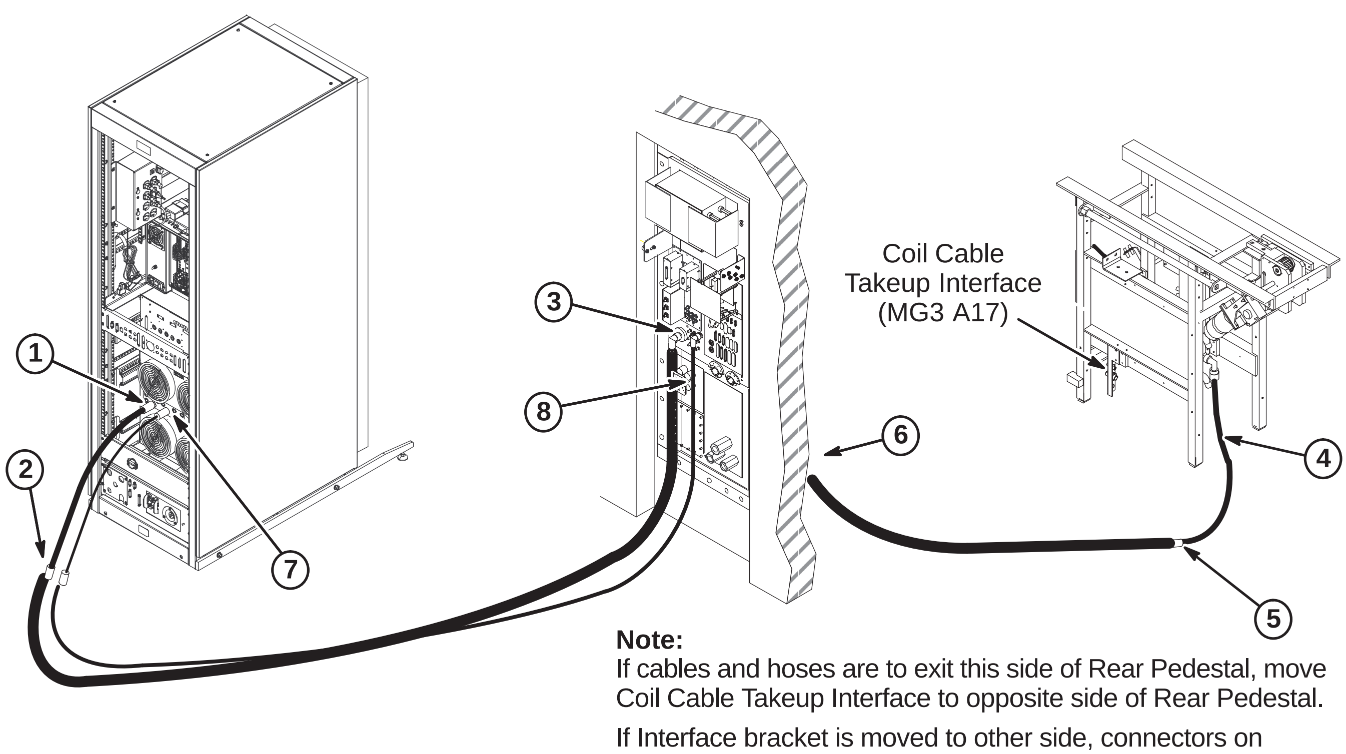
Run 1238

The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

6–4–3–B Install Runs 235, 257, 1238, 1249, 1250, and 1259

RF cables will be routed and connected as indicated in the following steps.

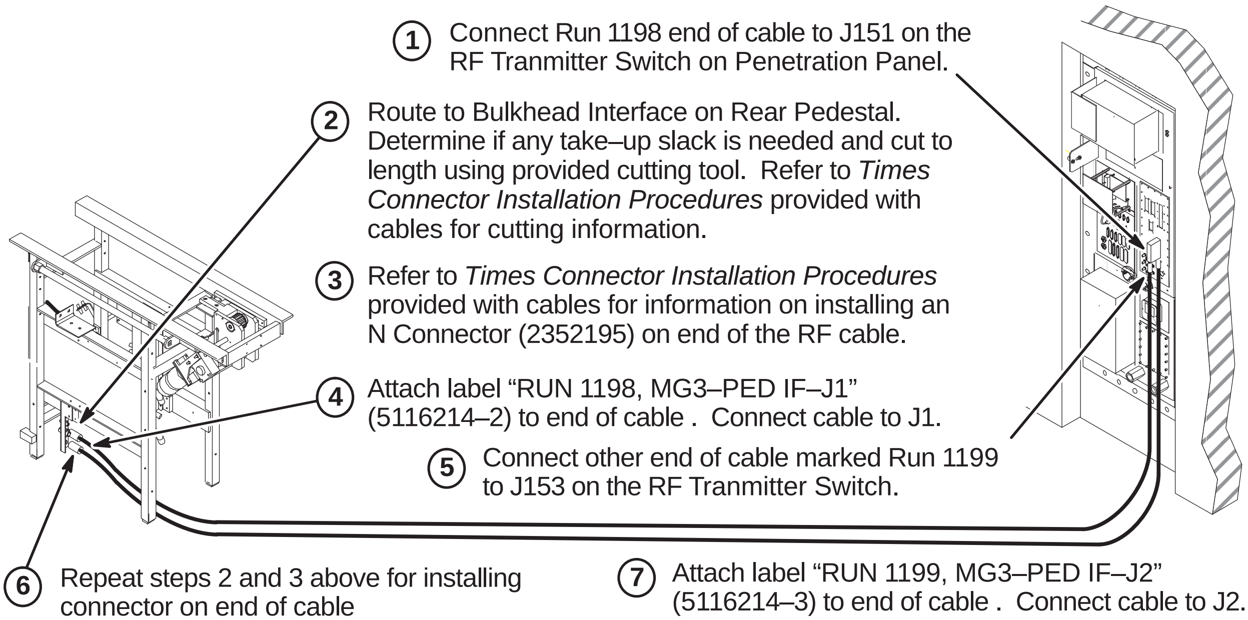
1. Connect extension cable Run 1259 to J4 Body Output at rear of RF Amp in RFS cabinet.
2. Connect end of Times RF Body cable marked “(RUN 235) FJ5” to Run 1259, Body extension cable. Attach label “RUN 235, Connect to Run 1259” (5121244) over existing label.
3. Route cable to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **6–4–3–E** for termination procedure of cut end.
4. Connect Run 1250 to J1 on Hybrid Splitter.
5. Move remainder of RF Body cable to Scan Room and connect other end of cable marked “(RUN 257) MG3A3–J1” to Run 1250 at Rear Pedestal as shown below. Attach label “RUN 257, Connect to Run 1250” (5121244–2) over existing label.
6. Route remainder of cable to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **6–4–3–E** for termination instructions of cut end.
7. Connect extension cable Run 1249 to J3 Head Output at rear of RF Amp IN RFS cabinet.
8. Connect end of Times RF Head cable marked “(RUN 1238) PP1–J4” to J4 at Penetration Panel. Route to RF Head extension cable at rear of SRF cabinet. Determine if any take–up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **6–4–3–D** for termination procedure of cut end.



Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

6–4–3–C Install Runs 1198 and 1199



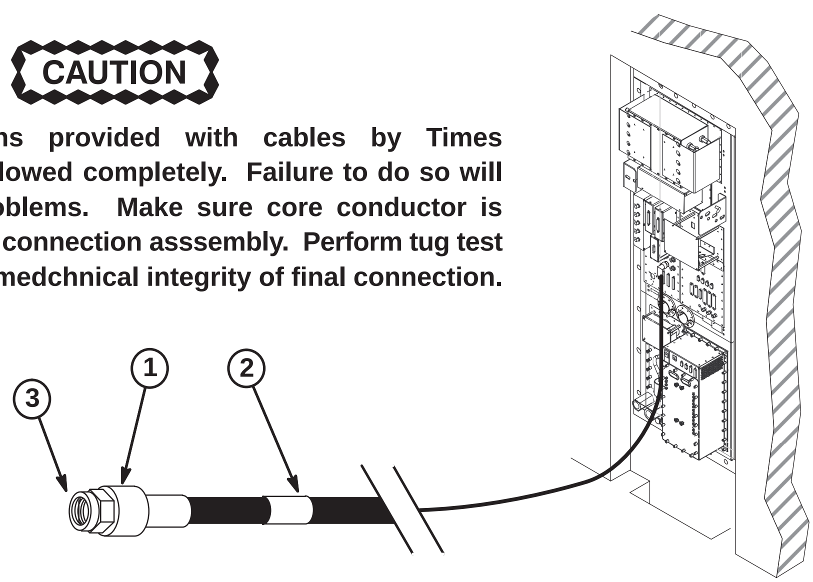
6–4–3–D Terminate and Connect RF Head Coil Cable

Perform the following steps for terminating and connecting the Head RF cables in the Equipment room.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an Straight Male N Connector (2352195) on each end of the RF cable. Use provided cutting tool (5111565) and the provided LMR600 Stripping Tool (2352193).
2. Attach label "RUN 1238 CONNECT TO RUN 1249" (5116214) to cable as shown in Equipment room.
3. Connect cable to Run 1249.

CAUTION

Installation instructions provided with cables by Times Microwave must be followed completely. Failure to do so will cause connectivity problems. Make sure core conductor is de–burred prior to final connection asssembly. Perform tug test and twist test to verify medchnical integrity of final connection.



6–4–3–E Terminate and Connect RF Body Cables at Penetration Panel

Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44. Two people are required to complete installation of cables.

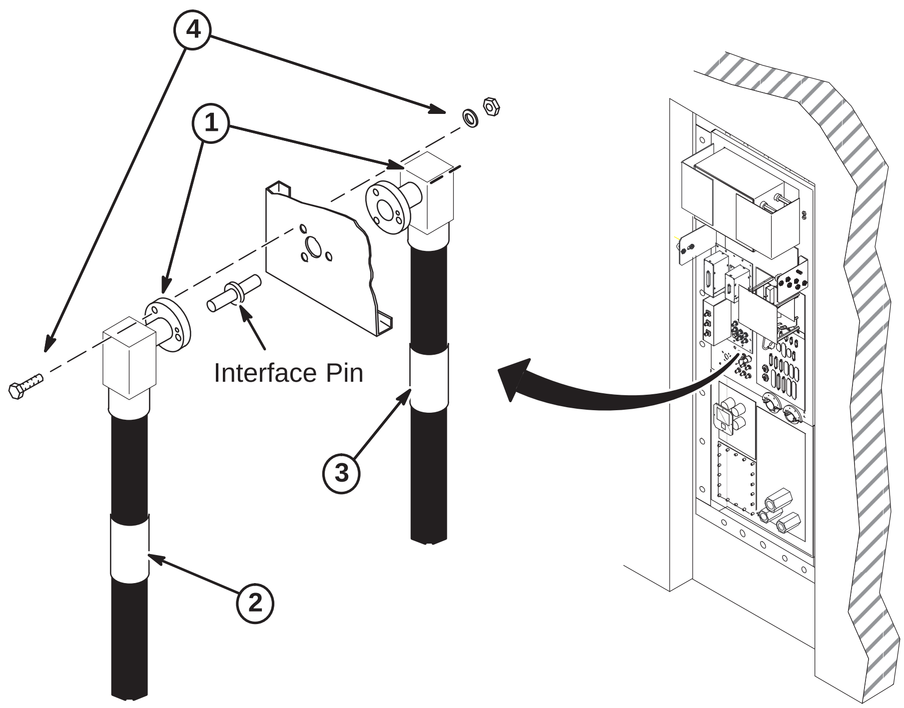
CAUTION

Installation instructions provided with cables by Times Microwave must be followed completely. Failure to do so will cause connectivity problems.

1. Refer to *Times Connector Installation Procedures* provided with cables for instructions on installing an EIA Right Angle Connector (2319986) on each end of the RF cables. Use provided cutting tool (5111565) and the LMR900 Stripping Tool (2384858).

   Note
   Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

2. Attach label "RUN 235, PP1–J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1–J44" (2357230–2) to cable as shown on Scan room side of Penetration Panel.
4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
   - 1/4–20 x 1.25 Hex Head Bolt (brass)
   - 1/4 Screw Size Washer (phos bronze)
   - 1/4–20, 7/16" Hex x 3/16" thk Nut (brass)



6–4–4 PNEUMATIC PATIENT ALERT INSTALLATION

The customer and users should be involved in the following decisions:

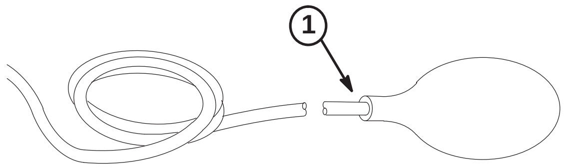
- Routing of Pneumatic Tubing
- Location of Control Box

CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

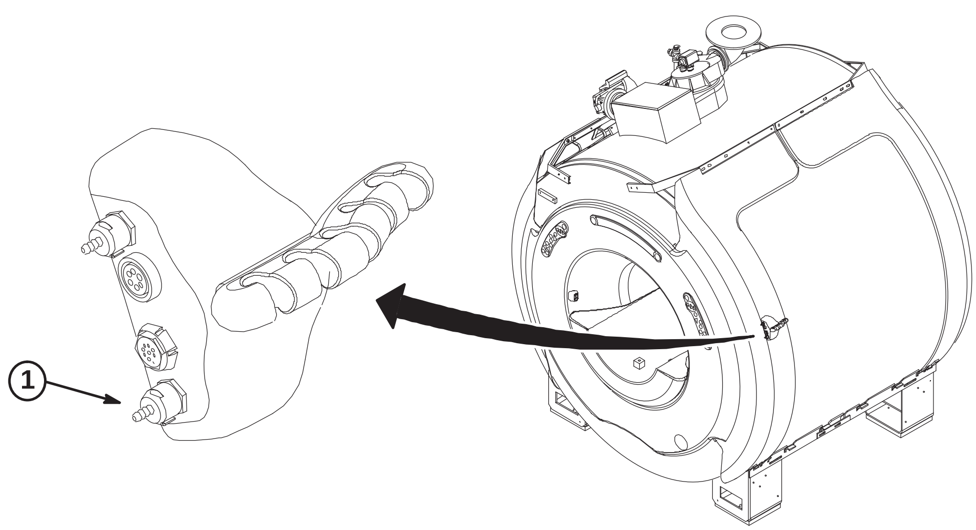
6–4–4–A SQUEEZE BULB INSTALLATION

1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the “CALL” connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.



6–4–4–B CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

1. Connect end of tubing to ”CALL” connection on Remote PAC Interface Assembly.



6–4–4–C MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

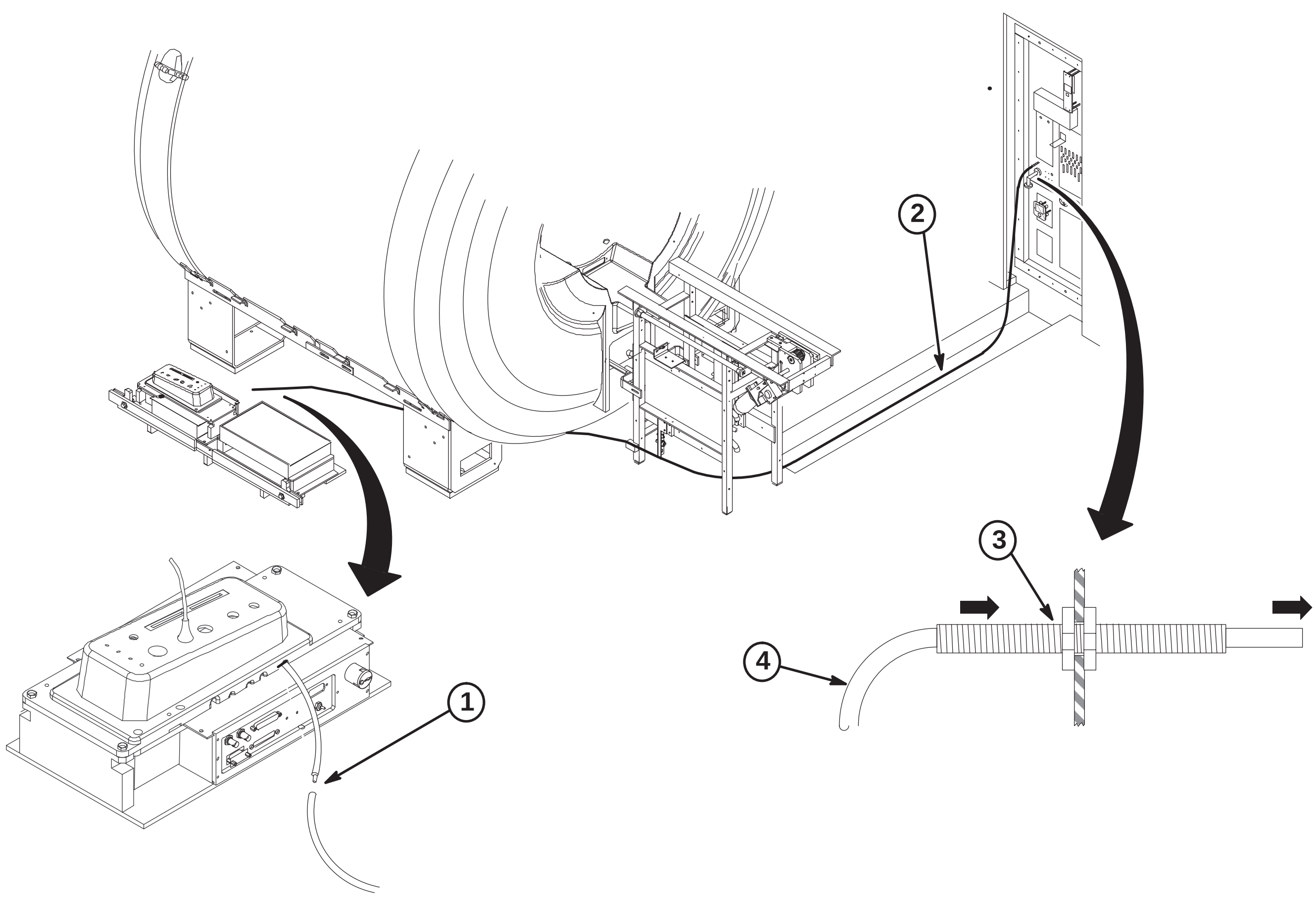
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

1. Push end of pneumatic tubing onto PAC–II tubing connector.

Note: If the pneumatic tubing is to be pulled thru fiber–optic conduit, coat with baby powder, not grease, to allow easier pulling.

2. Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.
3. Insert Wave Guide thru “ALERT AIR LINE” hole and fasten to Penetration Panel with nut on each side.
4. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



6–4–4–D DETERMINING NEED OF EXTENDER BOX

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46–317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:**
  The extender is <u>not</u> needed. Proceed to Procedure **6–4–4–F**, CONTROL BOX INSTALLATION.
- **If the pneumatic tubing is <u>not</u> long enough to reach control box**:
  The extender is needed. Proceed to Procedure **6–4–4–E**, INSTALLATION OF EXTENDER KIT.

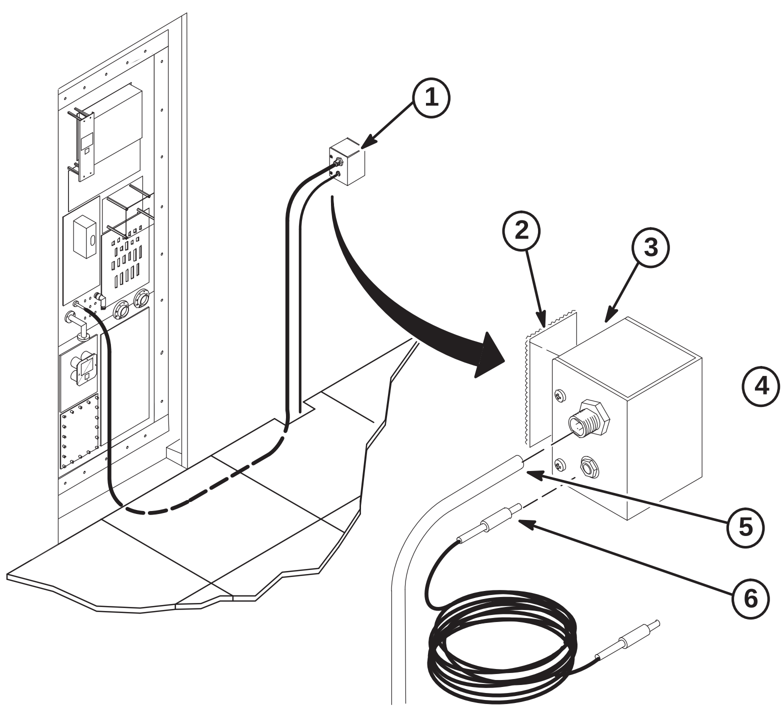
6–4–4–E INSTALLATION OF EXTENDER KIT (46–317758P2)

CAUTION

Do not mount Extender Box in Magnet Room because the box contains ferrous parts.

Install Extender Box

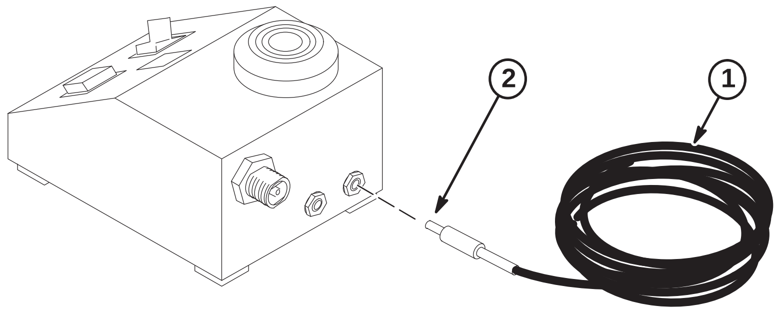
1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed–thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.



6–4–4–E Installation of Extender Kit (46–317758P2) (Continued)

Connection of Extender Wire to Control Box

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.



6–4–4–F CONTROL BOX INSTALLATION OPTIONS

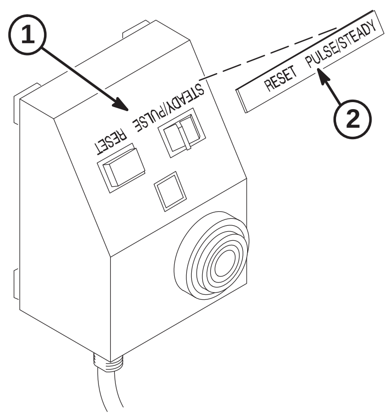
Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

- Mount control box on a wall or other vertical surface per Procedure **6–4–4–G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Mount control box under shelf per Procedure **6–4–4–G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed <u>within sight of operator</u>. Proceed to Procedure **6–4–4–K**, FINAL CONNECTIONS TO CONTROL BOX.

Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

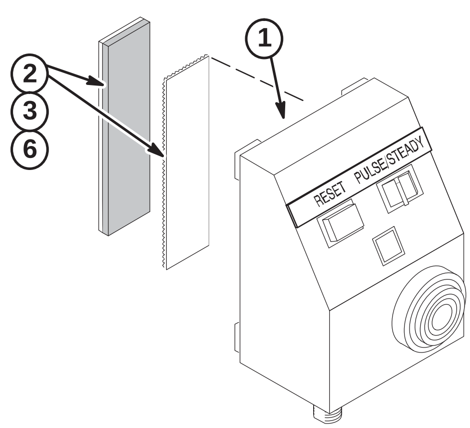
6–4–4–G CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate and apply over silk screened lettering as shown.
3. Select one of the following for installation of Control Box:
   - If control box will be mounted to WALL or SHELF with Velcro, go to Procedure **6–4–4–H**
   - If control box will be mounted to WALL with screws, go to Procedure **6–4–4–I**
   - If control box will be mounted to SHELF with screws, go to Procedure **6–4–4–J**



6–4–4–H Control Box Mounted With Velcro

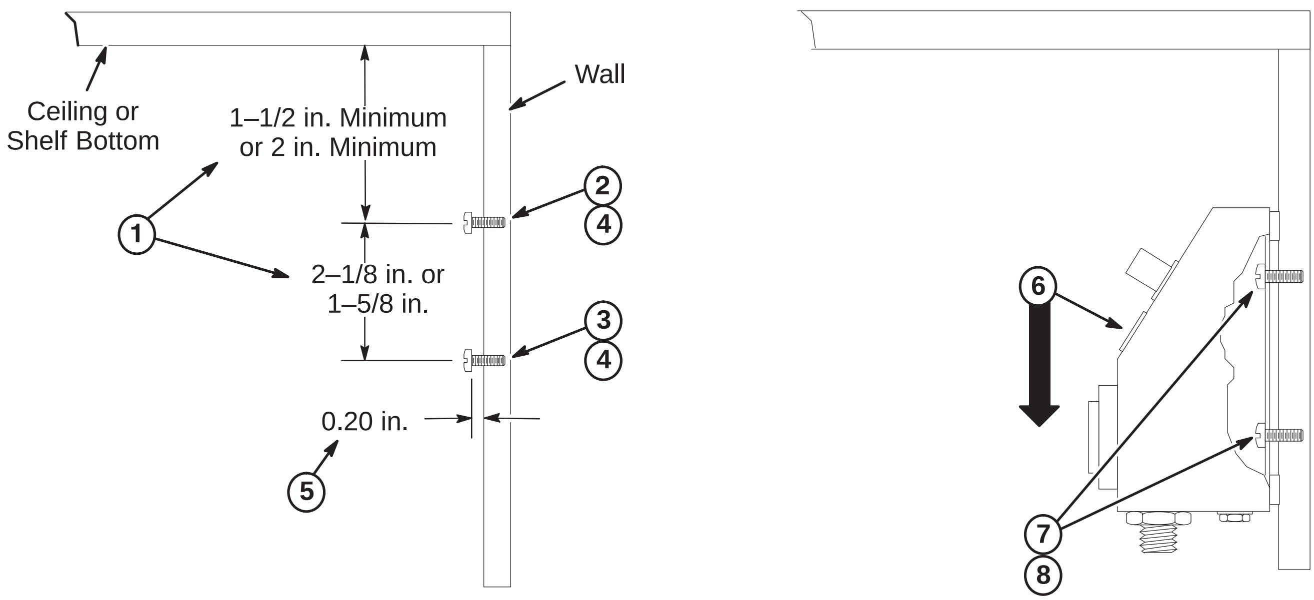
1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. jMount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to Procedure **6–4–4–K**, FINAL CONNECTIONS TO CONTROL BOX.



6–4–4–I Installing Control Box with Mounting Screws on Vertical Surface

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

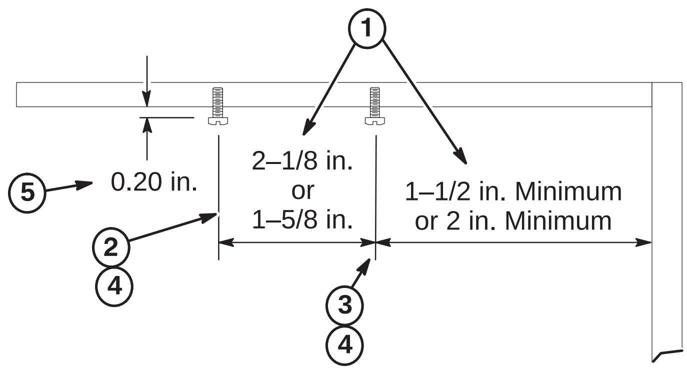
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **6–4–4–K**, FINAL CONNECTIONS TO CONTROL BOX.

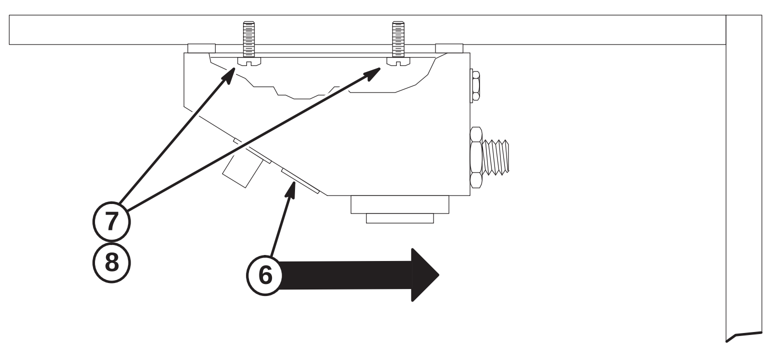


6–4–4–J Installing Control Box with Mounting Screws Under Shelf

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

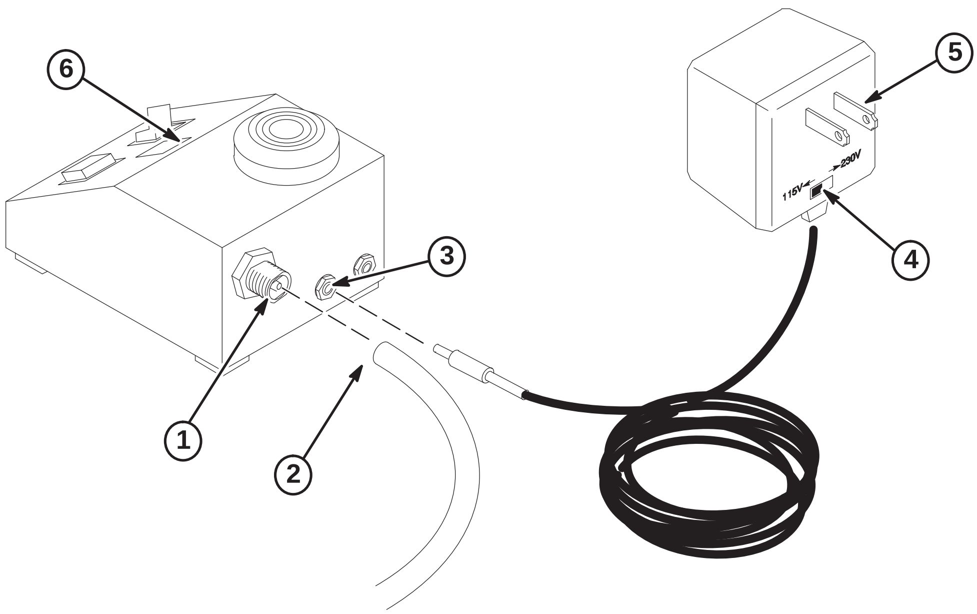
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **6–4–4–K**, FINAL CONNECTIONS TO CONTROL BOX.





6–4–4–K FINAL CONNECTIONS TO CONTROL BOX

1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed–thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is on.



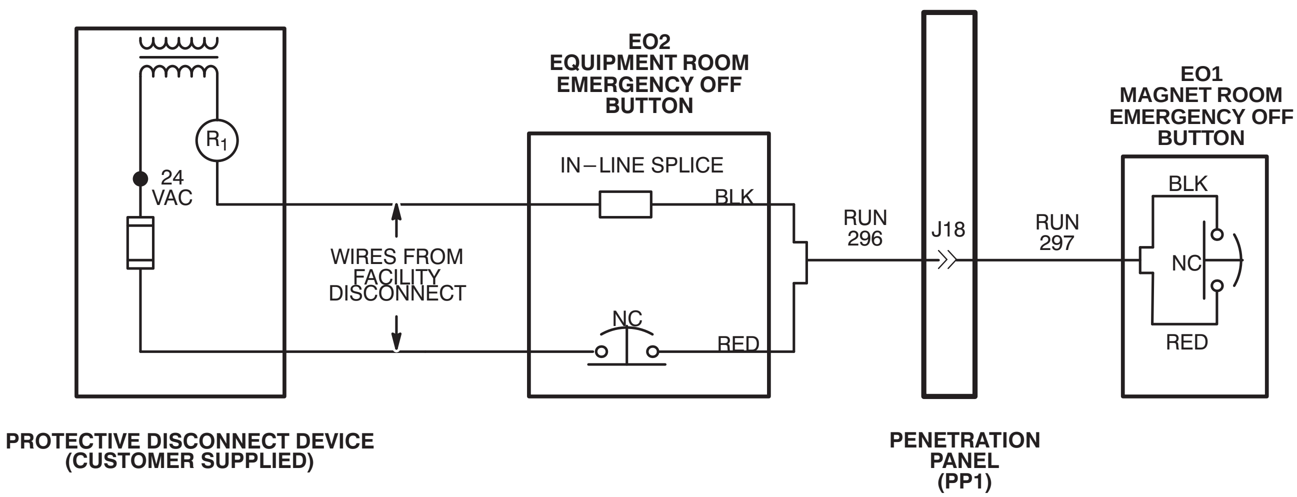
6–4–5 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. The Facility Disconnect Device and wires from the device are installed by qualified customer electrician. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Pre–Installation* manual, Section 5, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

Note: In Illustration below, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room “Emergency Off” (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2).
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2).
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push–on terminal.
5. Terminate black wire at end of Run 296 with local supplied “push–on” terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in–line splice.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1).



SECTION 7 – FINAL ENCLOSURE INSTALLATION

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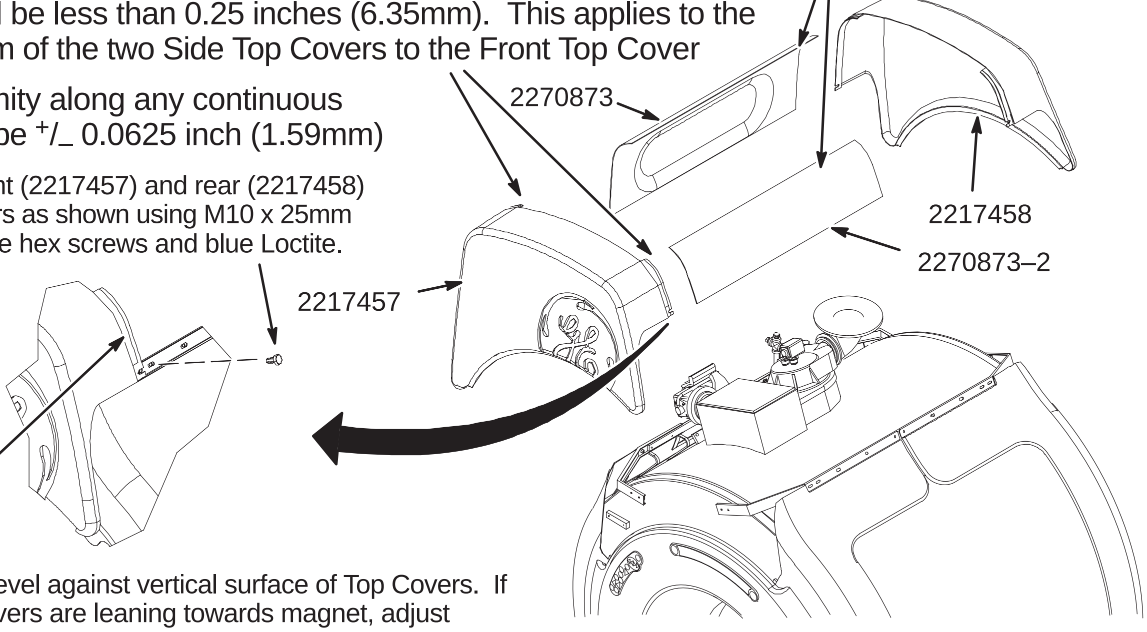
7–1 INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

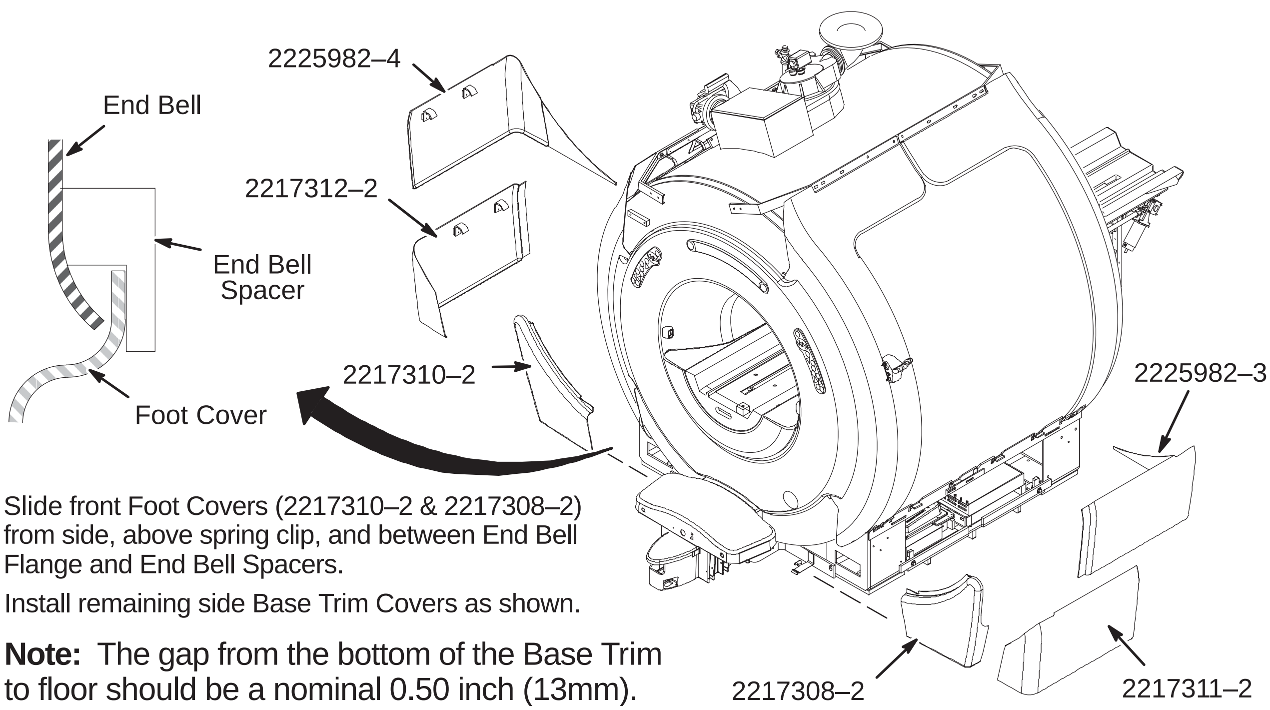
Gap uniformity along any continuous seam shall be $^{+}/_{-}$ 0.0625 inch (1.59mm)



1. Install front (2217457) and rear (2217458) Top Covers as shown using M10 x 25mm Low Profile hex screws and blue Loctite.
2. Place a level against vertical surface of Top Covers. If top of covers are leaning towards magnet, adjust Cosmetic Cover Ring and/or left and right Arc Covers.

2. Install side Top Covers as shown.

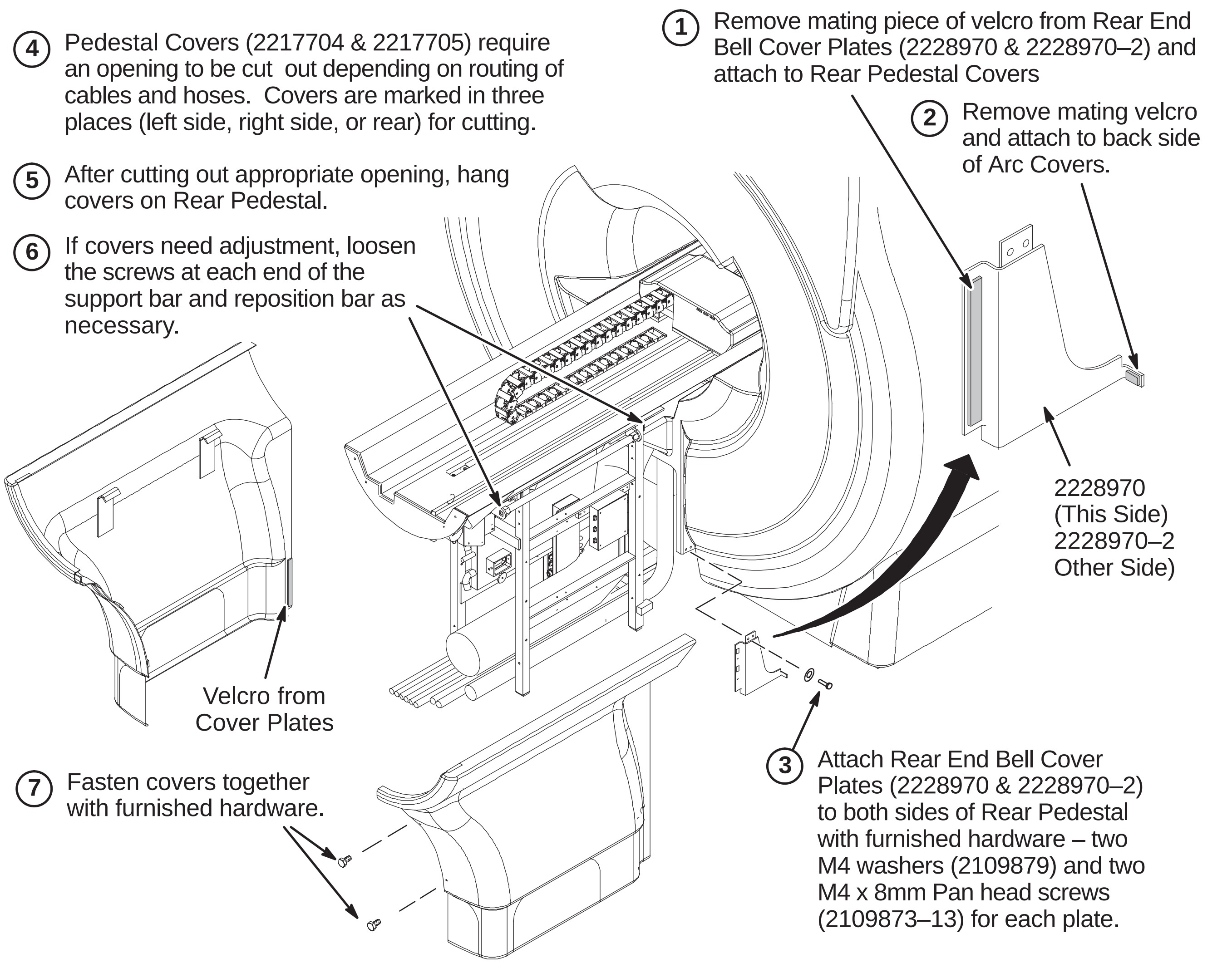
7–2 INSTALL BASE TRIM COVERS



1. Slide front Foot Covers (2217310–2 & 2217308–2) from side, above spring clip, and between End Bell Flange and End Bell Spacers.
2. Install remaining side Base Trim Covers as shown.

Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

7–3 INSTALL REAR PEDESTAL COVERS

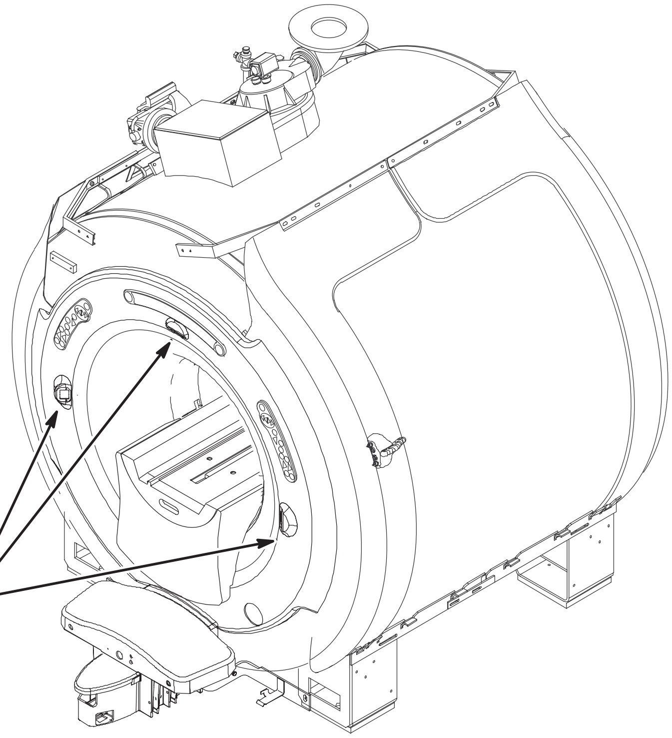


7–4 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

English



1. For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.

French

German

Portuguese

Spanish

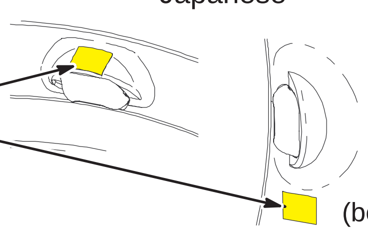
Italian

Swedish

Japanese

Chinese

2. If enclosure does not have labels attached, attach new labels as indicated.



(both sides)

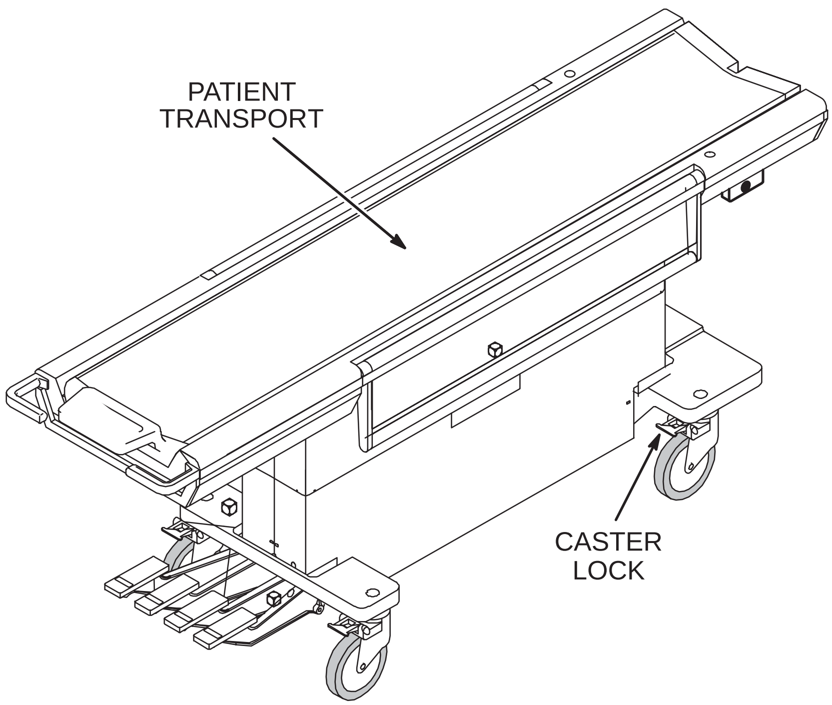
7–5 PATIENT TRANSPORT INSTALLATION

Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

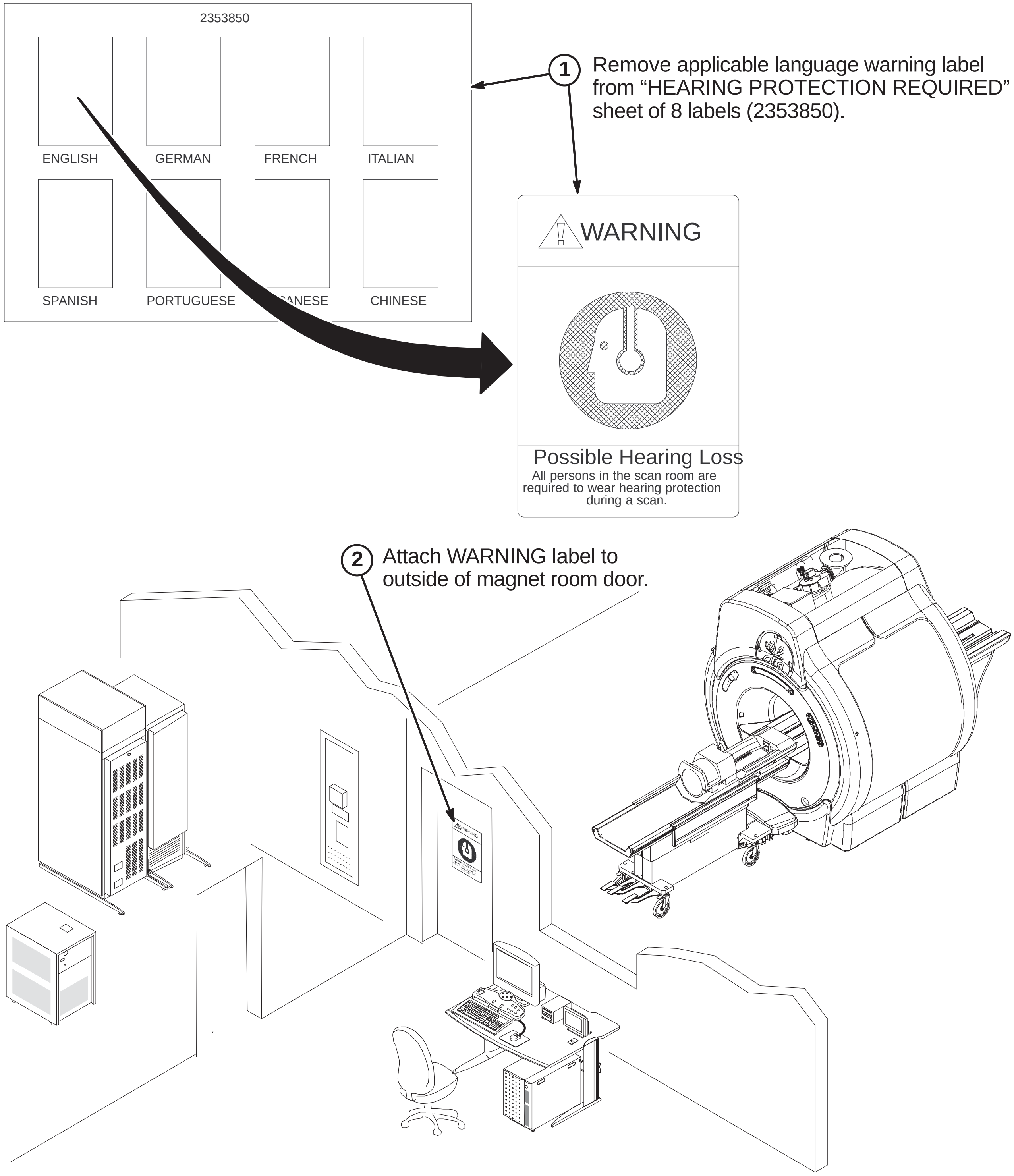
1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanial procedures in MR CD–ROM, Set up and Calibration: Patient Transport, LIGHTWEIGHT PATIENT TRANSPORT.

Note: If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.

4. If additional Patient Transports are supplied, refer to *Direction 15477, Second Patient Transport Option Installation*, (delivered with M1020BD option).



7–6 INSTALLATION OF HEARING LOSS WARNING SIGN



SECTION 8 – OPERATOR WORKSPACE

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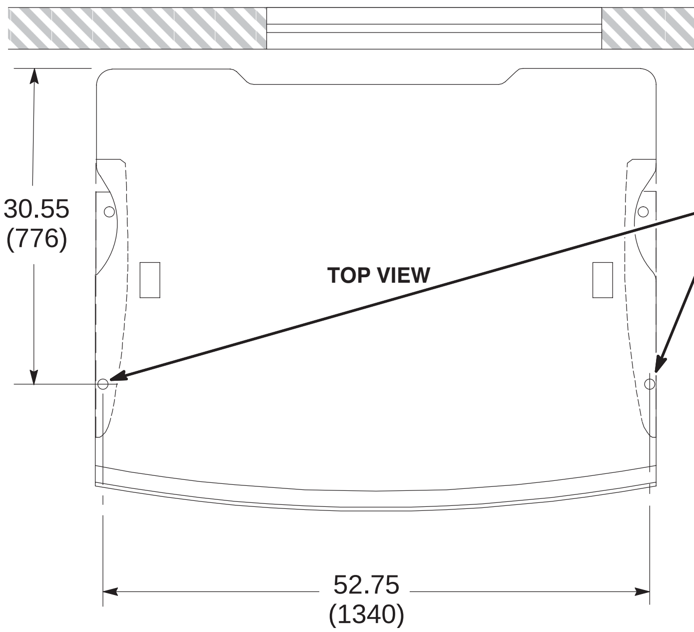
8–1 OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION

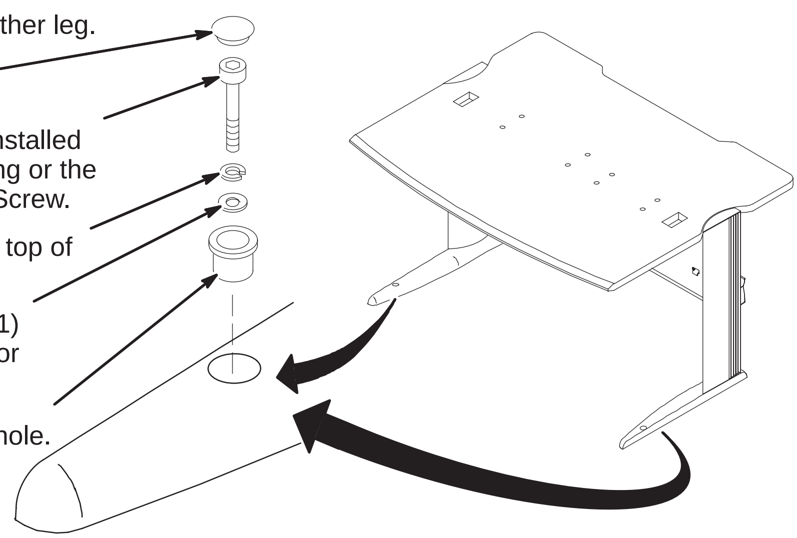


1. Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.
2. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
3. Move table to prepare for installing the floor anchors in Steps 4 and 5.
4. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4–20 x .375 diameter anchor (2168253).
5. Secure anchor in place with supplied Setting Tool (2168781).

8–2 SECURE OPERATOR WORKSPACE TABLE TO FLOOR

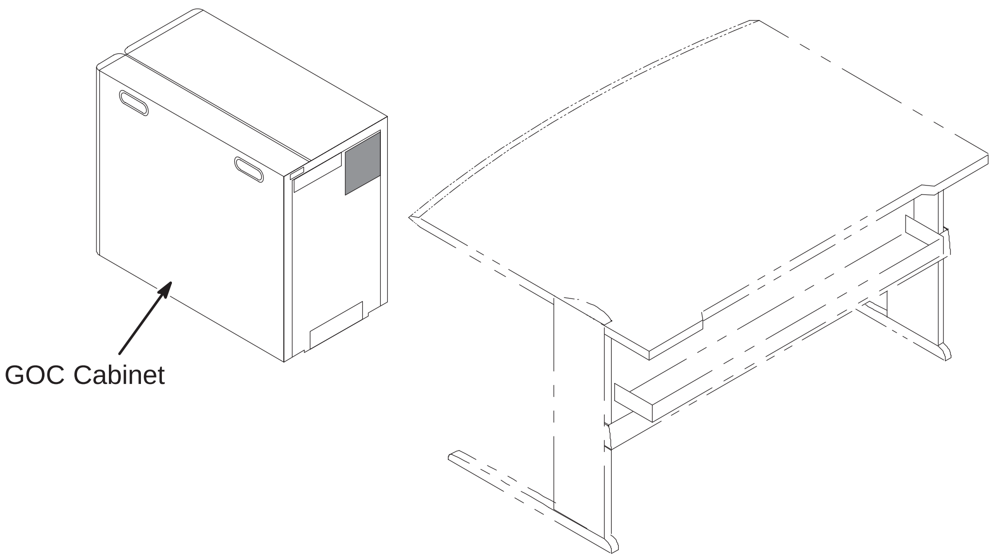
Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

6. Repeat anchoring process below for other leg.
5. Install black nylon Plug (2168254).
4. Depending upon depth of previously installed anchor, install either the 1–3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
1. Insert Bushing (2151454) in table leg hole.



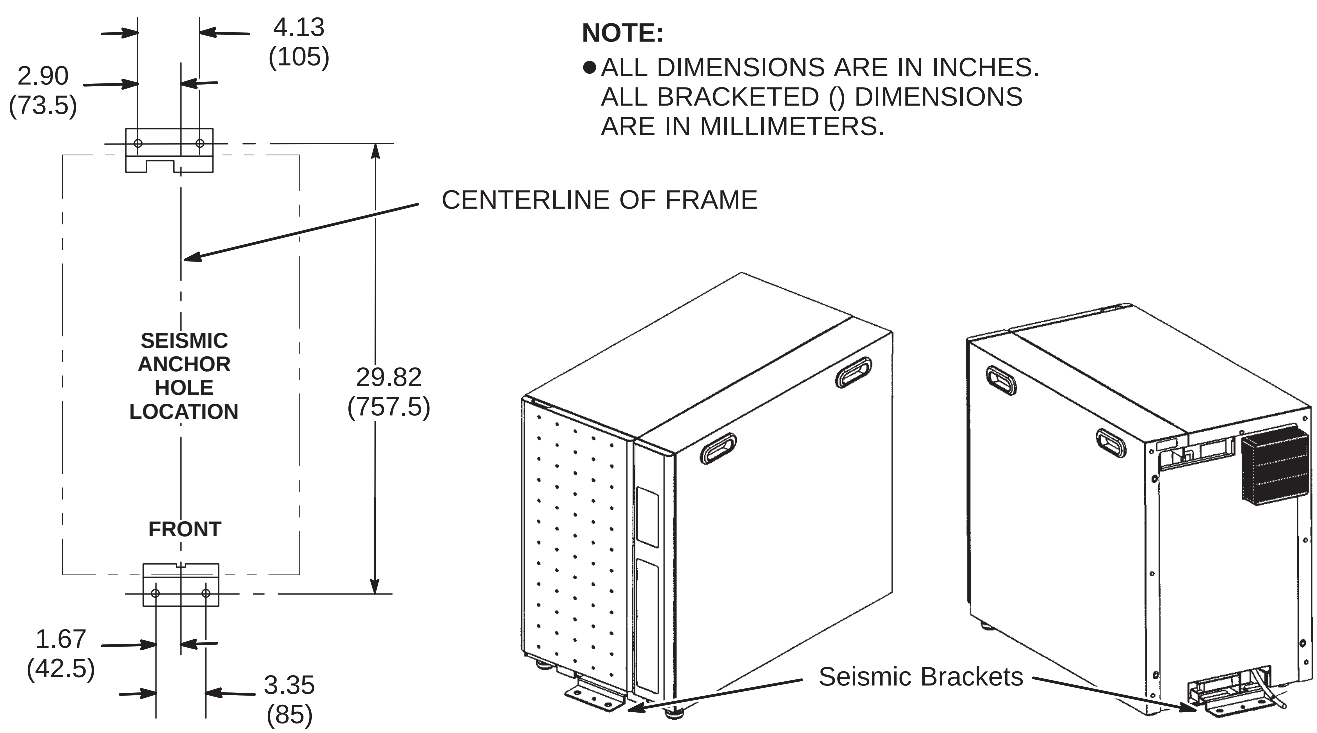
8–3 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



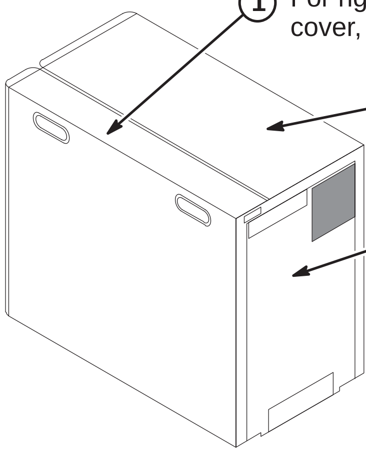
8–4 SEISMIC ANCHORING (IF REQUIRED)

If seismic anchoring is required, refer to site architectural drawings for fastener type details. Illustration below indicates hole location for installing brackets.

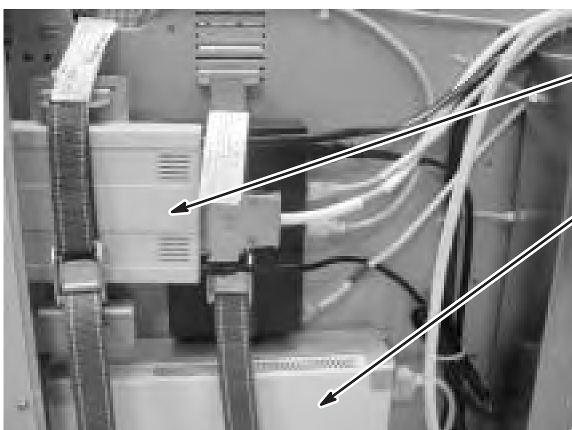


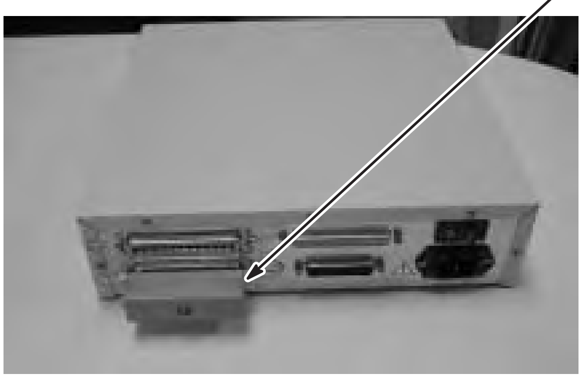
8–5 GOC CABINET COVER REMOVAL

1. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
2. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
3. Loosen four screws holding the back cover. Lift and remove cover.

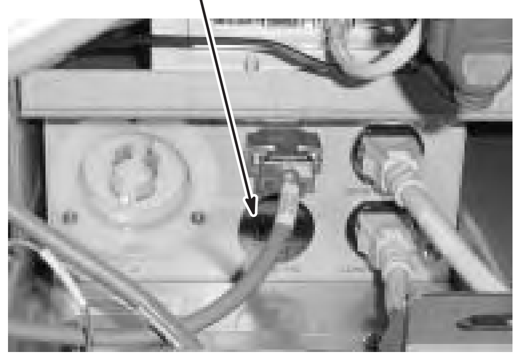


8–6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)





1. Mount Modem to GOC so that the underside is facing out to cover. Route customer provided phone lineand connect to Modem.
2. Mount DASM at this location. Route and connect cable from filming device.
3. Cable routed between DASM or Modem to the Linux PC are provided. Connect Run 1066 to Modem and Run 1067 to DASM.
4. If applicable, install the SCSI Terminator onto the bottom port of the DASM.
5. Verify that the **Active SCSI Terminator** located on the bottom of the DASM is in the **OFF** position.
6. Modem power cord must be connected to J5 on GOC PDU Module

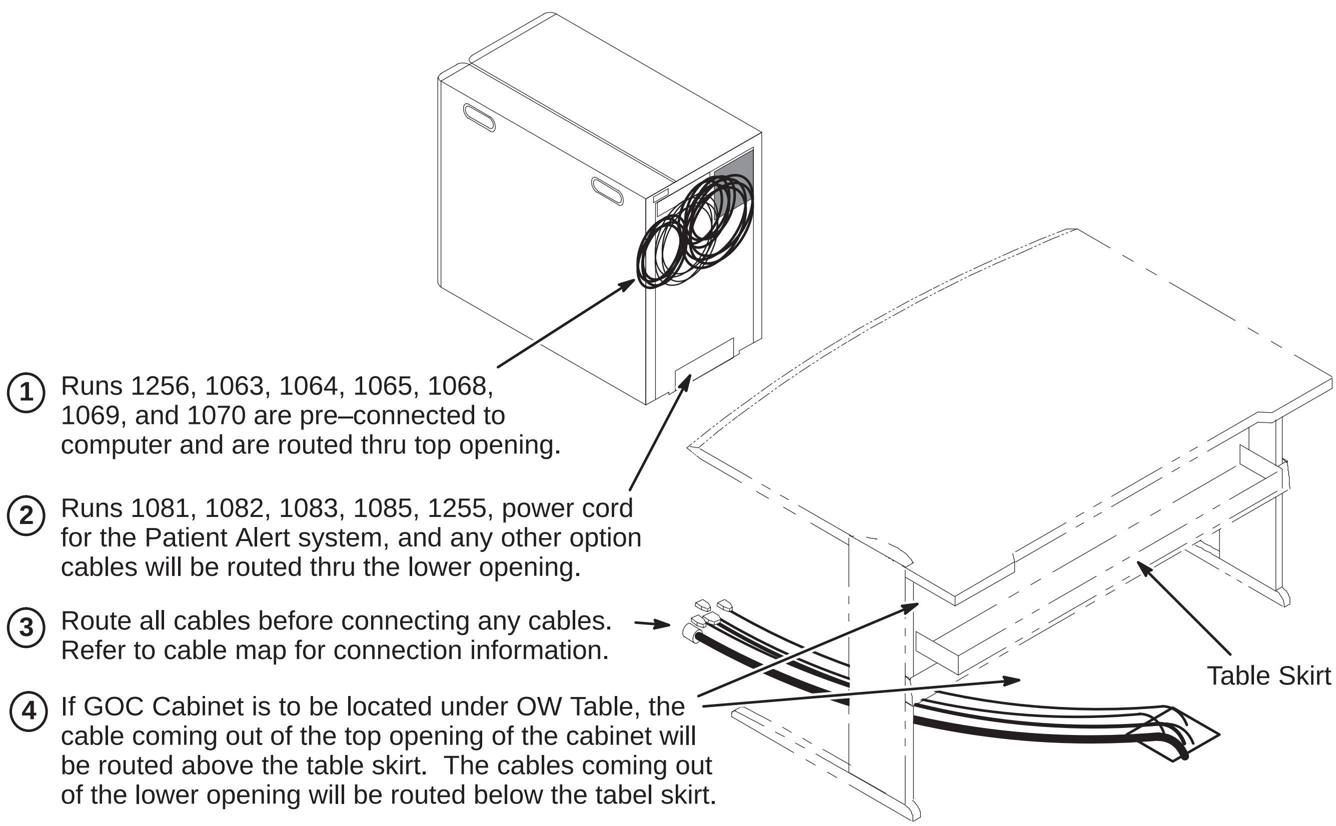


8–7 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Three power cords
- Two video ouput cables
- Keyboard output cable
- SCSI output cable

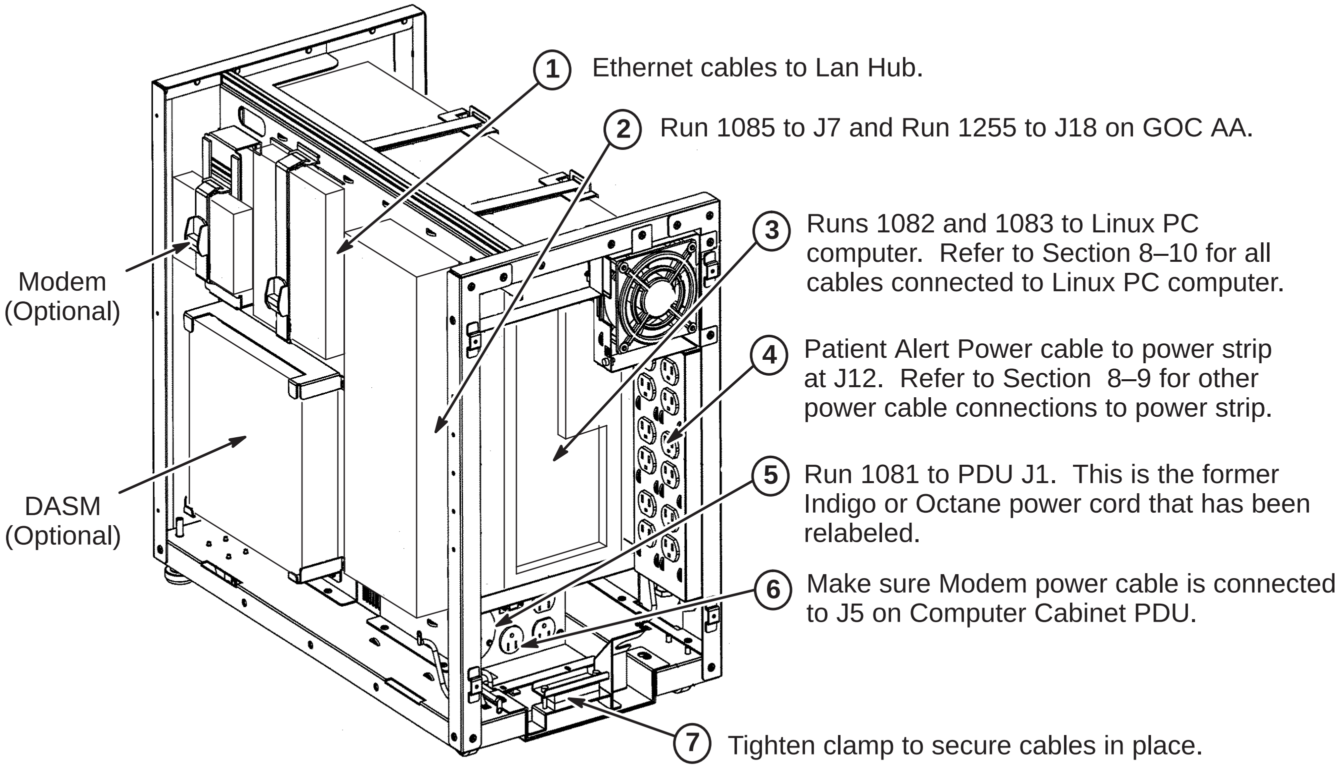
These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.



8–8 ROUTE CABLES FOR LOWER OPENING

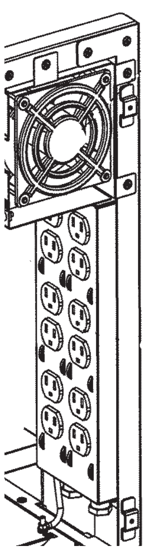
Refer to cable map for information on cables to be connected thru lower opening.

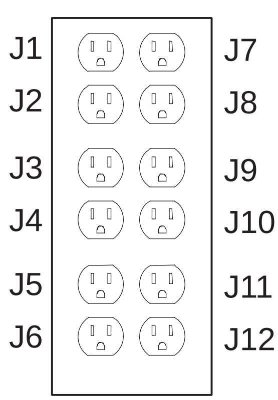
NOTE: The USB to Serial Converter that is included in the GOC Cabinet is **NOT USED** in this system.



8–9 POWER STRIP CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



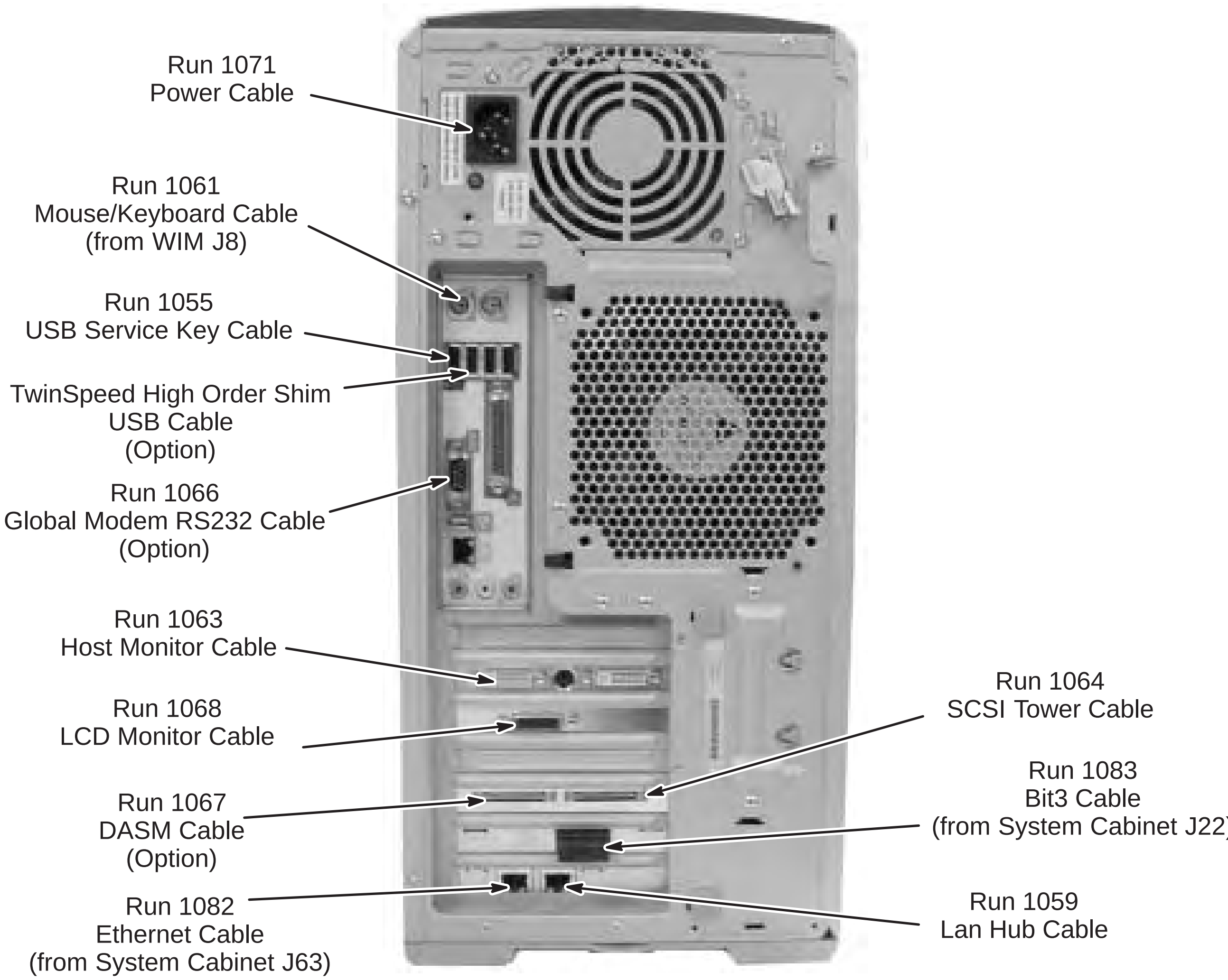


POSITION	NAME	MAX AMP CIRCUIT 1	MAX AMP CIRCUIT 2
J1	Host Comp	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	LAN Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

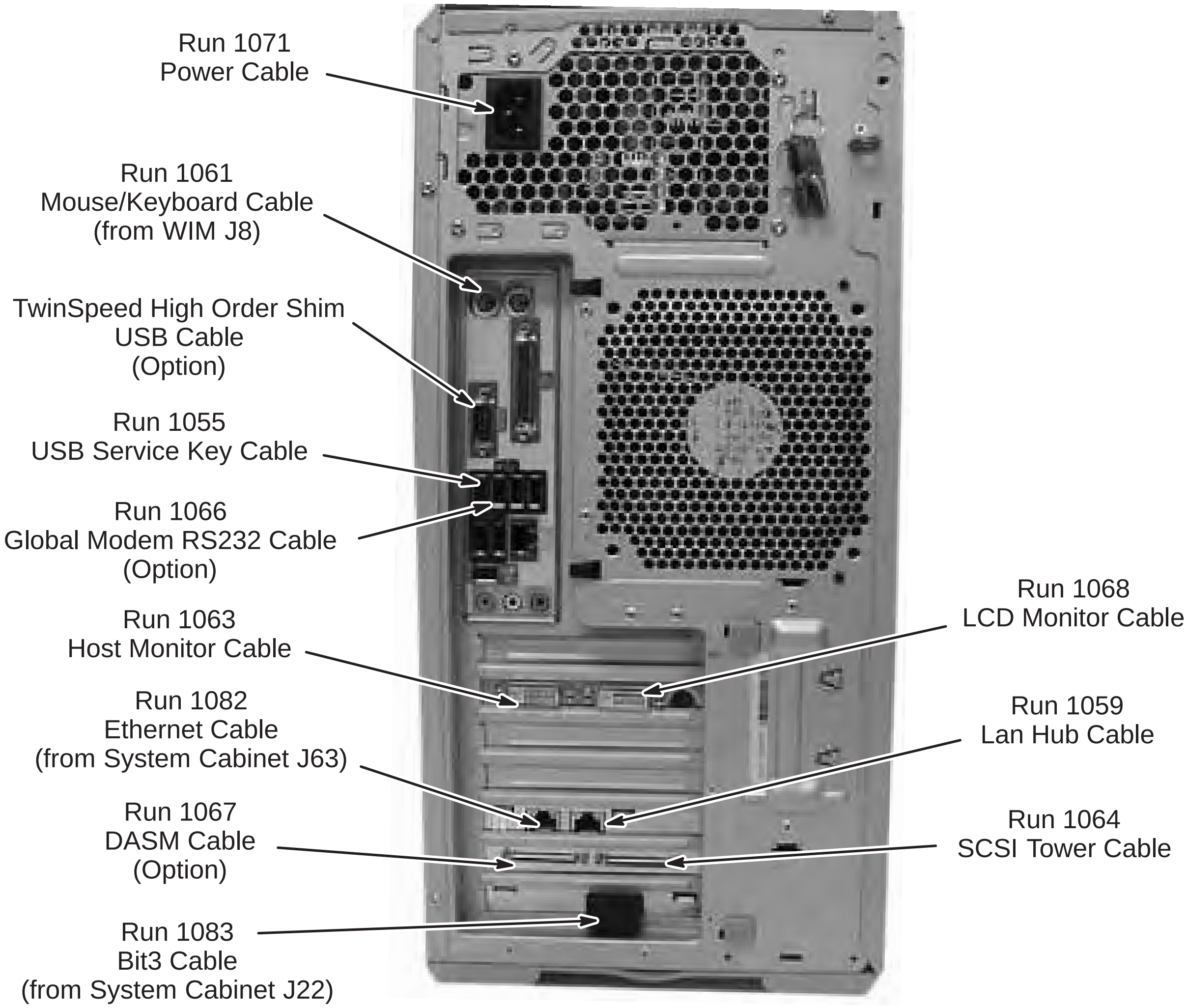
8–10 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the HP 8000 and the HP 8200 PC's are described in the following two sections. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site. Refer to the applicable PC for your site.

8–10–1 HP 8000 Connection Points



8–10–2 HP 8200 Connection Point



8–11 INSTALL SCIM/KEYBOARD

The Scan–Control Intercom Module (SCIM) is the current keyboard on all Signa LX and OpenSpeed systems. Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

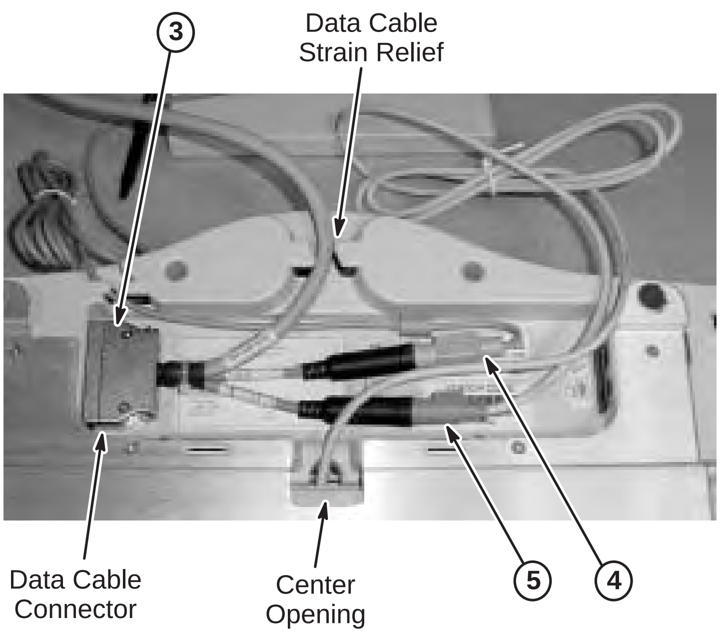
1. Remove the SCIM, Data Cable, and Keyboard From the box.

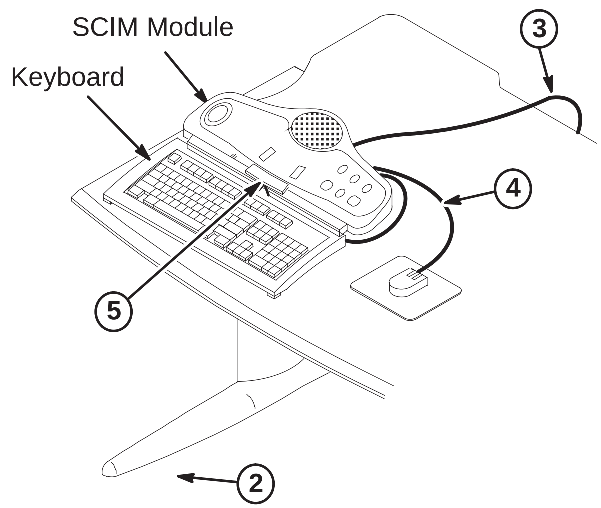
SCIM Module

Data Cable

Keyboard

2. Turn the SCIM unit up side down exposing the cable connections.
3. Route data cable Run 1262, preconnected to the GOC Cabinet and is located on back side of cabinet, from Computer Cabinet to bottom side of SCIM Module. Route thru data cable strain relief and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE)
4. Attach Run 1262 cable marked with mouse icon to the Mouse. Route the cable thru the strain relief notches at the rear of the SCIM.
5. Guide the keyboard cable thru the center opening and attach to the jack on Run 1262 with the keyboard icon. (Store excess cable in the base of the SCM by tucking it in the depressed area).

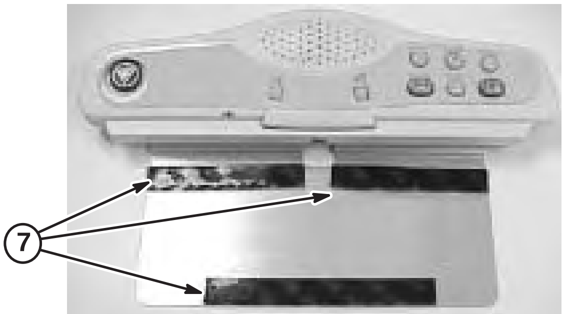


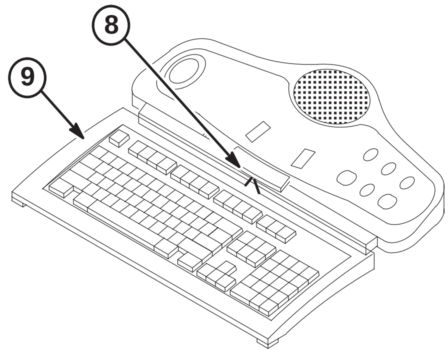


6. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
7. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).

8–11 INSTALL SCIM/KEYBOARD (Continued)

8. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
9. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

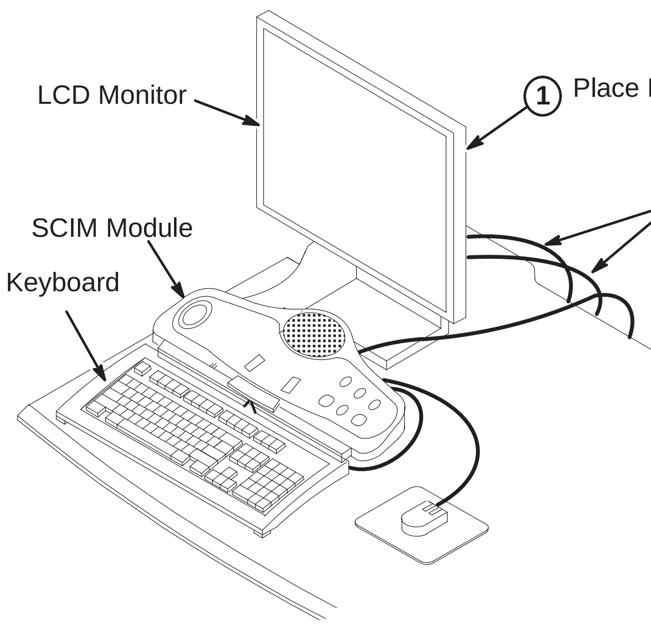




Note

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

8–12 INSTALL MONITOR

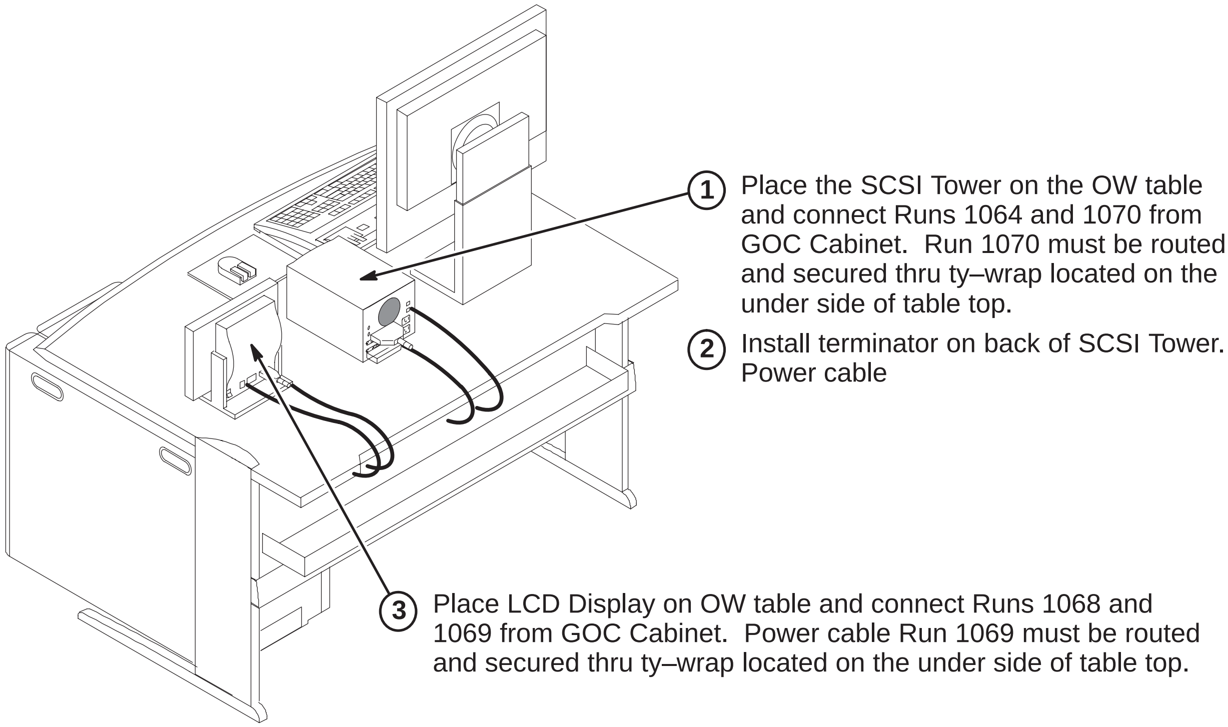


1 Place LCD Host Monitor on table in back of SCIM.

2 Route Runs 1063 and 1065 from Computer Cabinet and connect to monitor.

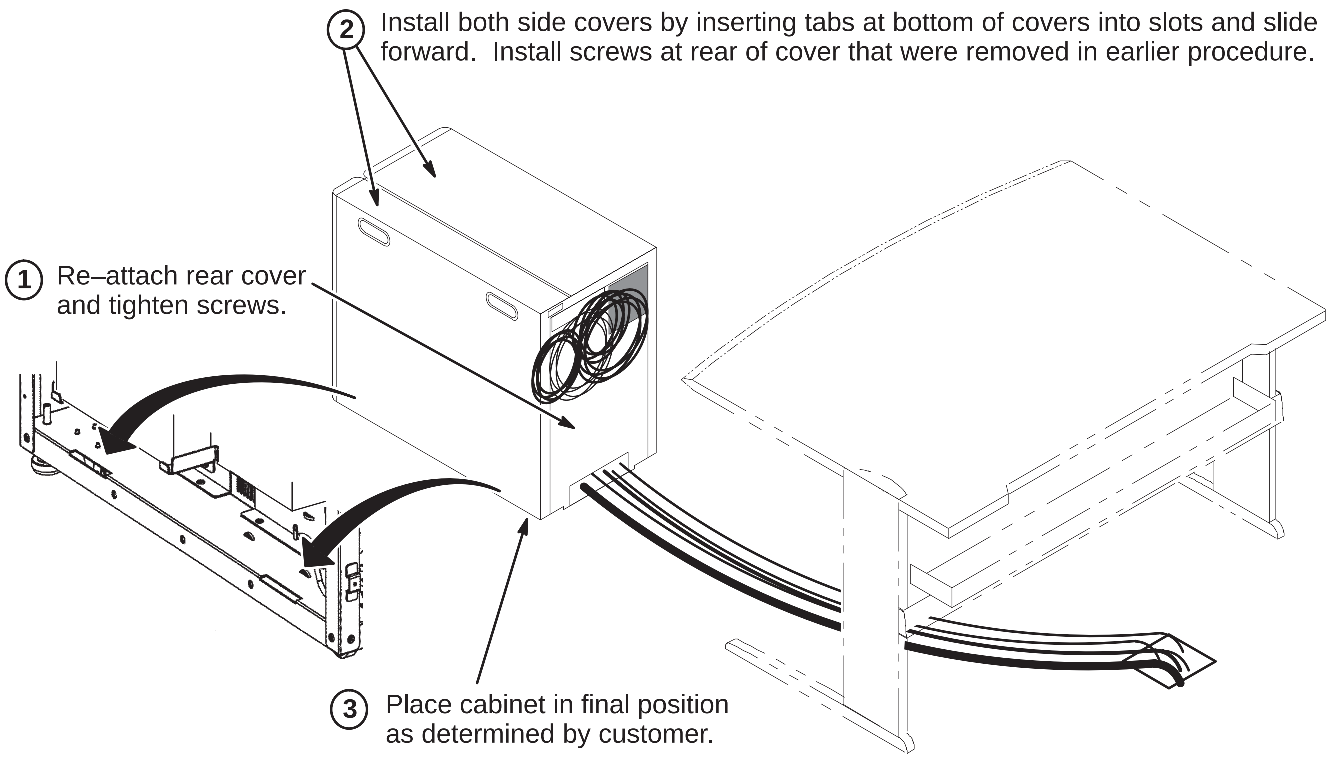
3 Route power cable Run1065 from Computer Cabinet and connect to monitor. Power cable must be routed and secured thru ty–wrap located on the under side of table top.

8–13 INSTALL SCSI TOWER AND LCD DISPLAY



8–14 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.



8–15 POWER ON INFORMATION

IMPORTANT
When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note
When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the WIM and Modem power.

SECTION 9 – COMPLETION CHECKLIST

TABLE OF CONTENTS

9–1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non–GE personel. Check off boxes at the front of each item below when complete.

- ❑ Check mechanical flowchart to make sure all steps are complete
- ❑ Resolve shipment shortages
- ❑ Resolve ommissions by mechanical contractors
- ❑ Have the on–site GE Healthcare Field Engineer (FE), Service Engineer (SE), or Project Mangager of Installation (PMI) to confirm that the blower box has been anchored properly and installed outside the minimum service area.
- ❑ Make sure all rating plates are installed
- ❑ Recheck all wiring connections using cable map
- ❑ Correct for wiring errors if necessary
- ❑ Complete Section 9–1–1.
- ❑ Complete Section 9–1–2.

9–1–1 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies and carts. These can be included with the “Return pile”, Recycling Operation will route them to appropriate location.

9–1–2 FINAL MECHANICAL INSTALLATION STEPS

1. Complete assembly of PPG cable and probe head.
2. Make sure all Cabinet, Magnet Enclosure, and Rear Pedestal covers have been installed.
3. Make sure Patient Transport is in position at front of Magnet.

9–2 GE FIELD ENGINEER RESPONSIBILITIES

9–2–1 PHANTOMS AND SERVICE KITS

1. Locate Phantom Kit and SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. Locate the multilingual (provided in eight languages) phantom attention labels shipped with each of the phantoms.

CAUTION

Three types of multilingual phantom attention labels are provided and each type must be applied to the correct phantom. The directions at the top of each label sheet describe which phantoms that can receive that label.

9–2–1 PHANTOMS AND SERVICE KITS (Continued)

4. Read the directions provided at the top of each of the multilingual attention label sheets and apply the appropriate language attention label for the site to each of the phantoms that should receive that type of label.
    a. Each phantom will receive the correct type of attention label written in the appropriate language for that site.
    b. Note that the LVshim phantom assembly will receive 3 labels, one for each phantom component.
5. Place service phantoms into SPT Phantom Cart or service storage cabinet to be organized and ready to be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.
6. Locate TPS RF I/F Kit (if applicable) and Signa Spares Kit. Place kits into service storage cabinet for ready use.

9–2–2 FINAL GE FIELD ENGINEER INSTALLATION STEPS

1. Record and enter applicable data into applicable site configuration files and records.
2. Complete Product Locator information for all installed serialized components, new or updated, via either:
    - (**U.S. Only)** *FE Site Verification Web Site*
      or
    - Process and return product locater installation cards for all serialized components to:
      Product Locator File, P.O. Box 414, W–523, Milwaukee, WI 53201–0414

Refer to Section 1–3 for details on submitting Product Locator information.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

3. Process the **Supplier Performance Report** located in Common Forms and Section 9–2–3 (reference copy).
4. Store the delivered site's set of service tools and spares kit in service cabinet at site.
5. Locate surface coil ("quick disconnect") adapters and any optional surface coils and positioning accessories and place on Patient Transport. Be sure only the correct polarity (normal or reverse **but not both**) head coil adapter is left at site.
6. Locate Box "To be opened by Applications Specialist" and set aside for later visit by Applications Specialist.
7. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals.
8. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
9. Verify that coordination of application support for instruction of site users on the appropriate Software Release has been made before returning system to the users. Turn site over to applications who will instruct users.

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9–2–3 SUPPLIER PERFORMANCE REPORT

GE Medical Systems Field Service

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: ____ Date: ____________________

Work Performed / Evaluated: ____

Commodity / Item Purchased: ____

GE Customer Name: ____

FDO Number: ____ Test Equipment Bar–code No; _

GEMS Evaluator Name: ____

Please rate suppliers as 1 to 5 on each of the 15 questions below:
1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

	Supplier schedules well and:					
1	starts on time?	1	2	3	4	5
2	finishes on time?	1	2	3	4	5
3	avoids disruptions?	1	2	3	4	5
	Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4	5
5	cooperates with changes in schedule?	1	2	3	4	5
6	deals with unexpected problems?	1	2	3	4	5
	Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4	5
8	adequate for contracted work?	1	2	3	4	5
9	calibrated and in good working order?	1	2	3	4	5
	Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4	5
11	always keep job sites clean and free of debris?	1	2	3	4	5
12	ensure both function and appearance are as expected?	1	2	3	4	5
	Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4	5
14	behavior?	1	2	3	4	5
15	attitude?	1	2	3	4	5

Examples / Details: ____

____

____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

1. Senior Operations Specialist for your LCT
2. The contractor who performed the work
3. ISS (already in default distribution)
4. Sourcing Supplier Quality (already in default distribution)

SECTION 10 – APPENDIX

TABLE OF CONTENTS

10–1 SYSTEM CONFIGURATION

The tables in this section list the catalogs for the hardware equipment which comprise the Signa EXCITE HD 1.5T (Release 12.x) system. Illustrations 10–1 and 10–2 show the major equipment of the system.

10–1–1 BASIC SYSTEM

The basic system is configured from several catalogs. The following tables indicate what catalog(s) has been installed for this site. For those tables which offer more than one catalog, select the catalog which has been delivered to the site.

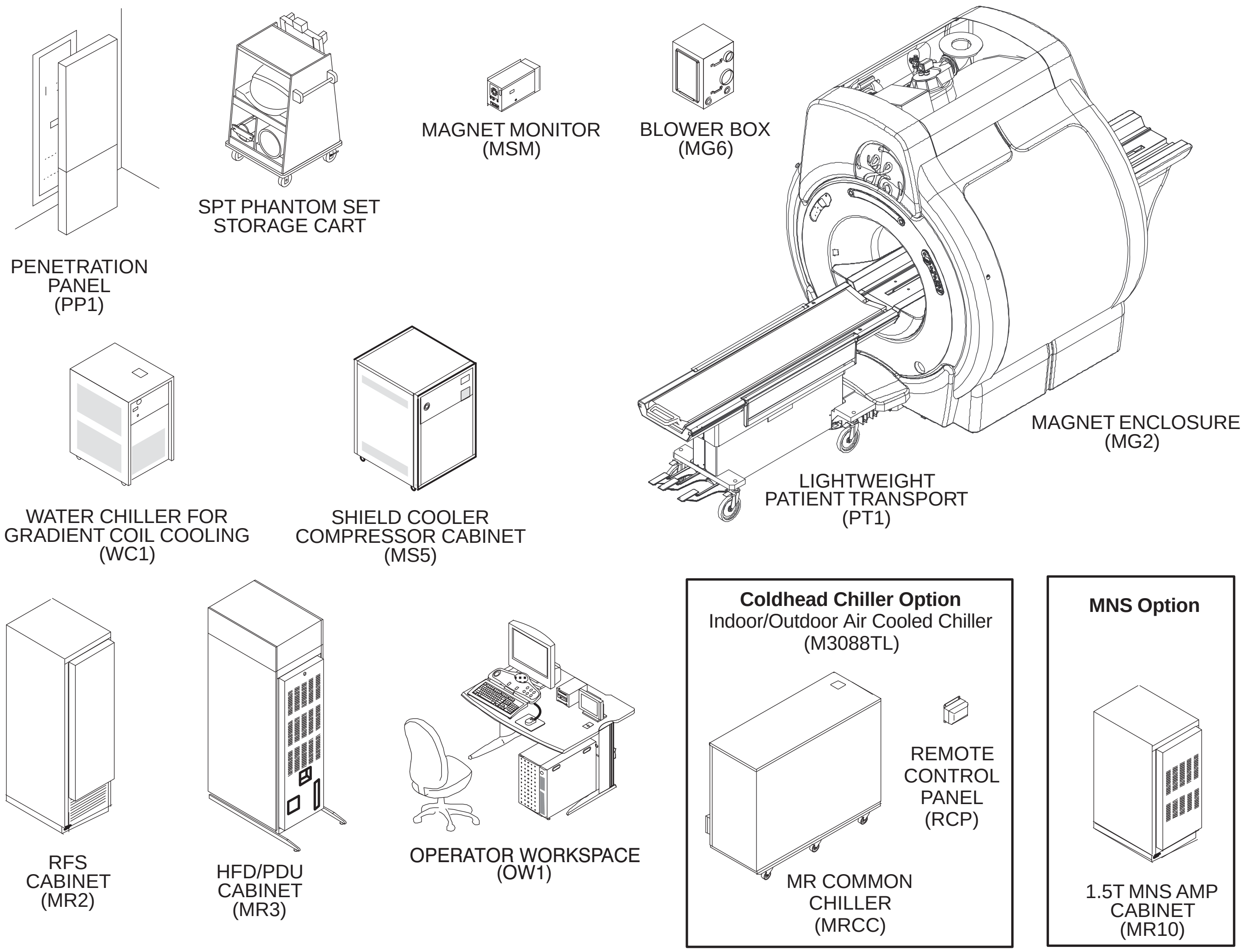


ILLUSTRATION 10–1

SIGNA EXCITE HD 1.5T SYSTEM

10–1–1 BASIC SYSTEM (Continued)

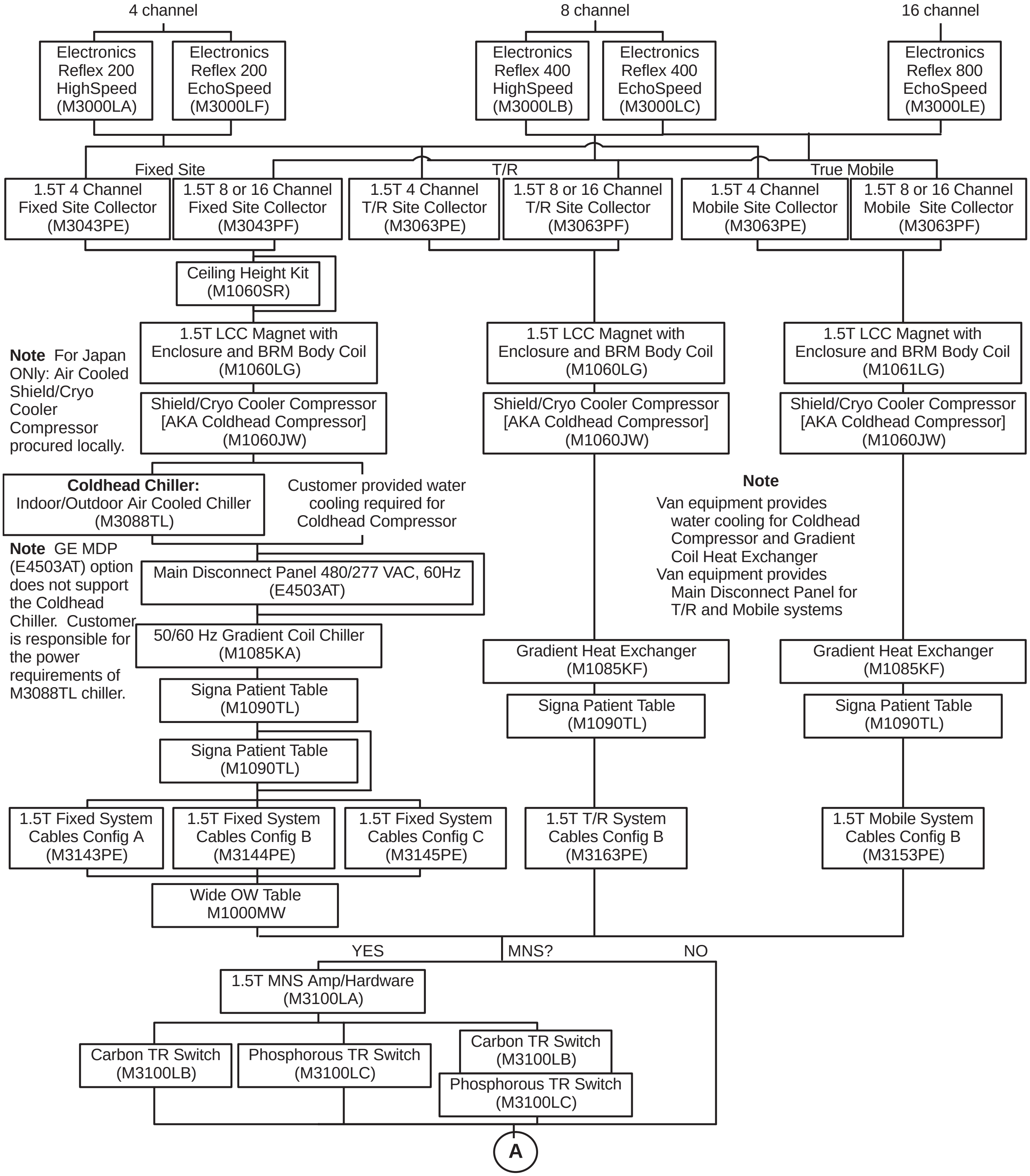


ILLUSTRATION 10–2

SIGNA EXCITE HD 1.5T SYSTEM CATALOGS

10–1–1 BASIC SYSTEM (Continued)

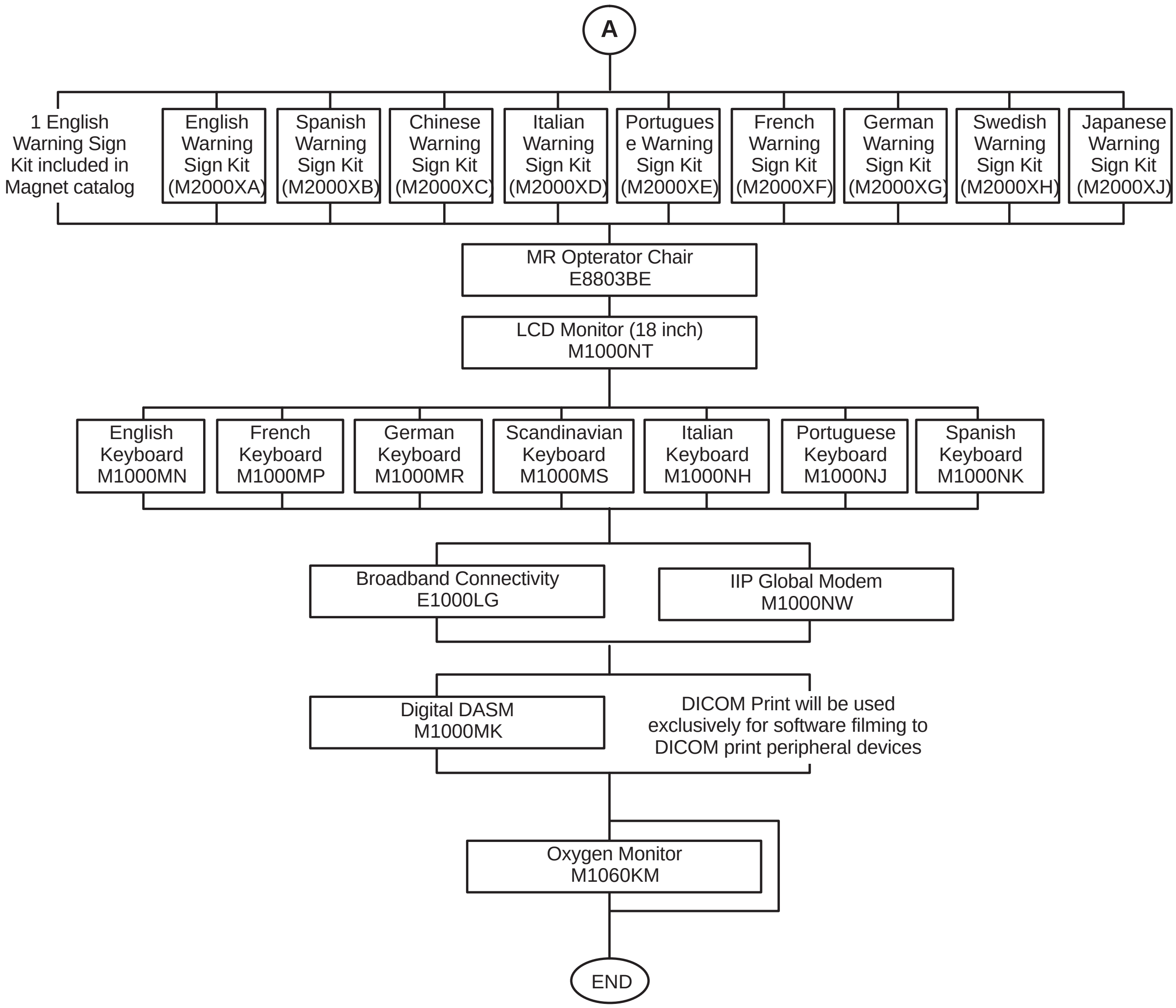


ILLUSTRATION 10–2 (Continued)

SIGNA EXCITE HD 1.5T SYSTEM CATALOGS

10–2 INSTALLATION FLOWCHART PROCEDURE DETAILS

The following table breaks down the installation procedures with detailed steps and the approximate lengths of time for completing the step. The majority of steps are for components installed in either the magnet room or equipment room.

"**Step Number**" definition for table below is as follows:

Mechanical Installer

MI 1 – E 1

Set of tasks to complete.
List of tasks that are associated with Installation Flow Chart Procedures located in Section 1–13 .

Session# 1 to 5.
Each session = 5 hours applied time for each MI or 10 hours total for a two–person team.

Room where procedures are performed.
E = Equipment Room
M = Magnet Room
Blank = No specific room

10–2–1 Day 1 (MI 1): Session 1

Step Number	Operation Name	Time (minutes)
MI 1 – 1	**Contractor Coordination**	
1	Find Print & Verify Layout	15
2	Verify All Conduits & E–Off	45
3	Day 1 Checklist	10
4	Meet Contractor	35
5	Meet Electrician	20
	Total Time for 2 people	**105**
MI 1 – 2	**Shipping Material**	
6	Remove Duct Covers / Floor Tiles	10
7	Verify with Customer	30
8	Remove Shipping Material fro Magnet	15
	Total Time for 2 people	**80**
MI 1 – 3	**Delivery**	
9	Meet Driver	10
10	Verify Route	5
11	Mover Coordination	5
12	Inventory Equipment	25
13	Set equipment Elec Room / Verify 480 Connection	20
14	Cabinet Placement	45
15	Remove Rear Pedestal from Crate	25
16	Layout & Cut Out Sub–Floor	10
17	Place cooling fan on top of PDU cabinet	15
18	Remove Magnet Covers	10
	Total Time for 2 people	**165**

(Continued on next page)

10–2–1 Day 1 (MI 1): Session 1 (continued)

Step Number	Operation Name	Time (minutes)
MI 1 – 4	Reroute Gas Lines	
19	Orange Tubing / Gas lines	50
20	Disconnect Mag Monitor	10
21	Lower Pen Panel	10
22	Cold Head Power	15
23	Blower Box Power	10
24	Blower Cable Ground	10
	Total Time for 2 people	**105**
MI 1 – 5	Pen Panel Install	
25	Stage Pen Panel	10
26	Installl Lower / Upper Pen Panel	15
	Total Time for 2 people	**25**
MI 1 – E1	Equipment Room Magnet Cables	
1	Hang magnet monitor & cable connects	55
	(runs 823,824,825,826, 827, 829, 831, 833)	
2	Attach run 1183 (patient power)	10
	1 person	**65**
MI 1 – M1	Magnet Room Pen Panel Set–up	
1	Hardware Install	10
	1 person	**10**
	MI 1 Elapsed Time Totals	**555**
	Minutes each for 2 person team	**240**
	Minutes for electronics room person	**65**
	Minutes for magnet room person	**10**
	Per Person Totals	
	Total Minutes Electronics Room Person	**305**
	Total Minutes Magnet Room Person	**250**

10–2–2 Day 1 (MI 2): Session 2

Step Number	Operation Name	Time (minutes)
MI 2 – M1	**Magnet Set–up**	
1	Stage E–stop cable (Run 297)	10
2	Hang Emergency Run Down Box	10
3	Emergency Run Down Cable to Box	15
4	Emergency Run Down Cable to Magnet	15
5	Stage Bore light / gorund / patient alert	10
6	Clean–up Magnet wires/gas lines/ Install Cover	20
	1 person	**80**
MI 2 – M2	**Dock Assembly Installation**	
7	Stage Dock station (Runs 387/778/779)	10
8	Install Dock Bracket & Dock Assembly	40
9	Run Dock station cables (Runs 387/778/779)	15
	1 person	**65**
MI 2 – M3	**Bore Light / Main Ground / Patient Alert**	
10	Mount Bore lite & Run cable (Run 300–in box)	20
11	Route Magnet ground (040)	20
12	Route / Connect Patient Alert	20
	1 person	**60**
MI 2 – M4	**Magnet Cover Installs**	
13	Magnet Top Covers	15
14	Magnet Front and Back Covers	15
15	Magnet Left and Right Covers	10
19	Magnet Base and Trim Covers	5
	1 person	**45**

(Continued on next page)

10–2–2 Day 1 (MI 2): Session 2 (continued)

Step Number	Operation Name	Time (minutes)
MI 2 – E1	**Pen Panel Power Wiring**	
1	Meet Electrician / Contractor	25
2	Attach U–bolts	5
3	Detach gradient plexiglass cover	5
4	Remove gradient lug nuts	5
5	Prep for gradients. Get cables, stage, measure, remove floor	25
6	Measure, cut gradient to length	45
7	Attach run 762 to gradient filter (X)	10
8	Attach run 763 to gradient filter (Y)	10
9	Attach run 764 to gradient filter (Z)	10
10	Attach gradient filter cover	10
	1 Person	**150**
MI 2 – E2	**PDU Wiring**	
11	Stage run 1254 to ground. *Get, measure, cut, run.*	15
12	Stage run 968 (PDU to system cabinet). *Get run.*	10
13	Stage run 1234 (PDU to RF amp). *Get run.*	10
14	Stage run 970 (gradient to water chiller). *Get run, plug into chiller.*	10
15	Stage run 1182 (power cable) to PDU. *Get run, plug into pen panel.*	10
16	Stage run 037 (PDU to system cabinet ground). *Get run.*	5
	1 Person	**60**
	MI 2 Elapsed Time Totals	**460**
	Per Person Totals	
	Total Minutes Magnet Room Person	**250**
	Total Minutes Electronics Room Person	**210**

10–2–3 Day 2 (MI 3): Session 3

Step Number	Operation Name	Time (minutes)
MI 3 – M1	Magnet Gradient Installation	
1	Attach Phenolic Blocks to Gradient cables	65
2	Term. Gradient cables to Pen Panel	55
3	Term. Gradient cables to Pen Panel	50
4	Term. / Crimp Gradient cables to Magnet	90
5	Verify Pen Panel Connections	10
	1 person	**270**
MI 3 – M2	Cooler Set–up	
6	Run / Connect Glycol Lines	10
7	Apply teflon Tape to Hose Barbs	5
	1 person	**15**
MI 3 – E1	Console Pull	
1	Pick cables and organize	15
2	1081 Pwr Cable for Console J1 to HP Compressor. *Rollout Cable, Remove Plug & Mark, Reattach Plug Cover, Attach Guide with Tape*	35
3	1085 Signal Cable from J12 Pen Panel to J7 HP Comp. *Rollout Cable*	5
4	1255 Console J18 to System Cabinet J26. *Rollout Cable*	5
5	1082 Console to Sys. Cab. Network cable. *Rollout Cable*	5
6	1083 Console to System Cabinet J22. *Rollout Cable*	5
7	No Run # Patient Air Alert. *Rollout Cable*	5
8	Tape all console cables & pull through conduit using guide wire	30
	1 person	**105**
MI 3 – E2	Dress and Connect Cables	
9	Dress cables and connect (1085, air hose) to Pen Panel	20
10	Dress cables and connect (1255, 1082, 1083) to system cabinet	20
	1 person	**40**
MI 3 – E3	Signal Cables / Bias Video Cables	
11	1188–1190 (J78, J203, J77) Pen Panel to Sys. Cabinet Driven Mod. J5, J10 & J3. *Get Cables, Attach between Pen Panel and System Cabinet, Dress.*	30
12	1184, Sys. Cabinet to Pen Panel J32	5
13	1185 – 1187 (J200, J201, J202) Sys. Cab. To Pen Panel J200, J201, J202	15
14	Connect Runs 1191–1195 Pen Panel (J92–J96) – Driven Module J22–J26 Get cables, Attach, Dress	15
15	1181 Pen Panel (J154) to PDU J8. *Get, Attach PP/PDU, Dress*	10
16	971 J2 PDU to J23 System Cabinet	10
17	1180 Sys. Cabinet to PDU	5
	1 person	**90**
	MI 3 Elapsed Time Totals	**510**
	Per Person Totals	
	Total Minutes Magnet Room Person	**285**
	Total Minutes Electronics Room Person	**235**

10–2–4 Day 2 (MI 4): Session 4

Step Number	Operation Name	Time (minutes)
MI 4 – M1	**Rear Pedastal Set–up**	
1	Connect Mag Ground (Run 040)	15
2	Remove Pedestal Wheels and Connect to Bridge	20
3	Connect Rear Ped Drive Belt	30
4	Stage, Run, Connect Rear Ped Cables	75
5	Dress Pen Panel Cabling	20
6	Verify Rear Ped Cable Connections	25
	1 person	**185**
MI 4 – M2	**Heliax and Video Connections**	
7	Heliax Cables Run and Terminate (Body/Head)	40
8	Run Video Cables (1200/1201/1202)	40
	1 person	**80**
MI 4 – M3	**Patient Blower Install**	
9	Run Cabling	5
10	Run Ductwork (4 inch and 8 inch)	10
11	Set Blower in position	5
	1 person	**20**
MI 4 – M4	**Cover Installs**	
12	Pen Panel Cover	10
13	Stage Rear Ped Covers	10
14	Level Rear Ped Feet	10
15	Cut / Install Rear Pedestal Covers	10
	1 person	**40**

(Continued on next page)

10–2–4 Day 2 (MI 4): Session 4 (continued)

Step Number	Operation Name	Time (minutes)
MI 4 – E1	**Fiber Optic / Door Switch**	
1	Fiber optic for injectors	20
2	Run glycol lines (2 people). *Get lines, measure, cut, terminate ends*	35
3	296 J18 Pen Panel to Main Disconnect E–Stop. *Remove floor tiles, stub out to disconnect for electrician to connect*	15
4	701 Door Switch to System Cabinet J15. *Run Cable from system cabinet to patient door*	35
5	J13 system cabnet to Pen Panel	15
6	J17 system cabnet to J2 PDU	20
	1 person	**140**
MI 4 – E2	**Heliax / Main PWR Termination / Pen Panel**	
7	Main PWR to sys cabnet from PDU, run 1234. *Terminate at system cabnet*	10
8	Body Heliax J3 Sys Cabnet to Pen Panel J4, run 1249 (whip cable). *Get, Run, Cut, File, Connect (whip cable)*	20
9	Body Heliax J44 Pen Panel to Sys Cabnet J4 (whip cable). *Get, Run, Cut, File, Connect to J44 Head*	20
10	Head J44 Pen Panel. *Connect head*	5
11	Feed Through of fiber optics from magnet side	10
12	Runs 1187 – 1189 to Pen Panel (”snake” cables)	5
13	Dress Pen Panel	10
14	Connect Heliax Head to Pen Panel (2 people)	5
	1 person	**85**
	MI 4 Elapsed Time Totals	**590**
	Per Person Totals	
	Total Minutes Magnet Room Person	**325**
	Total Minutes Electronics Room Person	**265**

10–2–5 Day 3 (MI 5): Session 5

Step Number	Operation Name	Time (minutes)
MI 5 – M1	Console Area Install	
1	Set Operator Workstation Components	15
2	Stage Operator Workstation Peripherals	5
3	Terminate GOC to Pen Panel Cables	15
4	Connect Peripherals to GOC	20
	1 person	**55**
MI 5 – M2	Final Checks	
5	Verify Console Area Connections	15
6	Final Room Clean–up. *Cable Cover Install / Floor Tile Install & Trash Removal*	80
7	Service Engineer Handoff (Computer Check Sheet)	65
8	Tool Pack–up and Final Review	25
	1 person	**185**
MI 5 – E1	Terminations	
1	Cable Runs 762 – 764. *Strip all 3 gradients to 18 inches, Terminate Connection*	60
2	1183 (blower). *Strip, Terminate Connection*	10
3	1081 (PC Post)	10
4	970 (GWC)	5
5	1234 (RFI Amp). *Strip, Terminate Connection*	10
6	1182 (SSM)	5
7	968 (RF System). *Strip, Terminate Connection*	10
8	1179 (Fiber Optic), sys to PDU cabinet. *Attach Plastic Head, Terminate Connection*	5
9	971 (sys interconnect)	5
10	1180 – 1181	10
	1 person	**130**
MI 5 – E2	Final Electronics Room Set–up	
11	Organize Room (floor tiles, cabinets)	40
12	Verify Equipment Room Connections	30
13	Install Pen Panel Cvrs – Equipment Rm	30
14	Tool Pack–up and Final Review	25
	1 person	**125**
	MI 5 Elapsed Time Totals	**495**
	Per Person Totals	
	Total Minutes Magnet Room Person	**240**
	Total Minutes Electronics Room Person	**255**

10–3 CABLE IDENTIFICATION

Cables delivered to site for installation have specifications dependent on its usage. The following tables contain information on cable runs for this system as it was originally installed.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
“FROM” & “TO”	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material my be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material’s ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable’s use in the MR system	

10–3–1 PDU POWER AND GROUND CABLES

The cables listed in Table 10–1 are connected at one end to the PDU module in the ACGD/PDU Cabinet.

TABLE 10–1
PDU POWER AND GROUND CABLES

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW–1	System Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	0	105	VW–1	Magnet Ground Cable
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW–1	Blower Cabinet Separate ground
968	MR1 A12 J1	MR3 PD1	35 ft Power (5–#8)	SO	600	208Y	75	FT–4	System Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3–#10)	STO	600	208	90	FT–4	ACGD Water Chiller Power
1081	OW1 A19 J1	MR3 PD1	65.5 ft Power (4–#10)	SO	600	120	75	FT–4	Operator Workspace Power
1182	PP1 PHPS PWR	MR3 PD1	61 ft. Power (3–#10)	AWM	600	120	90	FT–4	PHPS Module Power
1183	PP1 F7, F8	MR3 PD1	41 ft Power (3–#10)	SO	600	208	105	VW–1	Blower Cabinet Power
1234	MR2 RF AMP	MR3 PD1	35.5 ft Power (4–#8)	SO	600	208	90	FT–4	RF Amplifier Power
1254	PD1 GND	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	0	105	VW–1	RF Shield Ground Cable
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5–#8) With 30A Plug and Receptacle						Cable for Compact PDU to GE Magnet Power Supply or SC Shim Supply Power

10–3–2 MAGNET ROOM CABLES NOT CUT TO LENGTH

The cables listed in Table 10–2 are routed and connected in the Magnet Room and are not cut to length.

TABLE 10–2
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Table Elevation motor power
414	MG3 A35	MG3 A2 J9	30 ft. 9–pin Sub–D	CI3	300	<30	80	FT–4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Long Drive Encoder/Limit Switch
458	PP1 J17	OM3	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional Remote O2 Sensor
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	PAC Power
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F “N”	STO	2500 VDC	950	85	VW–1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F “N”	STO	2500 VDC	950	85	VW–1	Body Coil Drive
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT–4	Patient cooling blower power
786	PP1 F7, F8	MG6 J2	40 ft. 2 cond power cable	SO	600	120	80	FT–4	Gradient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub–D/rt. angl	CI3	300	<30	80	FT–4	Operator Display to PAC II
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW–1	Spliiter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub–D/26 pin	CI3	300	<30	80	FT–4	Operator Display to SRI

10–3–2 MAGNET ROOM CABLES NOT CUT TO LENGTH (Continued)

TABLE 10–2 (CONT'D)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1197	PP1 RX TX SW J150	MG3 16CHSW J25	5.8 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	
1200	PP1 J200	MG3 16CHSW J1	8–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1201	PP1 J201	MG3 16CHSW J2	8–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1202	PP1 J202	MG3 A17 J3	2–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1203	PP1 MC SW FLTR J203	MG3 16CHSW J11	49 ft. 37 pin Sub–D	CI3	300	<30	80	Ft–4	
1215	MG3 16CHSW J20	MG2 SRI J20	37 PIN SUB–D	CI3	300	<30	80	FT–4	
1216	PP1 A16 J77	MG3 16CHSW J5	25/15 pin Sub–D	CI3	300	<30	80	FT–4	
1217	PP1 A15 J78	MG3 16CHSW J10	37 pin Sub–D	CI3	300	<30	80	FT–4	
1250	MG3 A3 J1	CONNECT TO RUN 257	Times Microwave LMR 600FR	CATV R	6000 VDC		8	FTR	
1260	MG2 A33 J10	MG2 16CHSW	35 ft 3 cond w/ 9–Sub D	CI3	300	<30	105	FT–4	Bore Temperature Sensor
1261	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80–#28)	CI2	300	<30	80	VW–1	SRI Splitter to Oper Display

10–3–3 MAGNET MONITORING CABLES

The Magnet Monitoring cables listed in Table 10–3 are delivered with the Magnet.

TABLE 10–3
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8–#22) & Data Cable (4–#22)	CI3	300	<30	80	FT–4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT–4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8–#24)	CI3	300	<30	80	FT–4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8–#22)	CM	300	<30	80	FT–4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6–#24)	CM	300	<30	80	FT–4	Compressor to Compressor I/F
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor to Modem
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

10–3–4 GE MAGNET CABLES

The Magnet cables listed in Table 10–5 arrive on site with the magnet and are installed at the time of Magnet delivery.

TABLE 10–4
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12–#22)	CI3	150	60 VDC	80	FT–4	Switch Heaters
605	MS1	MR2 A8	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Instrumentation Cable, 80 feet
606	MS4	MS1 A3 A1	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Emergency Run Down Unit
608	MS1	MR2 A8	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Instrumentation Cable, 15 feet
609	MS7 OR MS8	PDU	Power Cable (5–#16)	SO	300	208Y	105	FT–4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Cold Head Control Cable

10–3–5 FIBER OPTIC CABLES

The Fiber Optic cables listed in Table 10–5 are routed and connected in the Magnet Room and the Equipment Room.

TABLE 10–5
FIBER OPTIC CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
711/ 712	PP1–J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC–II
1045	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1083	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1179	MR2 A11 J17	MR3 A11 J2	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic cables

10–3–6 EQUIPMENT ROOM AND MAGNET ROOM CABLES AND HOSES CUT TO LENGTH

The Gradient cables and the RF Head and Body cables listed in Table 10–6 will be routed, cut, and terminated at designated cable ends according to instructions found in the CABLE INSTALLATION section of this manual.

TABLE 10–6
EQUIPMENT ROOM AND MAGNET ROOM CABLES & HOSES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
235	CONNECT TO RUN 1259	PP1 J44	Times Microwave LMR–900FR	CATVR	5000 VDC		85	FT–4	1.5T Body Coil Run created from kit
257	PP1 J44	CONNECT TO RUN 1250	Times Microwave LMR–900FR	CATVR	5000 VDC		85	FT–4	1.5T Body Coil Transmit
762	MR3 CABI/F J3	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
762	MG3 A32 1/2	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
763	MR3 CABI/F J4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
763	MG3 A32 3/4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
764	MR3 CABI/F J5	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output
764	MG3 A32 5/6	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output
1198	PP1 RFTXSW J151	MG3 PEDI/F J1	Times Microwave LMR–600FR	CATVR	4000 VDC		85	FT–4	RF Head Transmit
1199	PP1 RX TX SW J153	MG3 PEDI/F J2	Times Microwave LMR–600FR	CATVR	4000 VDC		85	FT–4	RF Head Transmit
1238	PP1 J4	CONNECT TO RUN 1249	Times Microwave LMR–600FR	CATVR	6000 VDC		85	FT–4	RF Head Transmit

10–3–7 EQUIPMENT AND OPERATOR ROOM CABLES

The cables listed in Table 10–7 are routed and connected between Penetration Pnael, cabinets in equipment room and Operator Workspace.

TABLE 10–7
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional O2 Monitor cable
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	RF Door switch
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	PDU Status (ACGD)
1056	OW1 A17 PRT6	OW1–A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT–4	Laptop Access cable
1059	OW1 A17 PRT2	OW1–A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT–4	Lan Hub cable
1060	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	WIM Power cord
1061	OW1 A15 KYBD/ MS	OW1 A21 J8	2.9 ft. 25 pin Sub–D 2 PS/2s	CI3	300	<30	90	CL–2	Keyboard/Mouse cable
1063	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW–1	Host LCD Display Video cable
1064	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW–1	SCSI Tower Data cable
1065	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	Host LCD Display Power cable
1066	OW1 A15	OW1 A13	3.9 ft. 25/9 pn Sub–D	CM	300	<30	80	FT–4	Modem I/O cable
1067	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW–1	DASM I/O cable
1068	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15–#28)	CI3	300	<30	80	VW–1	LCD Display Video cable
1069	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	LCD Display Power cable
1070	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	SCSI Tower Power cable
1071	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	Host Computer Power cable
1074	OW1 A19 J2	OW1 A11	3.3 ft. Power (3–#14)	SVT	600	120	105	VW–1	Power Strip cord
1075	OW1 A19 J4	OW1 A11	3.3 ft. Power (3–#14)	SVT	600	120	105	VW–1	Power Strip cord
1082	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT–4	EXCITE Ethernet C'cable
1085	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Intercom
1180	MR3 J7	MR2 A11 J56	26 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Can
1181	MR3 J8	PP1 PHPS J154	65 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Can
1184	MR2 A11 J125	PP1 PHPS J32	65 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	
1185	MR2 A11 J200	PP1 J200	8–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1186	MR2 A11 J201	PP1 J201	8–Coax 8W8 Combo D	CATV E	1500	<30	85	FT–4	
1187	MR2 A11 J202	PP1 J202	2–Coax 8W8 Combo D	CATV R	1500	<30	85	FT–4	
1188	MR2 DRV J3	PP1 A1 J77	65 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	16 Channel SW Cntrl Power
1189	MR2 DRV J5	PP1 A17 J78	65 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	MC Drive
1190	MR2 DRV J10	PP1 J203	65 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	MC Drive

10–3–7 EQUIPMENT AND OPERATOR ROOM CABLES (Continued)

TABLE 10–7 (CONT'D)
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1191	MR2 DRV J21	PP1 A11 J92	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1192	MR2 DRV J22	PP1 A11 J93	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1193	MR2 DRV J23	PP1 A11 J94	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1194	MR2 DRV J24	PP1 A11 J95	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1195	MR2 DRV J25	PP1 A11 J96	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable
1196	PP1 A17 J1	PP1 PHPS J35	2 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	Repeater Board Power
1249	MR8 J3	CONNECT TO RUN 1238	5 ft. Times Microwave LMR	CATV R	6000 VDC		85	FT–4	RF Extension cable
1259	MR2 RF AMP J4	CONNECT TO RUN 235	5 ft. Times Microwave LMR 400FR	CATV R	2500 VDC		85	FT–4	RF Extension cable
1255	OW1 A21 J18	MR2 A11 J26	60 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	E–stop
1256	OW1 A7, A8, A20	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub–D	CI3	300	<30	80	VW–1	SCIM, Keyboard, Mouse cable

10–3–8 CABLE SPARES (FRU'S)

The generic cables listed below are designated as replacement cables for the associated cable Runs.

CABLE PART NUMBER	CABLE RUN
46–244710G30	Run 1227 Run 1236
46–244712G30	Run 1188
46–244713G30	Run 1181
46–244714G30	Run 1180 Run 1181 Run 1233
46–258877G30	Run 1189 Run 1190

10–4 FIBER OPTIC TERMINATION INSTRUCTIONS

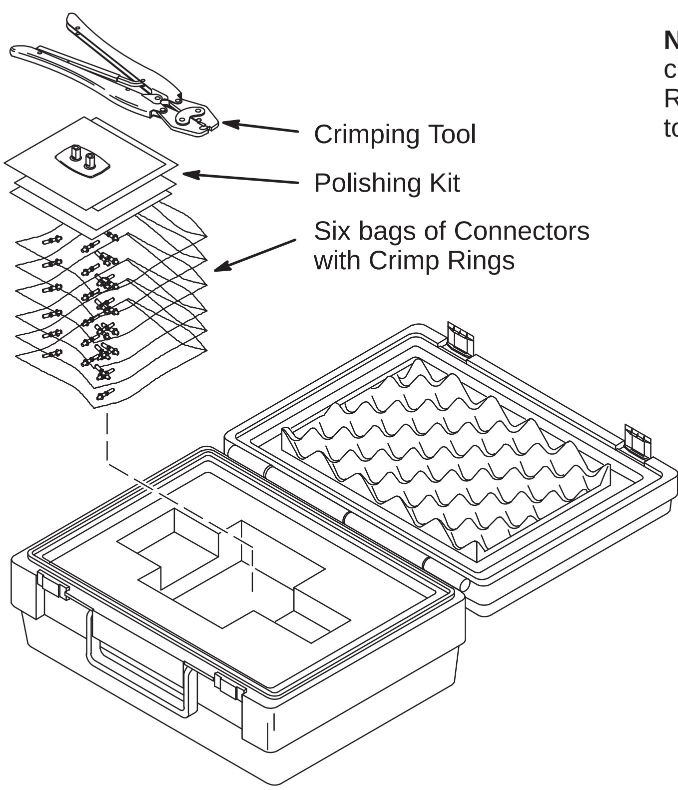
WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

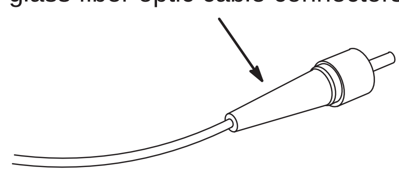
10–4–1 FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

1. The Fiber Optic Repair Kit, 46–301450G1 shown below, contains all the necessary materials to replace a damaged connector.
2. The “Spares Kit”, delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.
3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. f it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.



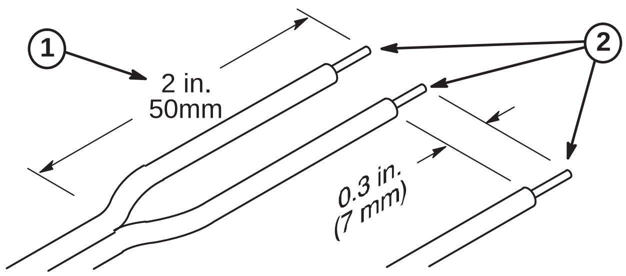
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



10–4–2 STRIPPING FIBER OPTIC OUTER JACKET

The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

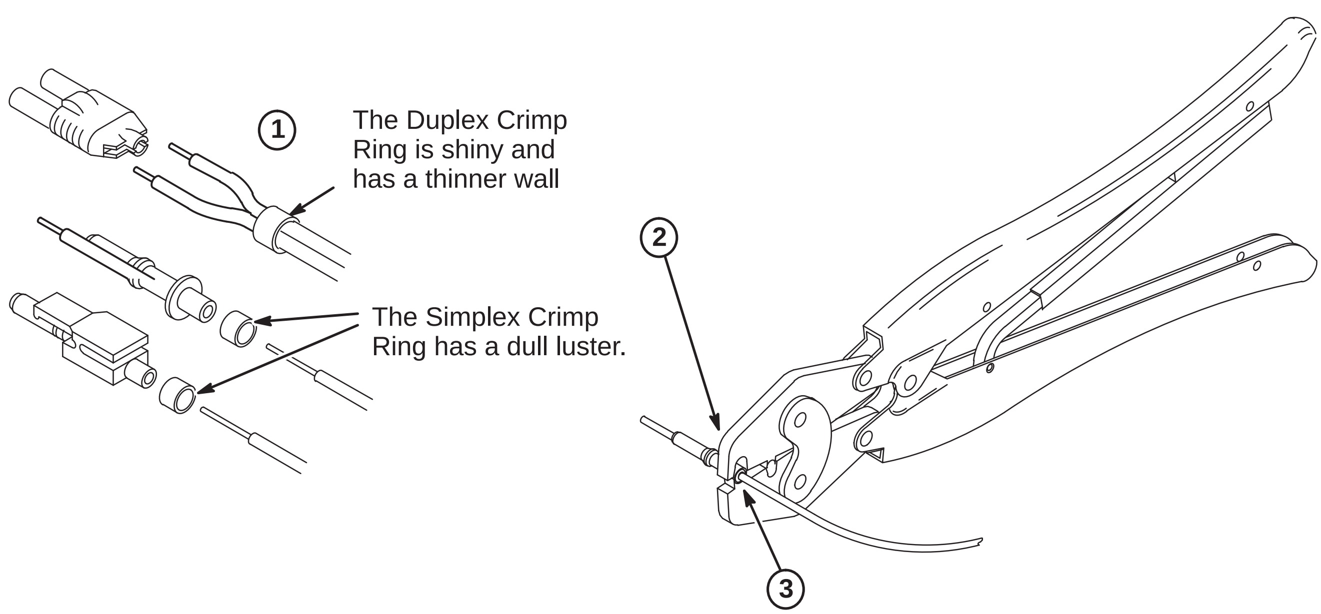
1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



10–4–3 CRIMPING CONNECTOR TO FIBER OPTIC CABLE

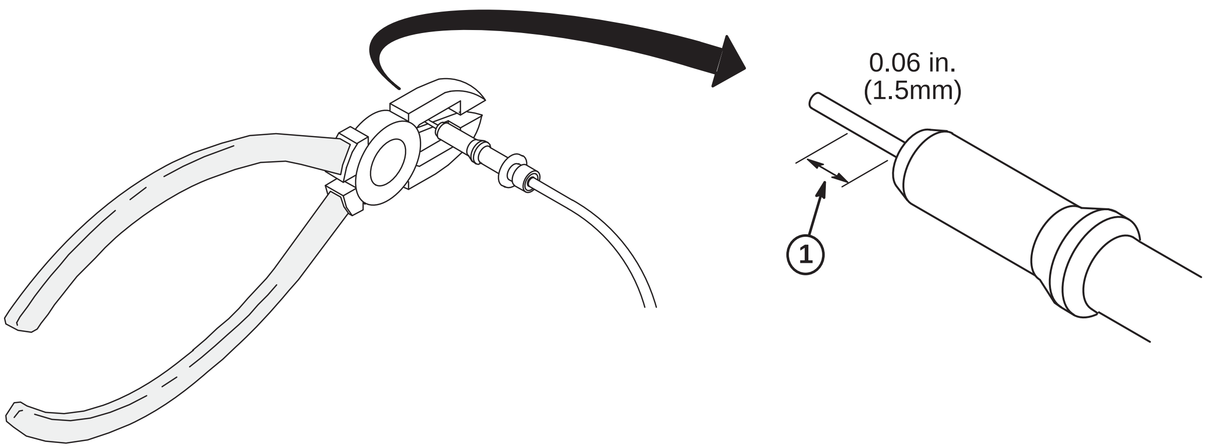
The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

1. Place the Crimp Ring and Connector over the end of the cable.
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.
3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.



10–4–4 TRIMMING EXCESS FIBER FROM CONNECTOR

1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 inch (1–1/2mm).



10–4–5 POLISHING FIBER OPTIC END

The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.